Demand-System Asset Pricing: Theoretical Foundations*

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Abstract

Recent asset pricing methods estimate investor-level demand for financial assets. We show that leading demand systems omit the central mechanism of portfolio choice: substitution depends on payoff similarity under a latent measure reflecting the curvature of utility. Consequently, they do not enforce no arbitrage and may be severely biased—estimated elasticities can be near one even if true elasticities approach infinity. This reconciles low estimates with higher theoretical benchmarks. Controls eliminate bias only under global mean-variance optimization with exactly measured payoffs, or when payoffs never overlap. Estimated elasticities are not structural parameters and do not show that markets are inelastic.

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1 Introduction

Recent approaches to asset pricing following Koijen and Yogo (2019) (hereafter, KY19) aim to estimate investor-level demand functions for financial assets—namely, the functions which determine how portfolios respond to ceteris paribus variation in asset prices.[1] Proponents of this approach argue that granular measurement of demand functions offers new insights into the functioning of financial markets, including equilibrium price determination, policy transmission and a wide array of counterfactuals.

Perhaps the most striking claim of this literature is that demand elasticities for financial assets—namely, the percentage change in asset holdings given a ceteris paribus percentage change in a given asset price—are orders of magnitude smaller than in calibrations of canonical models. For example, Koijen and Yogo (2021) report a cross-asset median elasticity of 0.4, which is "three orders of magnitude smaller than those implied by calibrations of standard asset pricing models."

The validity of this claim—and the invariance of estimated demand functions in counterfactual experiments more generally—hinges on whether estimation methods indeed recover investors' true demand elasticities from observed portfolio holdings. The main challenge is that observational data reflect *equilibrium* behavior, and thus do not directly reveal investor responses to counterfactual prices. As such, the mapping of observational data into demand functions requires a priori assumptions on investor behavior, and it is these assumptions which ultimately determine the recovered parameters.

In this paper, we show that leading demand estimation frameworks for financial assets, such as the logit framework of KY19, effectively assume away a foundational property of modern portfolio theory, which is that substitution between assets depends on their (utility-weighted) payoff overlap across states of the world. We give this notion a precise form. The relevant overlap is the second-moment matrix of payoffs under what we call the *substitution measure*: a reweighting of states by the slope of the investor's marginal utility (i.e., $u''(\cdot)$), which plays for substitution a role analogous to that of the risk-neutral measure for pricing. Two assets are substitutes to the extent that their payoffs covary under this measure, not under the physical measure. We term this property *payoff-based demand complementarities*, and show that it is absent from KY19's logit demand framework and its close cousins, such as the nested logit model. Instead, these frameworks assume that differences in substitution across assets are determined purely

[1] This price response can be jointly determined by multiple parameters, such as preferences and constraints. Beyond questions of identification, an additional issue is whether these demand functions remain stable under counterfactuals (Hurwicz, 1962; Marschak, 1953). We focus primarily on whether demand functions are properly estimated within a given equilibrium; Section 6.1 discusses whether they can be interpreted as stable parameters. See also Fuchs, Fukuda, and Neuhann (2026) for complementary results.

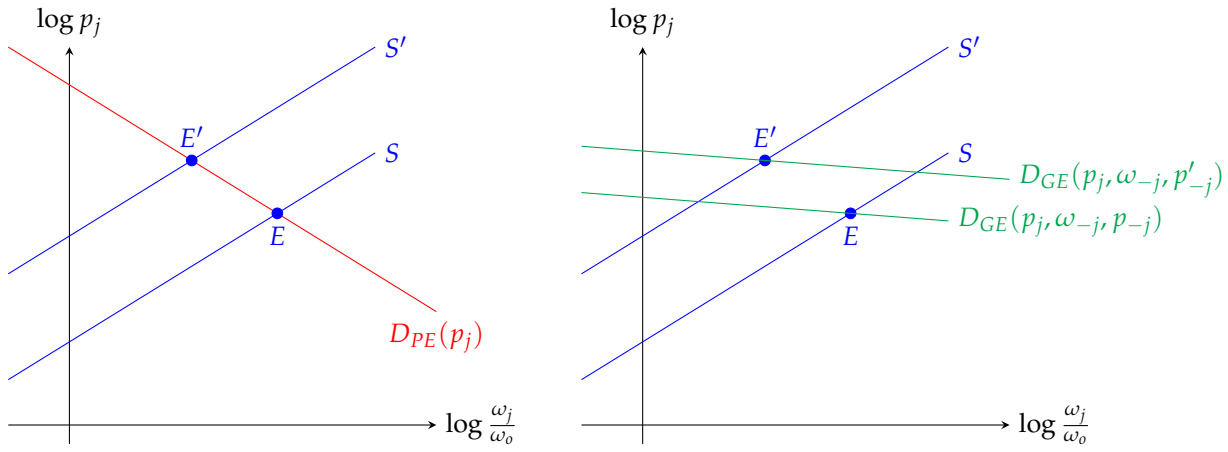
by their portfolio shares. That is, an investor who holds two assets in equal weights must treat them as *equally good substitutes for any other asset, no matter their payoff correlations.* This is not an approximation, but an exact consequence of logit's Independence of Irrelevant Alternatives (IIA) property. Using equilibrium models, we then show that this omission can produce severe bias in estimated elasticities. For example, the estimation method in KY19 can yield estimated elasticities near one even if the true elasticity is unbounded. The surprisingly low estimated elasticities from this literature—which deviate by *orders of magnitude* from canonical benchmarks—can thus be explained by a large bias from improperly accounting for payoff-based demand complementarities.

A simple thought experiment is instructive. Consider an investor who chooses among three assets: two stocks with highly correlated payoffs, say Microsoft and Apple, and a bond that is a poor substitute for either, each representing one third of her portfolio.[2] Now consider a supply shock to Microsoft that raises its price relative to Apple and the bond. If all other prices were to remain fixed, the investor would like to shift sharply from Microsoft to Apple, leaving her bond holdings roughly unchanged. But if many investors make the same trade, the equilibrium price of Apple must increase, reducing the incentive to switch from Microsoft to Apple. The equilibrium change in portfolios is thus much smaller than if Apple's stock price had remained fixed.

What is the stock-level elasticity recovered from this episode? It depends on the substitution patterns considered by the econometrician. Suppose she assumes, as logit does, that Apple and the bond are equally good substitutes for Microsoft—after all, their portfolio shares are the same. Then an elastic investor would have shifted out of Microsoft into Apple and the bond alike, and the rise in Apple's price would not affect her holdings of Microsoft relative to the bond. Since relative holdings barely changed, the econometrician concludes that the investor is very inelastic. Of course, this inference is mistaken because substitution is asset-specific and occurs mainly *between* Microsoft and Apple. We use *spillover bias* to describe the gap between the elasticity inferred from equilibrium data and the *structural elasticity* that governs behavior when all other prices remain fixed.

Figure 1 illustrates. Both panels depict the same supply shock and the same equilibria E and E'; they differ only in the assumed demand function. Under logit (left), substitution is symmetric conditional on shares: any third price moves demand for asset j and for an outside asset alike and cancels in their ratio, so relative demand depends on p_j alone. The observed change is thus interpreted as a move along an inelastic curve. Un-

[2] Equal shares serve only to simplify the exposition. Under logit, the investor reallocates toward each other asset in proportion to its portfolio share, irrespective of payoffs; with unequal shares she would therefore substitute into Apple and the bond in proportion to their shares rather than equally. However, the inference that follows is unchanged.



(a) Logit demand

(b) Payoff-based demand complementarities

Figure 1: Elasticity measurement based on different demand specifications. The left panel corresponds to logit (KY19): relative demand for a given asset is invariant in the prices and quantities of other assets. The right panel has payoff-based demand complementarities. We use ω_j to refer to the portfolio share of asset j, and ω_{-j} to denote the vector of portfolio shares of assets other than j. Analogous definitions hold for asset prices p_j. Demand is measured in portfolio shares relative to ω_o, the portfolio share of the outside asset.

der payoff-based substitution (right), other prices enter relative demand and, since they respond to the shock, the same observation reflects a *shift* of a highly elastic curve.

Spillover bias is not merely statistical omitted variable bias to be purged, it reflects the very demand parameters one seeks to estimate. When substitution is payoff-based, market clearing must reprice every asset whose payoffs overlap with the shocked asset. The spillover's magnitude depends on how elastic investor demand is. Low measured elasticities thus carry little information: the more elastic investors are, the larger the spillover, and the greater the downward bias. Since the problem is misspecified substitution, the bias does not depend on whether one uses time-series or cross-sectional data, or on the asset characteristics used to proxy for expected returns—under logit, any set of characteristics produces the same substitution matrix conditional on portfolio shares.

A demand model which fails to recover the correct price elasticity may nevertheless perfectly match observed price and quantity data: one can simply adjust the location of the demand function. In KY19 this can be done by varying "latent demand," which captures cross-investor heterogeneity in asset valuations *conditional* on prices and asset characteristics. KY19 estimate that this variable accounts for roughly 81% of the cross-sectional variance of stock returns, dwarfing the contributions of characteristics and supply. They interpret this as implying that sentiment and disagreement are central to the cross section of returns. We offer a different interpretation: if one ignores the demand complementarities that actually shape portfolio choice, these are (incorrectly) attributed to latent demand as a residual. Large and variable latent demand is then a symptom

of identification failure rather than an independent economic force, and should not be interpreted as indicating that markets are indeed slow to respond to price changes.

The absence of payoff-based complementarities also raises a more foundational issue. KY25 argue that a demand system—the set of all investor demand functions combined with market clearing—encodes the same information as the stochastic discount factor, because no arbitrage is equivalent to the existence of a solution to the canonical portfolio choice problem (Duffie, 2001). But that equivalence is specific to the canonical problem, which exhibits payoff-based demand complementarities. Logit, by contrast, cannot enforce the no arbitrage condition used to derive it: substitution based on portfolio shares cannot induce the pair-specific trading strategy that corrects mispricing.

One method of accommodating payoff-based demand complementarities is to condition on common characteristics, so that "similar" assets substitute differently. Haddad, He, Huebner, Kondor, and Loualiche (2025) (hereafter, HHHKL25) formalize this approach: controls can shift the slope of demand, and the symmetry restrictions that logit imposes on all assets are imposed only on assets with identical observables. This is a valuable step beyond logit, adding flexibility while reducing the $N \times N$ elasticity matrix to a relative elasticity and a $K \times K$ substitution block. The question is what makes those observables sufficient. Since similarity is latent, characteristics or loadings on risk factors must serve as proxies. These are typically built from realized payoffs under the physical measure, whereas substitution is governed by second moments *under the substitution measure*, which is itself unknown. The second moments that govern substitution agree under the two measures only if the investor's problem is quadratic in her portfolio—quadratic utility, or CARA with Gaussian payoffs—or if the assets never pay off in the same state. Controlled regressions thus cannot substitute for a priori restrictions on preferences and payoffs; they *require them* to produce well-defined conditional demand functions.

The substitution measure may also help to develop new approaches to asset demand identification. For example, assuming only CARA preferences (but not normality) turns the substitution measure into the risk-neutral measure. Some required moments can then be measured using option prices, bringing new data to bear on demand estimation.

Lastly, we ask whether asset demand elasticities, even if well-measured, are policy-relevant structural parameters. In a dynamic setting, demand elasticities alone cannot distinguish whether an investor's demand is due to her own tastes or her expectations of others' future demand. A related result holds for constraints: tastes for a given asset can be observationally equivalent to unobserved mandates that constrain portfolio choice. Yet constrained and unconstrained investors respond differently to counterfactual prices.

Related literature

Demand system asset pricing is concerned with investor-level demand functions estimated from asset quantity data. This is a sharp contrast to neoclassical asset pricing, which focuses on price data disciplined by no arbitrage (Ross, 2004). It also differs from existing approaches that do emphasize quantities, such as classical theories of portfolio balance (Tobin, 1969), convenience yields in Treasury markets (Krishnamurthy and Vissing-Jorgensen, 2012), and intermediary asset pricing (He and Krishnamurthy, 2013; Adrian, Etula, and Muir, 2014), which emphasize aggregate demand effects in certain asset classes or markets without estimating specific demand functions. The benefit of the structural approach is that estimated demand functions—to the extent that they recover invariant parameters—can be used to assess counterfactuals of interest.[3]

Similar in spirit to the demand system approach, Harris and Gurel (1986) and Shleifer (1986) use index inclusions to argue that demand curves for individual stocks slope down.[4] However, interpreting the price-quantity response to an inclusion event in terms of a demand slope requires that other asset prices—and the demand curve for the included stock itself—remain fixed. There is evidence against both. Greenwood (2005) documents cross-asset price spillovers: index events reprice stocks outside the event in proportion to their return covariance with the event portfolio. Barberis, Shleifer, and Wurgler (2005) show that inclusion alters a stock's comovement with index and non-index assets, altering the nature of demand complementarities that shape demand functions. The observed response conflates movements along a demand curve with shifts of the curve due to general-equilibrium repricing—much like the logit demand system. It is perhaps not surprising that both approaches yield estimated elasticities of similar magnitudes. Subsequent to our work, Binsbergen, David, and Opp (2025) and He, Kondor, and Li (2026) use dynamic models to show that supply shocks may also change future expected returns and volatility, thereby shifting the very demand curve that one seeks to identify.

Several papers model richer substitution within the logit framework. Nested logit allows assets to substitute differently *across* and *within* predetermined groups (e.g., Koijen

[3]Several papers build on the logit demand system to study substantive questions, including the effects of counterfactual wealth distributions (Koijen, Richmond, and Yogo, 2024), global imbalances and currencies (Jiang, Richmond, and Zhang, 2024), corporate bond markets (Bretscher, Schmid, Sen, and Sharma, 2026; Darmouni, Siani, and Xiao, 2023), asset purchase programs (Breckenfelder and De Falco, 2023), bond market substitution (Nenova, 2025), and stock market competitiveness (Haddad, Huebner, and Loualiche, 2025). While these studies extend the scope of asset demand systems in different ways, they do not address the specific issues of complementarities and spillovers we discuss here.

[4]See also additional studies of index inclusions (Chang, Hong, and Liskovich, 2015; Pavlova and Sikorskaya, 2023; Greenwood and Sammon, 2025), and research on fund flows (Gompers and Metrick, 2001; Coval and Stafford, 2007; Lou, 2012; Ben-David, Li, Rossi, and Song, 2022; Hartzmark and Solomon, 2025; Li, 2025) and central bank interventions (Krishnamurthy and Vissing-Jorgensen, 2011; Selgrad, 2023).

and Yogo, 2026; Chaudhary, Fu, and Li, 2023), which relaxes IIA across nests but preserves it within them. HHHKL25 instead let observable characteristics shift the *slope* of demand, so that observably similar assets are closer substitutes; An and Huber (2024) model substitution along a small number of factors in foreign exchange, and Nenova (2025) uses a rich set of covariates in bond markets. These approaches measure similarity in a space of observables. However, the similarity governing portfolio choice—payoff overlap under the substitution measure—is not a primitive of any such space, and controls recover it only in the CARA-Normal model with exactly measured payoffs, or with disjoint payoffs.

There are limited methods for dealing with complementarities in structural demand estimation (Berry and Haile, 2021).[5] In the context of financial markets, Allen, Kastl, and Wittwer (2025) estimate a model of complementarities in simultaneous auctions of treasury bills. However, their setting features a small number of assets and data on bid *schedules*, not just portfolio holdings, and relies on parametric assumptions. Davis, Kargar, and Li (2025) share our interest in accounting for low measured elasticities but use a partial equilibrium approach without endogenous price spillovers.

Koijen and Yogo (2025) (hereafter, KY25) provide a direct response to an earlier version of this paper, arguing that spillover bias is due to the specific estimator we employ, not the structure of the logit demand system. Section 4.5 shows that this claim rests on a mathematical error: in our benchmark model, their proposed cross-sectional estimator is unbiased only under a knife-edge symmetry restriction. Outside of this non-generic restriction, their estimator *has the wrong sign* and does not vary with the true elasticity even when the true elasticity diverges. Appendix B shows analogous results for the CARA-Normal model studied in a different section of KY25. We conclude that spillover bias due to demand complementarities is a pervasive concern for asset demand estimation.

By allowing for tastes over assets, our paper also contributes to a growing literature in which investors hold securities because of non-pecuniary values. Fuchs, Fukuda, and Neuhann (2025a) show that "tastes" are orthogonal to returns if investors have deontological preferences, but not if they are consequentialist.

2 Framework

We study the properties and identification of asset demand systems using a canonical general equilibrium economy based on Lucas (1978). To capture heterogeneous portfolios and permit identification, we enrich this model with investor "tastes"—heterogeneous

[5] In industrial organization, existing research typically considers small choice sets with limited complementarities. Gentzkow (2007) studies newspaper demand with a choice between print, online, or both.

valuations for financial assets. We begin by reviewing the rationale for such tastes.

2.1 The importance of heterogeneous valuations

The basic problem of demand estimation is that quantities and prices are jointly determined in equilibrium. To identify demand functions, one must therefore rely on instruments which provide as-if random variation in asset supply. Since aggregate supply is typically considered fixed and any given investor may choose to sell or buy an asset depending on the price, the literature on demand-system asset pricing typically focuses on variation in the *residual supply curve* faced by an individual investor, defined as aggregate supply minus the demand of some subset of market participants.

For this approach to work, one must find demand shocks to a subset of investors which are uncorrelated with the demand of other investors. These demand shocks must therefore not be driven by correlated shocks to discount rates, common changes in expected payoffs, or belief revisions in response to the trades of others. Since financial assets are investment goods whose current value generically depends on their resale value, these requirements extend to future prices.

Two microfoundations for such shocks have been proposed. The first is investor heterogeneity in *tastes* for particular assets, holding fixed expected cash flows. These might arise from differences in preferences over the *provenance* of cash flows, such as when some investors prefer to invest in socially responsible firms. Or, investors may have dogmatic beliefs about returns that are orthogonal to the beliefs of other investors.[6] The second is constraints or investment mandates that prevent some market participants from investing in a particular asset for *exogenous* reasons. We therefore use a framework that allows for both heterogeneous tastes (or dogmatic beliefs) and flexible constraints. Nevertheless, heterogeneous valuations are in some tension with the organizing principle of no arbitrage. Appendix C shows that no arbitrage might fail given heterogeneous tastes, so that equilibrium prices may fail to exist (without additional constraints).

2.2 Formal model

We consider a one-shot portfolio choice problem in which an investor can choose to consume at date 0 and/or at date 1. Section 6.1 considers a simple dynamic extension.

There is a unit continuum of investors indexed by i. Investor i has a strictly concave and twice-differentiable von Neumann–Morgenstern utility function defined over lotter-

[6] One can also formally link tastes and heterogeneous beliefs. Specifically, the state space over which payoffs are defined can be enriched to include "taste-based payoffs." Heterogeneous tastes then map into heterogeneous beliefs if investors differ in their subjective probabilities over this augmented state space.

ies which determine her consumption. A random state of the world $z \in \mathcal{Z} \equiv \{1, \ldots, Z\}$ is realized at date 1, and the probability of state z is $\pi_z \in (0,1)$. The set of assets is $\mathcal{J} \equiv \{1, \ldots, J\}$. Asset $j \in \mathcal{J}$ offers state-contingent cash flows $y_j(z)$ in state z. Investor i is endowed with e_j^i units of asset j and non-asset endowment w_0^i and $w_1^i(z)$ at dates 0 and 1, respectively. The aggregate endowment of asset j is $E_j = \int_i e_j^i di$. While the main text focuses on a finite state space, Appendix B presents a complementary CARA-Normal framework where cash flows follow a continuous distribution.

Within this framework, we introduce payoff-augmenting tastes, defined as additional "consumption-equivalent" value that is generated by a particular asset.[7] These taste parameters are designed to capture the heterogeneity in asset valuations that is required for a meaningful notion of exogenous shocks to residual demand. Formally, we say that investor i evaluates her payoffs from holding portfolio $a^i \equiv (a_j^i)_{j \in \mathcal{J}}$ by both the cash flows it generates and her *tastes* $\theta^i \equiv (\theta_j^i)_{j \in \mathcal{J}}$ over assets, where $\theta_j^i > 0$. Preferences are then defined over *effective units of consumption* delivered by portfolio a^i, defined as

$$\tilde{c}_1^i(z) \equiv \sum_{j \in \mathcal{J}} \theta_j^i y_j(z) a_j^i + w_1^i(z).$$

Investors may also be subject to portfolio constraints such as short-sale constraints, investment mandates, or restrictions on portfolio weights on particular asset classes. To capture these considerations in a flexible manner, we assume that investor i faces $N \geq 0$ investment constraints on portfolio choices. The n-th investment constraint is defined as

$$M_n^i(a^i, p) \geq 0,$$

where a^i is investor i's portfolio, $p \equiv (p_j)_{j \in \mathcal{J}}$ is the price vector, and the function $M_n^i(\cdot)$ is twice continuously differentiable in a_j^i for all j. We assume that the set of feasible portfolios induced by these constraints is convex.

[7] In our approach, tastes multiplicatively augment consumption. As we will show, tastes thus operate like "latent demand" shifters in KY19. An alternative approach is to specify additively separable tastes, whereby an investor obtains some additional value (or disutility) from holding certain assets that is separable from risk-return considerations. Overall, both formulations deliver essentially identical conclusions.

Given these assumptions, investor i's portfolio choice problem is:

$$\begin{aligned} \max_{a^i} \quad & u^i(c_0^i) + \delta \sum_{z \in \mathcal{Z}} \pi_z u^i(\tilde{c}_1^i(z)) \\ \text{s.t.} \quad & c_0^i = w_0^i - \sum_{j \in \mathcal{J}} p_j (a_j^i - e_j^i) \\ & \tilde{c}_1^i(z) = \sum_{j \in \mathcal{J}} \theta_j^i y_j(z) a_j^i + w_1^i(z) \text{ for all } z \\ & M_n^i(a^i, p) \geq 0 \text{ for all } n, \end{aligned} \tag{1}$$

where $\delta \in (0,1)$ is a discount factor and the first two constraints are budget constraints at time 0 and time 1 in state z. Throughout, $\lambda^i \equiv u^{i\prime}(c_0^i)$ denotes the marginal value of date-0 consumption, where $u^{i'}$ is marginal utility. Section 4.3 works with the pure portfolio choice variant in which the investor consumes only at date 1 and allocates her budget across assets; our results apply to that case with λ^i defined as the Lagrange multiplier on the date-0 budget constraint and no other binding constraints.

Our equilibrium concept is the standard notion of competitive equilibrium.

Definition 1 (Competitive equilibrium) *A competitive equilibrium consists of asset prices p and investor portfolios $(a^i)_i$ such that:*

1. *Given asset prices p, investor portfolio a^i solves decision problem (1) for each i.*
2. *The consumption goods market clears in every state.*
3. *Financial markets clear for every asset: $\int_i a_j^i di = E_j$ for each $j \in \mathcal{J}$.*

2.3 Demand functions and structural elasticities

Given a price vector p, portfolio choice problem (1) yields a set of J asset-level demand functions $a_j^i(p)$ which determine investor i's optimal portfolio position in each asset given prices p. (Demand functions can also be stated in terms of portfolio shares. In this case, investor i's portfolio share of asset j is denoted $\omega_j^i \equiv \frac{p_j a_j^i}{W^i}$, where $W^i \equiv \sum_{j \in \mathcal{J}} p_j a_j^i$ is the market value of investor i's portfolio.) We now describe the main properties of these demand functions that are important for our analysis.

Since preferences are defined over effective units of consumption, the marginal rate of substitution between state z and date 0 is

$$\tilde{\Lambda}^i(z) \equiv \frac{\delta \pi_z u^{i'}(\tilde{c}_1^i(z))}{\lambda^i},$$

where $\lambda^i = u^{i'}(c_0^i)$. The net benefit of buying an asset is the marginal value of additional effective consumption relative to its cost and impact on portfolio constraints. Accordingly, we define the *net marginal value of asset* j as

$$F_j^i(a^i, p) \equiv \theta_j^i \sum_{z \in \mathcal{Z}} y_j(z) \tilde{\Lambda}^i(z) + \sum_n \mu_n^i \frac{m_{n,j}^i(a^i, p)}{\lambda^i} - p_j, \tag{2}$$

where μ_n^i is the Lagrange multiplier associated with the n-th investment constraint, and $m_{n,j}^i(a^i, p)$ the partial derivative of constraint $M_n^i(a^i, p)$ with respect to a_j^i.

The vector of asset-level demand functions $a^i = (a_j^i)_{j=1}^J$ is determined by a set of optimality conditions requiring the net marginal value of every asset to be equal to zero:

$$F^i(a^i, p) \equiv \begin{bmatrix} F_1^i(a^i, p) \\ F_2^i(a^i, p) \\ \vdots \\ F_J^i(a^i, p) \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}. \tag{3}$$

Consistent with the literature, we focus on estimating the elasticity of demand for asset j with respect to variation in asset price p_s. This captures the percentage change in investor i's demand for asset j given a percentage change in the price of asset s, *holding all other prices fixed*. When demand is written in asset quantities, this elasticity is

$$\mathcal{E}_{js}^{a,i} \equiv -\frac{\partial \log a_j^i(p)}{\partial \log p_s}.$$

When demand is written in portfolio shares,[8] the analogous definition is

$$\mathcal{E}_{js}^{\omega,i} \equiv -\frac{\partial \log \omega_j^i(p)}{\partial \log p_s}.$$

Going forward, we refer to this object as the *structural elasticity* because it measures the demand change in response to a *ceteris paribus* change in a single asset price. We also refer to $\mathcal{E}_{jj}^i$ as the own-price elasticity and to $\mathcal{E}_{js}^i$ $(s \neq j)$ as the cross-price elasticity.

[8] The two definitions do not measure the same object: as (11) below shows, they differ by the own-price revaluation $\mathbf{1}_{j=s}$ and a common wealth term. We drop the superscript a or ω when only one convention is in play, and report elasticities in portfolio shares unless stated otherwise.

3 Demand Complementarities and Identification

We now turn to the main challenge of asset demand estimation, which is to develop a demand system which accurately identifies structural demand elasticities from observational data on portfolios and prices.

We emphasize two factors that make this problem particularly challenging: (i) portfolio choice exhibits *payoff-based demand complementarities* whereby asset demand functions are non-separable and substitution depends on payoff similarity between assets, and (ii) in market equilibrium, these complementarities generate *price spillovers* to other assets that alter observed substitution patterns. To show that these considerations affect demand inference even under ideal conditions, we assume that the econometrician observes a perfectly exogenous supply shock to a given asset.

3.1 Payoff-based demand complementarities

We begin by characterizing the demand functions derived from portfolio choice problem (1). The fundamental insight of modern portfolio theory is that the system of first-order conditions (3) exhibits cross-asset linkages, whereby the marginal value of a given asset depends on how its payoffs correlate with those of other assets held by the investor. Consequently, asset demand exhibits *payoff-based demand complementarities*, whereby asset-level demand functions are generically non-separable and substitution is driven by a utility-weighted notion of payoff overlap between assets.

To establish this, we compute the investor's optimal response to a *pure price change*, that is, to a change in one price holding all other prices fixed. This is the thought experiment underlying a demand elasticity. In general, the demand response to a price change depends on portfolio constraints. As a benchmark, we consider a class of constraints from KY19, short-sale constraints and investment-universe restrictions, which operate on the extensive margin and are therefore slack on the set $\mathcal{J}$ of assets the investor holds in strictly positive amounts.[9] The first-order condition (2) with respect to a^i_j reduces to:

$$F^i_j(a^i, p) = \theta^i_j \sum_z y_j(z)\tilde{\Lambda}^i(z) - p_j = 0, \quad \text{where } \tilde{\Lambda}^i(z) = \frac{\delta \pi_z u^{i\prime}(\tilde{c}^i_1(z))}{\lambda^i}. \tag{4}$$

Net marginal value F^i_j depends on the portfolio a^i through $\tilde{c}^i_1$, on the price vector

[9] Payoff-based demand complementarities remain relevant when incorporating intensive-margin constraints; unless the investor is entirely unable to change her portfolio, the desired portfolio change is still driven by payoff-based considerations.

through the term $-p_j$, and on λ^i. The direct effect of a change in the quantity of asset k is:

$$H^i_{jk} \equiv \frac{\partial F^i_j}{\partial a^i_k} = \frac{1}{\lambda^i} \sum_z \delta \pi_z u^{i\prime\prime}\big(\tilde{c}^i_1(z)\big) \theta^i_j y_j(z) \theta^i_k y_k(z), \tag{5}$$

while

$$\frac{\partial F^i_j}{\partial p_k} = -\mathbf{1}_{j=k} \quad \text{and} \quad \frac{\partial F^i_j}{\partial \lambda^i} = -\frac{1}{\lambda^i}\left(F^i_j + p_j\right) = -\frac{p_j}{\lambda^i},$$

where the last equality follows because $F^i_j = 0$.

Equation (5) is the main object of interest, and our measure of *payoff overlap*. It is the second-moment matrix of (taste-adjusted) state-contingent payoffs $\theta^i_j y_j(z)$ under the curvature weights $\delta \pi_z u^{i\prime\prime}(\tilde{c}^i_1(z))/\lambda^i < 0$. Its (j,k) element depends directly on the payoffs of assets j and k, and indirectly on all other assets through the curvature weights. Buying more of asset k lowers the marginal value of asset j depending on their payoff overlap: $H^i_{jk} = 0$ when payoff distributions are disjoint, and $H^i_{jk} < 0$ generically otherwise. Matrix H^i is symmetric, and negative definite if the payoff vectors are linearly independent.[10]

Substitution measure. The curvature weights in (5) can be normalized to a probability measure over states. We call it the *substitution measure* $\hat{\pi}^i$ because the payoff overlap matrix H^i is proportional to the second-moment matrix of taste-adjusted asset payoffs under this measure. Because the curvature weights are investor-specific, so is the substitution measure.

Definition 2 (Substitution measure) *Investor i's substitution measure is the probability measure on $\mathcal{Z}$ given by*

$$\hat{\pi}^i_z \equiv \frac{\pi_z \left(-u^{i\prime\prime}(\tilde{c}^i_1(z))\right)}{\sum_{z'} \pi_{z'} \left(-u^{i\prime\prime}(\tilde{c}^i_1(z'))\right)}. \tag{6}$$

Writing $\widehat{\mathbb{E}}^i$ for expectations under $\hat{\pi}^i$ and $\tilde{y}^i_j(z) \equiv \theta^i_j y_j(z)$ for taste-adjusted payoffs, let

$$K^i_z \equiv \frac{\delta \pi_z u^{i\prime\prime}(\tilde{c}^i_1(z))}{\lambda^i} < 0$$

be the curvature weights in (5) and $\kappa^i \equiv \sum_z K^i_z < 0$. The payoff overlap matrix is

$$H^i = \kappa^i \, \widehat{\mathbb{E}}^i\big[\tilde{y}^i \, (\tilde{y}^i)^{\mathrm{T}}\big]. \tag{7}$$

[10] Non-redundancy is required for H^i to be invertible, and it fails when some portfolio of assets has linearly dependent payoffs. In that case the first-order conditions pin down only the combined position in the redundant assets, individual demands are indeterminate, and compensated substitution is unbounded.

The substitution measure is the analogue of the risk-neutral measure with $-u''$ in place of u'. The risk-neutral measure $Q^i_z \propto \pi_z u^{i\prime}(\tilde{c}^i_1(z))$ prices payoffs, while the substitution measure governs how marginal values respond to holdings. This offers an intuitive formal notion of payoff similarity in portfolio choice: by Lemma 1 below, two assets are substitutes to the extent that their taste-adjusted payoffs covary *under* $\hat{\pi}^i$.

The substitution measure has two useful properties. First, the risk-neutral and substitution measures are related through absolute risk aversion $A^i(c) \equiv -u^{i\prime\prime}(c)/u^{i\prime}(c)$:

$$\hat{\pi}^i_z \;\propto\; Q^i_z \, A^i\big(\tilde{c}^i_1(z)\big). \tag{8}$$

In general, this mapping is latent because consumption is latent. Under CARA preferences, however, the substitution measure *is* the risk neutral measure. Section 5.2 uses this observation to sketch a method to recover demand from state prices.

Second, the substitution measure overweights the states in which marginal utility is steeper than on average. Writing $\bar{u}^{i\prime\prime} \equiv \mathbb{E}_\pi[u^{i\prime\prime}(\tilde{c}^i_1)]$ for the average slope of marginal utility across states, the wedge between the substitution and physical measures is

$$\hat{\pi}^i_z - \pi_z = \pi_z \, \frac{u^{i\prime\prime}(\tilde{c}^i_1(z)) - \bar{u}^{i\prime\prime}}{\bar{u}^{i\prime\prime}}. \tag{9}$$

Two assets with little payoff overlap under the physical measure may thus still be close substitutes if they both pay off in a state with particularly steep marginal utility.

Implications for structural demand. Since $F^i = \mathbf{0}$ holds along the demand function, totally differentiating the system yields the following characterization of demand.

Lemma 1 (Structural Jacobian) *Suppose the assets in $\mathcal{J}$ are unconstrained for investor i and non-redundant, in the sense that her taste-adjusted payoff vectors $\{(\theta^i_j y_j(z))_{z\in\mathcal{Z}} \mid j \in \mathcal{J}\}$ are linearly independent. Denoting by $\big(H^i\big)^{-1}_{jk}$ the (j,k) element of $(H^i)^{-1}$, a pure change in p_k moves the portfolio according to*

$$\frac{\partial a^i_j}{\partial p_k} = \underbrace{\big(H^i\big)^{-1}_{jk}}_{\textit{substitution}} - \underbrace{v^i_k b^i_j}_{\textit{wealth}}, \quad \textit{where} \quad b^i \equiv \big(H^i\big)^{-1} p \quad \textit{and} \quad v^i_k \equiv -\frac{\partial \log \lambda^i}{\partial p_k}. \tag{10}$$

Hence, the Jacobian is

$$\left(\frac{\partial a^i_j}{\partial p_k}\right)_{j,k} = (H^i)^{-1}\left(I - p(v^i)^{\mathrm{T}}\right),$$

where I is the identity matrix. The demand elasticities in quantities and portfolio shares are

$$\mathcal{E}^{a,i}_{jk} = -\frac{p_k}{a^i_j}\left[(H^i)^{-1}_{jk} - v^i_k b^i_j\right] \quad \textit{and} \quad \mathcal{E}^{\omega,i}_{jk} = \mathcal{E}^{a,i}_{jk} - \mathbf{1}_{j=k} + \frac{\partial \log W^i}{\partial \log p_k}. \tag{11}$$

Proof. See Appendix A, which also gives a closed form for v^i. ■

The first term in (10) is *substitution*: since asset k is more expensive, the investor rebalances across assets in the pattern dictated by payoff overlap under her substitution measure, $(H^i)^{-1} = (\kappa^i)^{-1}\widehat{\mathbb{E}}^i[\tilde{y}^i(\tilde{y}^i)^{\mathrm{T}}]^{-1}$. The second term is the *wealth* response: b^i is how the portfolio moves when the investor becomes richer at fixed prices, and v^i_k is the fall in the log marginal value of a date-0 dollar caused by a rise in p_k.

The payoff structure enters the demand Jacobian through H^i, allowing different assets to respond differently to the same price change. We thus compare how two assets respond to the price of a third. *Payoff-based demand complementarities* are present when an investor's optimal relative response depends on the payoff overlap between the two assets and the third asset. We state the definition in relative quantities a^i_j/a^i_ℓ, but it holds identically in portfolio shares, which divide by a common wealth term.[11]

Definition 3 (Payoff-based demand complementarities) *Investor i's demand for assets j and ℓ exhibit payoff-based demand complementarities with respect to a third asset $k \notin \{j, \ell\}$ if:*

(i) investor i's relative demand for assets j and ℓ responds to the price of asset $k \notin \{j, \ell\}$:

$$-\frac{\partial \log\left(a^i_j/a^i_\ell\right)}{\partial \log p_k} = \mathcal{E}^{a,i}_{jk} - \mathcal{E}^{a,i}_{\ell k} \neq 0; \tag{12}$$

(ii) the relative response $\mathcal{E}^{\omega,i}_{jk} - \mathcal{E}^{\omega,i}_{\ell k}$ varies with at least one payoff overlap term H^i_{jk} or $H^i_{\ell k}$.

Investor i's demand system *exhibits payoff-based demand complementarities if there exists at least one triple of assets for which conditions (i) and (ii) hold.*

In general, the presence of *some* assets subject to payoff-based demand complementarities are sufficient to bind the investor's demand system together. Moreover, in the canonical portfolio choice model, *every* triple of assets generically exhibits payoff-based

[11] For $k \notin \{j, \ell\}$, the common portfolio-value term that separates quantity from share elasticities in (11) cancels in the difference, while both own-price revaluation terms are zero, so the derivative of $\log(a^i_j/a^i_\ell)$ and that of $\log(\omega^i_j/\omega^i_\ell)$ coincide. Equivalently, $\mathcal{E}^{\omega,i}_{jk} = \mathcal{E}^{\omega,i}_{\ell k}$ for every third asset k if and only if the relative demand ω^i_j/ω^i_ℓ is a function of p_j and p_ℓ alone.

demand complementarities: combining equations (10) and (11) yields

$$\mathcal{E}^{\omega,i}_{jk} - \mathcal{E}^{\omega,i}_{\ell k} = \underbrace{-p_k \left[\frac{(H^i)^{-1}_{jk}}{a^i_j} - \frac{(H^i)^{-1}_{\ell k}}{a^i_\ell} \right]}_{\text{substitution at fixed } \lambda^i} - \underbrace{\left[\frac{\partial \log a^i_j}{\partial \log \lambda^i} - \frac{\partial \log a^i_\ell}{\partial \log \lambda^i} \right]_p \frac{\partial \log \lambda^i}{\partial \log p_k}}_{\text{wealth effect}}. \tag{13}$$

The first term is the response at a fixed marginal value of wealth: j and ℓ substitute toward or away from k according to their payoff overlap with k. The second term operates through wealth, and only the *difference* in the two positions' elasticities to λ^i enters. Under CRRA utility with no non-asset endowment at date-1 and no binding constraints, the wealth term vanishes, and every position has the same elasticity to λ^i. In this case, differences in relative demand responses are determined entirely by the substitution term, which depends on the pattern of payoff overlap across the portfolio. We later use this case to isolate the payoff-based mechanism we emphasize.

Payoff-based substitution follows directly from the fact that preferences are defined over (taste-adjusted) *consumption*, not asset holdings: since consumption is the *sum* of all asset payoffs, substitution depends on the joint distribution of asset payoffs across states—under the substitution measure, which weights each state by how much the investor's marginal valuation of consumption in that state responds to her portfolio. While this mechanism is generic in standard portfolio theory, it is precisely what KY19 rules out: Section 4.1 shows that their demand system sets (12) to zero for every asset triple, so that portfolio shares must serve as sufficient statistics for every substitution pattern.[12]

3.2 Identification from exogenous supply shocks

We now discuss how payoff-based demand complementarities complicate inference of structural demand parameters from observational data on asset prices and portfolios. To do so, we assume that we have an ideal shock in hand: a *purely exogenous supply shock* χ_s that directly affects only asset s. For example, in the spirit of the instrumental variable strategy of KY19, an outside investor may have decided to adjust her supply of asset s for investor-specific reasons. Importantly, the shock shifts neither the preferences nor payoff expectations of any other investor, but does shift prices. Contrary to the pure price shock, however, a supply shock can lead to endogenous repricing of other assets.

How does an investor respond to this shock? To see this, totally differentiate the first-order conditions (3) with respect to shock χ_s, denoting by $\mathcal{S}_{js} \equiv \frac{dp_j}{d\chi_s}$ the *price spillover*

[12] That complementarities depend on *intensive-margin* holdings distinguishes asset markets from discrete-choice settings with extensive-margin complementarities, such as Gentzkow (2007).

from asset s's shock to asset j's price. Assuming that demand and equilibrium prices are differentiable in χ_s near the initial equilibrium, with $\mathcal{S}_{ss} \neq 0$ and positive prices and holdings, the optimal change of i's portfolio is given by the system of equations

$$\begin{bmatrix} \frac{da_1^i}{d\chi_s} \\ \vdots \\ \frac{da_s^i}{d\chi_s} \\ \vdots \\ \frac{da_J^i}{d\chi_s} \end{bmatrix} = \begin{bmatrix} \frac{\partial a_1^i}{\partial p_1} & \cdots & \frac{\partial a_1^i}{\partial p_s} & \cdots & \frac{\partial a_1^i}{\partial p_J} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ \frac{\partial a_s^i}{\partial p_1} & \cdots & \frac{\partial a_s^i}{\partial p_s} & \cdots & \frac{\partial a_s^i}{\partial p_J} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ \frac{\partial a_J^i}{\partial p_1} & \cdots & \frac{\partial a_J^i}{\partial p_s} & \cdots & \frac{\partial a_J^i}{\partial p_J} \end{bmatrix} \begin{bmatrix} \frac{dp_1}{d\chi_s} \\ \vdots \\ \frac{dp_s}{d\chi_s} \\ \vdots \\ \frac{dp_J}{d\chi_s} \end{bmatrix} + \begin{bmatrix} \frac{\partial a_1^i}{\partial \chi_s} \\ \vdots \\ \frac{\partial a_s^i}{\partial \chi_s} \\ \vdots \\ \frac{\partial a_J^i}{\partial \chi_s} \end{bmatrix}. \tag{14}$$

While only asset s is directly affected by the shock, the observed response is the sum of endogenous portfolio adjustments for *all* assets in the choice set. These adjustments reflect two main channels. The first is the set of *structural responses* $\frac{\partial a_j^i}{\partial p_k}$ (shown in blue), which reflect the investor's ideal response to ceteris paribus variation in a single price, and thus map directly into structural elasticities. In canonical portfolio choice models, these responses reflect payoff-based demand complementarities (Definition 3). The second is the set of price spillovers to other assets $\frac{\partial p_j}{\partial \chi_s}$, which are necessary to clear markets when investors optimally adjust multiple asset positions in response to the shock. The last term on the right-hand side is the direct response to the shock at fixed prices. It is zero when the shock does not affect i's endowment, and captures the implied wealth transfer otherwise; the valuation effect of price changes is part of the price derivatives.[13]

To recover specific structural elasticities from observed behavior, the econometrician must therefore account for the cross-elasticities and price spillovers that occur in response to a shock. The problem is that, even with exogenous supply variation, observational data show only the *total* derivative with respect to all margins of adjustment. We call this the *observed elasticity* $\hat{\mathcal{E}}_{ss}^i$. When demand is written in asset quantities, it is

$$\hat{\mathcal{E}}_{ss}^{a,i} \equiv -\frac{\frac{da_s^i}{d\chi_s}}{\frac{dp_s}{d\chi_s}} \cdot \frac{p_s}{a_s^i}.$$

When demand is written in portfolio shares, we replace a_j^i with ω_j^i in the definition.

To see how the observed elasticity differs from the structural elasticity, we decompose it into its components: structural cross-elasticities—which reflect demand comple-

[13] The two decompositions in Proposition 1 differ only with respect to this direct term. We focus on demand complementarities and spillovers; under the condition of Proposition 1, the direct term vanishes when demand is written in portfolio shares, as in HHHKL25.

mentarities in portfolio choice—and price spillovers $\mathcal{S}_{js}$.

Proposition 1 (Decomposition of the observed elasticity) *When demand is written in asset quantities, the structural and observed own-price elasticities satisfy:*

$$\hat{\mathcal{E}}^{a,i}_{ss} = \mathcal{E}^{a,i}_{ss} + \underbrace{\sum_{j \neq s} \frac{\mathcal{S}_{js}\frac{1}{p_j}}{\mathcal{S}_{ss}\frac{1}{p_s}} \mathcal{E}^{a,i}_{sj}}_{\substack{\text{Cross-elasticities} \\ \text{and spillovers}}} - \underbrace{\frac{\frac{\partial a^i_s}{\partial \chi_s}\frac{1}{a^i_s}}{\mathcal{S}_{ss}\frac{1}{p_s}}}_{\text{Direct response}} . \tag{15}$$

When demand is written in portfolio shares and shares are independent of wealth,[14] *the difference between the structural and observed own-price elasticities has no direct term and satisfies:*

$$\hat{\mathcal{E}}^{\omega,i}_{ss} = \mathcal{E}^{\omega,i}_{ss} + \underbrace{\sum_{j \neq s} \frac{\mathcal{S}_{js}\frac{1}{p_j}}{\mathcal{S}_{ss}\frac{1}{p_s}} \mathcal{E}^{\omega,i}_{sj}}_{\substack{\text{Cross-elasticities} \\ \text{and spillovers}}} . \tag{16}$$

Proof. See Appendix A. ■

A given supply shock thus yields the demand Jacobian applied to one specific *combination* of price changes—the equilibrium price response—rather than the Jacobian itself. The difference between structural and observed elasticities is the *sum* of cross-elasticities weighted by the relative price responses $\frac{\mathcal{S}_{js}/p_j}{\mathcal{S}_{ss}/p_s}$. With many assets, this difference can be sizable even when the spillover to any given asset is small. A demand system aims to recover structural elasticities from observed portfolio changes. As such, it is critical that the demand system accurately captures the mechanisms of substitution and spillovers.

4 Limitations and Biases of Logit Asset Demand

We have shown that observed asset-level elasticities in response to supply shocks generically differ from structural elasticities in which all other prices are held fixed. To recover structural demand parameters from observational data, researchers must therefore rely on theoretical structure to derive a mapping from observed portfolio changes into parameters of interest. This yields useful estimates only if the proposed structure accurately captures the central determinants of investor behavior, including the payoff-based demand complementarities which govern portfolio choice.

[14] Wealth-independent portfolio shares obtain, for instance, under CRRA utility with no non-asset endowment and no binding constraints, as in the economy of Section 4.3.

We now evaluate whether leading frameworks appropriately account for these considerations, beginning with the influential logit demand system in KY19. We show that their model rules out payoff-based demand complementarities *by assumption*. This has stark effects: the demand system generally fails to enforce no arbitrage pricing, and estimated elasticities may be severely biased. Section 5 argues that similar issues apply to related methods, such as nested logit, and semi-structural approaches such as HHHKL25.

4.1 No payoff-based demand complementarities

KY19 describe asset demand in terms of $\omega_j^i(p)/\omega_o^i(p)$, the portfolio shares of asset j *relative* to the portfolio share of an outside asset o. The outside asset plays the role of a benchmark asset (or numeraire). Relative portfolio shares are specified to be log-linear in log prices and a set of factor loadings $(x_h(j))_h$ on asset characteristics; by Assumption 1 in KY19 these characteristics are taken to be sufficient to summarize expected returns and the variance-covariance matrix of returns.[15] In particular, the KY19 specification is

$$\frac{\omega_j(p)}{\omega_o(p)} = \frac{\omega_j}{\omega_o}(p_j) = \exp\left(-\beta_0 \log p_j + \sum_{h=1}^{K-1} \beta_h x_h(j) + \beta_K\right) \cdot \zeta(j), \tag{17}$$

where β_0 is the coefficient on the log price of asset j, $x_h(j)$ is asset j's loading on the h-th characteristics-based factor, $(\beta_h)_{h=1}^{K-1}$ are the associated demand coefficients, and β_K and $\zeta(j)$ are demand shifters for all inside assets and asset j, respectively.[16] The latent asset-specific demand parameter $\zeta(j)$ corresponds to tastes θ_j in the context of our model.

The key restriction is that p_j is the *only* price that influences $\omega_j^i(p)/\omega_o^i(p)$. Equivalently, asset characteristics, demand coefficients, and latent demand are all assumed invariant in the price vector p. This restriction is maintained even though characteristics and their coefficients are taken to summarize the expected returns process, which is naturally a function of p. Yet the restriction is necessary for (17) to be a well-defined demand *function* with fixed parameters: if these objects varied with p, the price response of demand would include their derivatives with respect to p, which a single price instrument per asset (the estimation method in KY19) cannot identify. Moreover, counterfactuals would require a specified mapping from prices to all demand parameters and character-

[15] Assumption 1 of KY19 is necessary but *not sufficient* to derive the logit demand specification. The proof of KY19's Corollary 1, which yields logit demand, makes additional assumptions, although these are not clearly stated in the main text. We thank Zhiguo He for clarifying this point.

[16] Technically, β_0 is the coefficient of log market equity, where market equity is the product of price and number of shares. This ensures neutrality to variation in the level of prices that is not economically meaningful, such as different choices for the initial number of shares in a firm. However, if the number of shares is constant, this term is another additive constant. Hence we focus simply on log prices.

istics. We summarize this assumption as follows.

Assumption 1 (Price-invariant characteristics and coefficients in KY19) *In demand specification (17), (i) asset characteristics other than price do not depend on prices: $\frac{\partial x_h(j)}{\partial p} = \mathbf{0}$ for every inside asset j and every characteristic $h = 1, \ldots, K-1$; and (ii) the demand coefficients β_0, $(\beta_h)_{h=1}^{K-1}$, β_K and latent demand $\zeta(j)$ do not depend on the price vector p.*

This structure yields sharp implications for substitution: given observed portfolio shares ω, the substitution matrix is entirely determined by the price coefficient β_0. Conditional on portfolio shares, any two assets must therefore be equally substitutable for any third asset, no matter their underlying returns process. This is the well-known Independence of Irrelevant Alternatives (IIA) property which defines the logit demand system, but now imported to portfolio choice. The problem is that canonical portfolio choice theory implies that substitution must be based on payoff similarity, which logit rules out. (Section 4.4 shows that this property holds conditional on *any* characteristics.)

Lemma 2 (No payoff-based demand complementarities in KY19's logit asset demand) *Under Assumption 1, the matrix of inside-asset portfolio-share elasticities $\mathcal{E}^i_{\text{logit}} \equiv (\mathcal{E}^i_{jk})_{jk}$ is*

$$\mathcal{E}^i_{\text{logit}} = \beta_0 \left(I - \mathbf{1}(\omega^i)^{\mathrm{T}} \right),$$

where $(\omega^i)^{\mathrm{T}}$ is the transpose of the vector of inside-asset shares and $\mathbf{1}$ is the vector of ones. Hence

$$\mathcal{E}^i_{jj} = \beta_0(1 - \omega^i_j) \quad \text{and} \quad \mathcal{E}^i_{jk} = -\beta_0 \omega^i_k, \tag{18}$$

where the own-price formula applies to inside assets j, and the cross-price formula applies to every inside asset k and every asset $j \neq k$, including the outside asset. Since $\mathcal{E}^i_{jo} = 0$ for every j, for every inside asset k and every pair $j, \ell \neq k$,

$$\mathcal{E}^i_{jk} - \mathcal{E}^i_{\ell k} = 0.$$

Hence logit exhibits no payoff-based demand complementarities in the sense of Definition 3.

The absence of payoff-based demand complementarities has two main implications: logit asset demand need not enforce no arbitrage, and it can produce biased elasticity estimates when demand exhibits payoff-based complementarities.

4.2 Failure to enforce no arbitrage

We first analyze whether the logit asset demand system—the set of logit demand functions combined with market clearing—is sufficient to ensure the absence of arbitrage. This is of interest because KY19 derive their demand system assuming arbitrage-free pricing, and KY25 claim that a demand system is *equivalent* to a stochastic discount factor derived from no arbitrage. This is incorrect: precisely because logit demand system does not exhibit payoff-based demand complementarities, it does not enforce no arbitrage. This means that the demand system may not be internally consistent under counterfactuals, or fails to incorporate the marginal investor who does enforce no arbitrage.

Consider two inside assets A and B with identical taste-adjusted cash flows, so that every investor regards them as perfect substitutes. No arbitrage requires $p_A = p_B$.[17] Now consider a perturbation to the price vector that presents an arbitrage. In the true model with demand complementarities, a price gap generates an unbounded, *asset-specific* long-short response which eliminates the arbitrage. Logit demand cannot produce the same corrective mechanism: under the IIA property, a change in the relative price of A and B creates a bounded reallocation of demand that is homogeneous across all remaining assets in proportion to portfolio shares only, irrespective of their cash flows.[18]

Proposition 2 (Logit demand cannot enforce no arbitrage) *Let assets A and B have identical taste-adjusted cash flows for every investor i: $\theta_A^i y_A(z) = \theta_B^i y_B(z)$ for all z, so that latent demand satisfies $\zeta(A) = \zeta(B)$ and characteristics satisfy $x_h(A) = x_h(B)$ for all h. Assume investors face no portfolio constraints on these assets, and that demand coefficients $\beta_0, (\beta_h)_h$ and latent demand $\zeta(\cdot)$ are common across investors. Write $c_j \equiv \sum_{h=1}^{K-1} \beta_h x_h(j) + \beta_K + \log \zeta(j)$. Then, in the KY19 demand system with the outside asset as numeraire, $p_o = 1$, market clearing determines the price of every inside asset j as:*

$$p_j = \left(\frac{E_o \exp(c_j)}{E_j} \right)^{\frac{1}{1+\beta_0}}, \tag{19}$$

[17] Identical taste-adjusted cash flows allows no arbitrage to retain its force: Appendix C shows that when taste-adjusted payoffs differ across investors, no pricing function is free of arbitrage in the conventional sense. We therefore evaluate KY19 in the most favorable case. Kocherlakota (2025) derives a complementary result showing that log-linear demand curves are generically inconsistent with no arbitrage.

[18] As in the proof of Proposition 2 in Appendix A, the perturbation that raises $\log p_A$ and lowers $\log p_B$ by equal amounts moves the share of *every* other asset ℓ by $d\omega_\ell = \beta_0 \omega_\ell (\omega_A - \omega_B) d \log p_A$, and moves the relative share of A and B by the bounded amount $d(\omega_A - \omega_B) = -2\beta_0 w d \log p_A$ when $\omega_A = \omega_B \equiv w$.

which is a function of asset j's own supply and own characteristics alone. Consequently

$$\frac{p_A}{p_B} = \left(\frac{E_B}{E_A}\right)^{\frac{1}{1+\beta_0}}, \tag{20}$$

and two assets that every investor regards as perfect substitutes trade at different prices whenever they are in different supply. For any heterogeneous supply vector, the resulting price vector admits an arbitrage: there exists a long-short portfolio of assets A and B with zero taste-adjusted payoff and a negative price. The true model instead enforces $p_A = p_B$ for any supply vector.

Proof. See Appendix A. ■

Note that latent demand $\zeta(\cdot)$ is interpreted as a structural parameter. If one instead treats latent demand as a residual, $\zeta(A)$ and $\zeta(B)$ can be chosen to produce *any* desired price ratio p_A/p_B. But then latent demand measures a violation of no arbitrage rather than a preference, and the demand system has no structural interpretation. More generally, because latent demand is a free intercept, the ability of a given demand system to fit observed portfolio holdings is not evidence that its cross-asset structure is correct.

4.3 Spillover bias

We now show that the absence of demand complementarities can lead to sharply biased elasticity estimates in standard portfolio choice settings. To establish this, we first show how demand elasticities can be identified from exogenous supply shocks under logit demand. By Assumption 1, our results hold conditional on any set of asset characteristics. Hence we treat characteristics as given for now, and discuss their role in 4.4.

We distinguish three notions of elasticity: (i) the *structural elasticity*, which is the true theoretical elasticity to a pure price shock, (ii) the *observed elasticity*, which is the total portfolio response to the supply shock used to create price variation, and (iii) the *measured elasticity*, which reflects the econometrician's estimate of the structural elasticity given observed behavior. The logit demand system offers a specific mapping from observed to measured elasticities, and we ask whether it accurately recovers the structural elasticity.

Consider the ideal scenario in which the econometrician observes an exogenous supply shock to asset j and the associated *changes* in prices and the investor's portfolio shares. Further assume that all demand shifters other than p_j are invariant to the shock which creates variation in p_j. Under this assumption, one can identify slope coefficient β_0 from the *observed* elasticity of relative portfolio shares in response to a supply shock to

asset j. Specifically, totally differentiating (17) yields

$$\hat{\beta}_0 = -\frac{d}{d\log p_j}\log\frac{\omega_j^i(p)}{\omega_o^i(p)}.$$

By the definition of elasticity, logit demand thus implies:

$$\hat{\beta}_0 = \hat{\mathcal{E}}_{jj}^i - \hat{\mathcal{E}}_{oj}^i.$$

Under the hypothesis of logit demand, the structural elasticity β_0 is thus identified from the *difference* in the observed elasticities of focal asset j and the outside asset, and β_0 determines the absolute elasticity of demand for asset j as well. In particular, Lemma 2 shows that the absolute elasticity under logit is $\beta_0(1-\omega_j)$, where ω_j is the observed portfolio share of asset j. (Appendix D shows in detail how β_0 and budget balance can be used to recover the full substitution matrix.) Of course, the problem with this identification strategy is that logit rules out payoff-based demand complementarities—and the associated price spillovers—by assumption. When the true demand functions depend on multiple asset prices, this assumption leads to potentially severe estimation bias. This is what we show next.

Remark 1 (Source of bias) *Spillover bias is not due to the assumption that the econometrician observes time series variation in response to a shock: Section 4.5 shows that spillover bias generically contaminates cross-sectional estimators as well. The simple reason is that cross-sectional and time-series data are both reflections of the same underlying equilibrium relations, and a demand system which fails to accurately model cross-asset linkages recovers biased estimators from either.*

Logit bias. To formally characterize spillover bias in logit demand estimation, define the *logit bias* $\mathcal{B}_{jk}^i$ as the difference between the true relative elasticity $\mathcal{E}_{jk}^i - \mathcal{E}_{ok}^i$ and its logit estimator, which is the difference in observed elasticities $\hat{\mathcal{E}}_{jk}^i - \hat{\mathcal{E}}_{ok}^i$:

$$\mathcal{B}_{jk}^i \equiv \left(\mathcal{E}_{jk}^i - \mathcal{E}_{ok}^i\right) - \left(\hat{\mathcal{E}}_{jk}^i - \hat{\mathcal{E}}_{ok}^i\right).$$

The next result offers a general decomposition of this bias into the underlying demand elasticities and price spillovers based on Proposition 1. We focus on the bias in the own-price elasticity of asset j, but analogous results hold for cross-price elasticities.

Proposition 3 (Spillover bias in logit demand systems) *Suppose portfolio shares are inde-*

pendent of wealth. For any inside asset j, the bias in the logit estimator is:

$$\mathcal{B}^i_{jj} = \sum_{k \neq j} \Bigg[\underbrace{\left(\frac{\partial \log \omega^i_j}{\partial \log p_k} - \frac{\partial \log \omega^i_k}{\partial \log p_k} \right)}_{\equiv \Delta^i_{jk}} - \underbrace{\left(\frac{\partial \log \omega^i_o}{\partial \log p_k} - \frac{\partial \log \omega^i_k}{\partial \log p_k} \right)}_{\equiv \Delta^i_{ok}} \Bigg] \underbrace{\frac{\mathcal{S}_{kj}/p_k}{\mathcal{S}_{jj}/p_j}}_{\textit{Price Spillover}} . \tag{21}$$

Proof. See Appendix A. ■

Decomposition (21) has a clear interpretation. Each term in the sum asks: when asset k's price changes, does asset j respond more or less like the outside asset? The difference $\Delta^i_{jk} - \Delta^i_{ok}$ measures the excess sensitivity of asset j relative to the outside asset with respect to each potential substitute k. The bias is zero if $\Delta^i_{jk} = \Delta^i_{ok}$ for all $k \neq j$. This is the IIA property: the outside asset controls for the spillover from p_k if and only if it responds to p_k in the same way as asset j, for every k. Given a violation of the IIA property, the magnitude of the bias scales with the price spillovers. Importantly, the spillover depends on the very demand parameters one seeks to estimate: by market clearing, the price spillover must be larger when assets are regarded as closer substitutes.

Bias in an equilibrium model. Logit's bias depends on price spillovers, which must be endogenously determined in asset market equilibrium. Hence we now compute the bias using a simple framework based on Lucas (1978). To sharply characterize the sources and consequences of spillover bias, we consider a tractable and highly transparent setting (described in Definition 4) in which there are only three assets: two *inside assets* which are the focus of demand estimation, and an *outside asset* which serves as the numeraire. Our results are not specific to this structure: Appendix B establishes analogous findings in a canonical general equilibrium CARA-Normal economy with an arbitrary number of assets and a factor structure in payoffs.

Definition 4 (Three-asset economy) *There are two aggregate states, $z \in \{1, 2\}$, and a distributional shock $\iota \in \{r, g\}$ which further affects asset payoffs. The probability of state z is π_z, and the probability of distributional shock ι satisfies $\Pr(\iota = r) = \rho$. There are three assets:*

(i) Tree 2, which pays $(1 - \phi) \cdot y(2)$ if and only if state 2 is realized, where $\phi \in [0, 1)$. We will refer to this asset as the outside asset, and normalize its price to $p_2 = 1$. The aggregate endowment of this asset is 1.

(ii) A green tree with price p_g which pays only in aggregate state 1, and pays more when $\iota = g$. The aggregate endowment of this asset is $\frac{1}{2} + \psi$, where ψ is an exogenous supply shifter.

(iii) A red tree with price p_r which pays more when $\iota = r$ in aggregate state 1, and pays $2\phi \cdot y(2)$ in state 2. The aggregate endowment of this asset is $\frac{1}{2}$.

The specific state-contingent payoffs of all three assets are summarized in Table 1.

		State 1 (π_1)		State 2 ($1-\pi_1$)
		Green shock ($1-\rho$)	Red shock (ρ)	
Tree 1	green tree	$y(1)+\epsilon$	$y(1)-\epsilon$	0
	red tree	$y(1)-\epsilon$	$y(1)+\epsilon$	$2\phi \cdot y(2)$
Tree 2 (outside asset)		0		$(1-\phi)\cdot y(2)$

Table 1: Payoff structure in the three-asset economy.

We also make the following assumptions on investor endowments and constraints:

(i) Investors have the same initial endowments: $e^i_j = E_j$ and $w^i_0 = 0 = w^i_1(z)$ for all i, j and z.

(ii) Investors care only about consumption at date 1: (1) reduces to the pure portfolio choice problem $\max_{a^i} \sum_z \pi_z u^i(\tilde{c}^i_1(z))$ subject to $\sum_j p_j(a^i_j - e^i_j) = 0$, and λ^i is the multiplier on the budget constraint.

Parameter ϵ modulates the substitutability between inside assets in state 1. If $\epsilon = 0$, green and red trees are perfect substitutes with respect to their state-1 cash flows. If $\epsilon \to y(1)$ then red and green assets are not substitutable within state 1 at all. Parameter ϕ modulates the overlap between the red asset and the outside asset. We maintain the following assumptions for the remainder of this section unless stated otherwise. Investors have common tastes: $\theta^i_j = 1$. There is no aggregate risk: $y(1) = y(2) = 1$. The red tree pays only in state 1, $\phi = 0$; Section 5.1 relaxes this. (Recall also from Definition 4 that investors consume only at date 1 and $w^i_0 = w^i_1(z) = 0$.) None of these assumptions drives our conclusions and Appendix B establishes analogous results without them.

We begin by verifying that this economy exhibits precisely the payoff-based demand complementarities that the logit demand system rules out by assumption.

Lemma 3 (Payoff-based demand complementarities in the three-asset economy) *Consider the three-asset economy with fixed endowments and $\phi = 0$. The outside asset has no payoff overlap with either tree: $H^i_{g2} = H^i_{r2} = 0$. The two trees overlap in both sub-states of state 1:*

$$H^i_{gr} = (1-\epsilon^2)\frac{1}{\lambda^i}\sum_{z\in\{g,r\}} \pi_z u^{i\prime\prime}(\tilde{c}^i_1(z)) \neq 0 \quad \text{for all } \epsilon \in [0,1). \tag{22}$$

In particular, under log utility and $\rho = \pi_1 = \frac{1}{2}$*, where* $p = (1,1,1)$ *and* $\omega^i = (\frac{1}{4}, \frac{1}{4}, \frac{1}{2})$ *for every* ϵ*, the structural elasticities with respect to the green price are:*

$$\mathcal{E}^i_{gg} = \frac{1-\epsilon^2}{2\epsilon^2}, \quad \mathcal{E}^i_{rg} = -\frac{1-\epsilon^2}{2\epsilon^2}, \quad \text{and} \quad \mathcal{E}^i_{2g} = 0. \tag{23}$$

Thus, payoff-based demand complementarities given a shock to the green price are:

$$\mathcal{E}^i_{rg} - \mathcal{E}^i_{2g} = -\frac{1-\epsilon^2}{2\epsilon^2}, \tag{24}$$

which is nonzero for every $\epsilon < 1$*, diverges as* $\epsilon \to 0$*, and is zero only at* $\epsilon = 1$*.*

Proof. See Appendix A. ■

The mechanism is transparent. The red tree responds to the green price and the outside asset does not, because the red tree overlaps with green in payoff space and the outside asset does not. Logit cannot accommodate this mechanism because it requires $\mathcal{E}^i_{rg} = \mathcal{E}^i_{2g} = -\beta_0 \omega^i_g$. The two are equal only at $\epsilon = 1$, where the trees have disjoint payoffs, all three elasticities in (23) are zero, and the only consistent logit has $\beta_0 = 0$. For every $\epsilon < 1$, however, no β_0 fits the underlying substitution patterns. Note also that prices and portfolio shares are the same for every ϵ while (23) is not. Contrary to logit, this means that portfolio shares are not a sufficient statistic for substitution.

We now derive the closed-form demand functions needed to quantify the bias. For this purpose, and for the remainder of the section, we further specialize preferences to log utility, $u^i(c) = \log c$. This is again only for tractability: it delivers demand functions in closed form, and Appendix B obtains analogous results under CARA preferences.

Lemma 4 (Optimal portfolio shares) *Given relative prices* (p_g, p_r)*, the optimal relative portfolio shares for investor* i *are given by*

$$\frac{\omega^i_g(p_g,p_r)}{\omega^i_2(p_g,p_r)} = \frac{\pi_1}{\pi_2} p_g \cdot \frac{(p_g+p_r)\epsilon^2 - (p_g-p_r) + 2p_r\epsilon(1-2\rho)}{(p_g+p_r)^2\epsilon^2 - (p_g-p_r)^2}; \tag{25}$$

$$\frac{\omega^i_r(p_g,p_r)}{\omega^i_2(p_g,p_r)} = \frac{\pi_1}{\pi_2} p_r \cdot \frac{(p_g+p_r)\epsilon^2 + (p_g-p_r) - 2p_g\epsilon(1-2\rho)}{(p_g+p_r)^2\epsilon^2 - (p_g-p_r)^2}. \tag{26}$$

Proof. See Appendix A. ■

These demand functions reflect the demand complementarity in Lemma 3: the relative demand for each inside asset depends on the prices of *both* inside assets, not only its own. It is exactly this dependence that the logit specification discards.

Now consider an exogenous shock ψ to green asset supply. Since this shock shifts

the green asset's price, it is a valid price instrument under the assumption of logit demand. We can thus apply the logit estimator to the observed equilibrium asset quantities and prices and compare it to the true structural elasticity.

Proposition 4 finds that spillover bias arises for all $\epsilon < 1$ and is *unbounded*: the logit estimator always understates the structural elasticity, and the bias diverges to infinity as inside assets become perfect substitutes ($\epsilon \to 0$). From here on we let aggregate states be equally likely, $\pi_1 = \pi_2 = \frac{1}{2}$, which is not necessary but yields transparent expressions.

Proposition 4 (Logit bias) *Consider any investor i. The logit bias is:*

$$\mathcal{B}^i_{jj} = -\underbrace{\left(\mathcal{E}^i_{j,-j} - \mathcal{E}^i_{2,-j}\right)}_{\substack{\text{Payoff-based}\\ \text{demand complementarity}}} \times \underbrace{\frac{\mathcal{S}_{-j,j}/p_{-j}}{\mathcal{S}_{jj}/p_j}}_{\text{Price spillover}}. \tag{27}$$

Given an exogenous supply shock ψ around $\psi = 0$, measured and structural elasticities satisfy:

$$\hat{\mathcal{E}}^i_{gg} - \hat{\mathcal{E}}^i_{2g} = \frac{(1-\epsilon^2)}{(1+\epsilon)^2 - 4\epsilon\rho} \quad \textit{and} \quad \mathcal{E}^i_{gg} - \mathcal{E}^i_{2g} = \frac{(1-\epsilon^2)(1-\epsilon(1-2\rho))}{8\rho(1-\rho)\epsilon^2}. \tag{28}$$

In the limit where green and red assets are perfect substitutes, measured and structural elasticities are:

$$\lim_{\epsilon\to 0}\left(\hat{\mathcal{E}}^i_{gg} - \hat{\mathcal{E}}^i_{2g}\right) = 1 \quad \textit{and} \quad \lim_{\epsilon\to 0}\left(\mathcal{E}^i_{gg} - \mathcal{E}^i_{2g}\right) = \infty. \tag{29}$$

For any $\epsilon > 0$, the logit bias is

$$\mathcal{B}^i_{gg} = \frac{(1-\epsilon^2)^2(1+(1-2\rho)\epsilon)}{8\epsilon^2\rho(1-\rho)((1+\epsilon)^2 - 4\epsilon\rho)}, \tag{30}$$

which decomposes as in (27):

$$\mathcal{B}^i_{gg} = \underbrace{\left(\frac{\overbrace{-\mathcal{E}^i_{gr} \to \infty \text{ as } \epsilon \to 0}^{\frac{(1-\epsilon^2)(1-\epsilon(1-2\rho))}{8\epsilon^2\rho(1-\rho)}} + \overbrace{\mathcal{E}^i_{2r} = 0}^{0}}\right)}_{\text{Payoff-based demand complementarity (cf. (27))}} \times \underbrace{\frac{(1-\epsilon^2)\left(1+(1-2\rho)\epsilon\right)}{\left(1-(1-2\rho)\epsilon\right)\left((1+\epsilon)^2 - 4\epsilon\rho\right)}}_{\substack{\text{Price spillover (cf. (27))}\\ \text{bounded, } \to 1 \text{ as } \epsilon \to 0}}. \tag{31}$$

The bias is positive, diverges as $\epsilon \to 0$, is strictly decreasing in ϵ, and is zero if and only if $\epsilon = 1$.

Proof. See Appendix A. ■

Figure 2 illustrates Proposition 4. Measured elasticities are small throughout and of the same order of magnitude as leading estimates in the demand-system literature. In contrast, the structural elasticity approaches infinity as $\epsilon \to 0$, but is zero when $\epsilon \to 1$. Since the bias is increasing in the underlying structural elasticity, the logit estimate can differ from the structural elasticity by orders of magnitude. This means that low estimated elasticities are uninformative about the underlying structural elasticity.

Intuition. Consider first the case where the two inside assets are close substitutes: $\epsilon \approx 0$. Any change in the green price then leads to an immediate near-arbitrage opportunity. Because investors would like to respond by selling green and buying red, the red price must adjust to clear the market. Given this spillover, it is less attractive to substitute across from green to red, leading to low *observed* elasticities. Importantly, the *reason* for this large price spillover is that each individual investor would have substituted very rapidly had the red price remained fixed. Hence low measured elasticities do not *indicate* that financial markets are indeed slow to respond to profitable trading opportunities.

Conversely, the bias is zero in the limit $\epsilon \to 1$ because inside assets are not substitutable, no spillover occurs, and logit accurately captures the underlying substitution patterns. Our analysis thus also admits a more positive interpretation: while the bias is severe when close substitutes are available, it is more muted when assets are not close substitutes. Nevertheless, substitutability ultimately is an unobserved object that must be recovered from the data, and logit does not accurately measure it: the measured elasticity ranges from zero to one while the structural elasticity ranges from zero to infinity.[19]

Remark 2 (Aggregation from "micro-logit" to asset elasticities) *We have derived spillover bias at the level of the individual investor ("micro logit"). One could also derive stock-level elasticities by averaging across investors. This does not solve the identification problem: since investor-level elasticities are underestimated for every investor in this economy, so are asset-level elasticities. In the setting of Proposition 4, asset-level elasticities are the same as at the asset-investor level. The main difference between "micro" and "macro" logit is that aggregated asset classes may have fewer close substitutes. However, this also makes pure supply shocks harder to find.*

4.4 The role of characteristics and latent demand

Our results on spillover bias and lack of no arbitrage enforcement in logit asset demand hold conditional on *any* asset characteristics and irrespective of latent demand. This is

[19] The range is one to two if measured in asset quantities, by (11). However, the bias, which is a difference of relative elasticities in common units, is the same whether demand is measured in shares or quantities. Appendix B further shows that, with a different utility function, the estimated elasticities can be larger than one but the bias still remains unbounded.

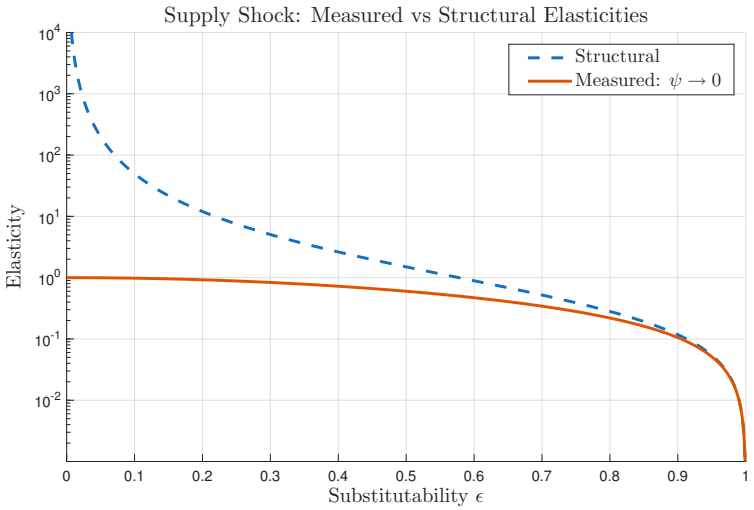


Figure 2: Measured vs. structural elasticity given an exogenous supply shock ψ as $\psi \to 0$. The measured elasticity is $\hat{\mathcal{E}}^i_{gg}$. The structural elasticity is $\mathcal{E}^i_{gg}$. Parameters: $\pi_1 = \frac{1}{2}$, $y(1) = y(2) = 1$, $\theta^i_j = 1$, and $\rho = \frac{1}{2}$.

perhaps surprising since the logit specification (17) prominently features these parameters. But there is a precise reason for this invariance: *conditional on portfolio shares*, characteristics and latent demand have no bearing on logit's substitution matrix.

Specifically, characteristics enter the KY19 demand equation as shifters of its *intercept*. They determine how much of each asset an investor holds at given prices and, under Assumption 1, they summarize expected returns and their covariances. But they do not shape substitution conditional on portfolio shares: By Lemma 2 the elasticity matrix is $\beta_0(I - \mathbf{1}(\omega^i)^\mathsf{T})$, no matter the underlying asset characteristics. That is, characteristics influence elasticities *only* through the shares they generate, but cannot induce substitution based on payoff similarity. Enriching the characteristic set therefore improves the fit of the level of demand while leaving untouched the substitution matrix. The same is true of latent demand, which is an *asset-specific* shifter of the intercept. That the model fits observed portfolio holdings thus offers no confirmation that the slope of demand is correct.

These issues bear directly on a headline finding of this literature. In Table 3 of KY19, latent demand accounts for roughly 81% of the cross-sectional variance of stock returns, dwarfing the contributions of observed characteristics and asset supply. KY19 interpret this as evidence that sentiment and disagreement are central to the cross section of returns. Lemma 2 suggests an alternative reading. If the price and characteristic terms of the demand system cannot represent the demand complementarities that actually shape portfolios, the resulting cross-elasticities are (incorrectly) attributed to the residual. Large and variable latent demand is then a symptom of omitted complementarities rather than an independent economic force, and does *not* indicate inelastic demand.

4.5 Spillover bias in cross-sectional estimators

We have characterized spillover bias when the econometrician observes the response of an asset's price and portfolio share to an exogenous supply shock. This *within-asset* approach is instructive because, *under the assumption of logit demand*, such variation is sufficient to identify the structural elasticity of interest; its failure thus reveals that logit demand does not appropriately account for demand complementarities and spillovers.

Despite this simple logic, KY25 claim that a cross-sectional estimator can identify the structural parameter β_0 from a single cross-section of prices and quantities, including in our benchmark model. The estimator is based on the log-linear demand function

$$\log(\omega_j) = -\beta_0 \log(p_j) + \alpha + \eta_j, \tag{32}$$

where η_j is latent demand and the portfolio share of the outside asset is absorbed into the constant α. As discussed in Section 4.2, the assumed demand function has a structural interpretation only if latent demand is a *parameter*, not merely a residual. We follow the notation of KY25. Given a vector of supply instruments z, the proposed estimator is

$$\hat{\beta}_0^{\text{KY}} = -\frac{\text{Cov}_{\text{cs}}[\log \omega_j, z_j]}{\text{Cov}_{\text{cs}}[\log p_j, z_j]},$$

where Cov_{cs} denotes the cross-sectional covariance. The estimator accurately recovers β_0 if and only if latent demand is orthogonal to the instrument in the cross-section:

$$\hat{\beta}_0^{\text{KY}} = \beta_0 - \frac{\text{Cov}_{\text{cs}}[\eta_j, z_j]}{\text{Cov}_{\text{cs}}[\log p_j, z_j]}.$$

This condition tightly restricts the set of permissible demand parameters. Consider first our benchmark model.[20] For two inside assets and any instrument which varies across assets, the covariance satisfies $\text{Cov}_{\text{cs}}[\eta, z] \propto \eta_g - \eta_r$. Identification thus requires the *exact identity of latent demand*, $\eta_g = \eta_r$. Combining demand function (32) with market clearing given pre-shock endowments $E_g = E_r = \frac{1}{2}$, latent demand must also satisfy

$$\eta_g - \eta_r = (1 + \beta_0) \log \frac{p_g}{p_r} + \log \frac{\frac{1}{2} + \psi}{\frac{1}{2}} = 0, \tag{33}$$

where prices and supplies are those at which the cross-section is observed, that is, after the

[20] KY25 use both our benchmark model and a CARA-Normal model to make their claims. Results for the CARA-Normal model are in Appendix B.

green supply shock ψ. Since the cross-section consists of two inside assets, the estimator is the slope between two points, $\beta_0 = -1 - \log(1+2\psi)/\log(p_g/p_r)$: the entire cross-sectional price difference is the response of relative demand to the supply difference.

Consider first the case most favorable to KY25, in which the two inside assets are identical in payoffs, probabilities and tastes ($\rho = \frac{1}{2}$, $\phi = 0$, common tastes) and in equal supply prior to the shock. Pre-shock prices are then equal, and the supply shock is the only source of a cross-sectional price difference. In this case, condition (33) holds to first order in ψ, and the estimator recovers the price coefficient β_0.

Now let the assets differ in anything other than supply. Any asymmetry in payoffs, probabilities or tastes creates a pre-shock price difference that is independent of supply, and it enters the estimator directly: as $\psi \to 0$ the supply term $\log(1+2\psi)$ vanishes while $\log(p_g/p_r)$ does not, so the estimator converges to -1, *invariant and of the wrong sign*, whatever the true elasticity.[21] For finite shocks, the estimate is a blend of the two sources of price variation. To first order around the benchmark with perfect symmetry, $\log(p_g/p_r) \approx 2\epsilon(1-2\rho) - \epsilon^2 \log(1+2\psi)$, so that

$$\hat{\beta}_0^{\text{KY}} \approx -1 - \frac{\log(1+2\psi)}{2\epsilon(1-2\rho) - \epsilon^2 \log(1+2\psi)} \quad \text{whereas} \quad \beta_0 = -1 + \frac{1}{\epsilon^2}.$$

which passes through $\pm\infty$ as the two sources of price variation offset.

Since observable characteristics are imperfect proxies for unobservable payoffs, exact symmetry cannot be ensured in data, and the condition under which the estimator is unbiased is *measure zero* in the space of payoff and price distributions.[22] Figure 3 illustrates this result by comparing the structural elasticity, the exact cross-sectional estimator from KY25, and our within-asset estimator given a small deviation from symmetry ($\rho = 0.495$) and a small supply shock. The KY25 cross-sectional estimator is biased and *has the wrong sign* for any value of the structural elasticity. Our within-asset estimator instead exhibits small biases when ϵ is close to one, but it is strongly biased if ϵ is close to zero. Outside of unrealistic knife-edge cases, the within-asset and cross-sectional estimators thus both fail because they are built on a misspecified model.

[21] Binsbergen, David, and Opp (2025) show that the estimation procedure in KY19 yields a large share of incorrect signs if the price coefficient is not restricted to be positive, and argue this may be due to momentum in returns. Our account of these errors is complementary to their momentum-based explanation.

[22] KY25's claim to the contrary is due to an error whereby they approximate the estimator—which is defined in levels—in changes. The approximation coincides with the estimator only in the symmetric economy. See Fuchs, Fukuda, and Neuhann (2025b) for details.

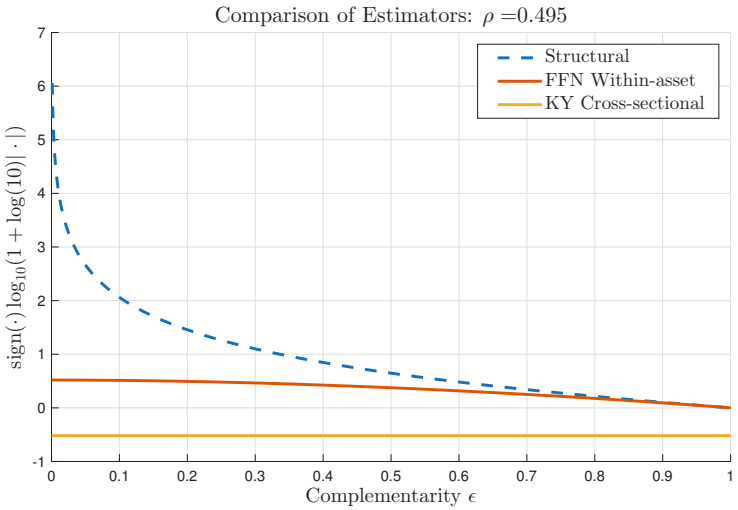


Figure 3: Structural elasticities, compared to our within-asset and KY25 cross-sectional estimators given a small perturbation of symmetry ($\rho = 0.495$). We use an adjusted $\log_{10}(1 + \log(10)|\cdot|)$-scale to accommodate negative values.

5 Alternative Approaches to Modeling Substitution

In the logit demand system, asset characteristics do not determine substitution conditional on portfolio shares. This leads to the pathology that all inside assets must be equally substitutable conditional on portfolio shares, even if some pairs appear much more similar to each other, such as two assets with identical market risk exposure. The standard logit model for product markets has the same flaw: with a single price coefficient, cross-price elasticities depend only on market shares, so a price cut on one product draws buyers from every other in proportion to its share, however dissimilar the products (Berry and Haile, 2021). In product markets, the standard remedies — nested logit, and random coefficients on characteristics — work by making substitution depend on product proximity, so that products with similar characteristics are closer substitutes.

These approaches have also been used in finance (e.g., Koijen and Yogo, 2026; Chaudhary, Fu, and Li, 2023). But they do not readily transfer to asset markets. The problem is that the notion of similarity that governs substitution in portfolio choice is not a *primitive* of the product space. By Definition 2, two assets are substitutes to the extent that their taste-adjusted payoffs covary under the investor's substitution measure, which is endogenously determined from latent consumption. A random-coefficients demand system would reproduce the portfolio-choice substitution matrix only if its characteristics spanned the payoff space and the distance between assets were measured under the substitution measure $\hat{\pi}^i$—in which case the characteristics simply *are* the payoffs, weighted by investor-level curvature. Since taste-adjusted payoffs and the substitution measure are

unobserved and investor-specific, neither can be included in the demand specification.

Asset markets thus require distinct methods to model heterogeneous substitution. We now discuss two potential alternatives. Section 5.1 considers the approach in HHHKL25. Subsequent to our work, they propose a method in which assets with different characteristics may substitute differently, leading to demand *conditional* on characteristics. This method has advantages over logit but requires stringent restrictions: in the empirically relevant case where payoff distributions are not disjoint, the only model with non-satiated utility in which controls address spillover bias is the CARA-Normal model.

Section 5.2 sketches a novel approach in which state prices are used to identify the substitution measure. This method also uses CARA preferences, but imposes no distributional assumptions on payoffs. We leave a full development to future work.

5.1 Substitution conditional on observables

We first consider the semi-structural approach in HHHKL25 in which observable asset characteristics can influence the *slope* of demand, not just the intercept. This is achieved by assuming that the substitution matrix itself depends on observables.

Assuming a functional form does not by itself eliminate spillover bias, and demand complementarities can induce spillover bias even *conditional* on controls. To account for this, HHHKL25 impose two assumptions. Assumption (A1) states that two assets with identical observables have the same cross-elasticity to *every* third asset, $\mathcal{E}^i_{j\ell} = \mathcal{E}^i_{k\ell}$ for $\ell \neq j, k$. This is the absence of third-price effects that logit imposes on every pair of assets, but here it must hold only *conditional on observables*. (Lemma 6 in the Appendix formalizes this restriction.) Removing common spillovers by comparing assets with identical observables then yields the *relative elasticity* $\mathcal{E}^i_{j|k} \equiv \mathcal{E}^i_{jj} - \mathcal{E}^i_{kj}$. Assumption (A2) turns the relative elasticity into a *parameter*: $\mathcal{E}^i_{j|k} = \mathcal{E}^{\text{rel}}$.

Required substitution matrix. Let $M \equiv H^{-1}$ denote the substitution matrix of Lemma 1, let $b_j \in \mathbb{R}^K$ denote the controls for asset j, and let B be the matrix that stacks the b_j^{T}. (We suppress the investor index and set $\theta^i_j = 1$; tastes enter only through taste-adjusted payoffs and change nothing below.) HHHKL25 state (A1) and (A2) in semi-elasticities of portfolio shares; we state them in the units of Lemma 1, compensated responses of quantities to prices.[23] When the dependence on observables is bilinear, as it is in HHHKL25,

[23] The matrix of share semi-elasticities is a known transformation of M involving prices, holdings, and the wealth term, $\mathcal{E} = -\mathrm{diag}(\omega) - W^{-1} D_p J_a D_p + \omega g^{\mathrm{T}}$ with $J_a = M - bv^{\mathrm{T}}$, $D_p = \mathrm{diag}(p)$, and $g_k = \partial \log W / \partial \log p_k$. It introduces no channel through which preferences enter other than M. We work with M because it is the object in which the exceptions in Proposition 5 are exact; the illustration below handles share units directly through Lemma 6.

(A1) and (A2) jointly imply that the substitution matrix has the form

$$M = \mathcal{E}^{\text{rel}} I + B\Gamma B^{\mathrm{T}}, \tag{34}$$

where the first block is the relative elasticity and the second measures spillovers *through* observables. The relative elasticity is not the asset-level elasticity: the own-price slope of asset j is $\mathcal{E}^{\text{rel}} + b_j^{\mathrm{T}}\Gamma b_j$, so the cross-section recovers an asset's own-price response net of the response of an asset with the same observables to the same price.[24]

Can controls eliminate payoff-based demand complementarities? By Definition 2, the substitution matrix of the canonical model is $M = \kappa^{-1}\widehat{\mathbb{E}}[yy^{\mathrm{T}}]^{-1}$: the inverse of the second-moment matrix of payoffs under the substitution measure. Payoff-based demand complementarities are generic because this matrix has no special structure. The key question is whether controls can deliver a *conditional* substitution matrix of the form in (34).

Since payoffs determine substitution, the most prevalent method for constructing controls decomposes asset payoffs into factor loadings and idiosyncratic risk. The econometrician thus chooses K risk factors and orthogonalizes payoffs under the physical measure π, so that payoffs follow a K-factor model with loadings b_j and homoskedastic, mutually uncorrelated residuals. Under the physical measure, asset payoffs thus satisfy

$$\mathbb{E}_\pi[yy^{\mathrm{T}}] = \sigma^2 I + B\Lambda_\pi B^{\mathrm{T}}. \tag{35}$$

While this has the structure of (34), it is obtained under the physical measure. Substitution instead is governed by second moments of payoffs *under the substitution measure*. Controls thus eliminate payoff-based demand complementarities if they deliver a substitution matrix with the same factor structure, *but under* $\hat{\pi}$. This is the following result.

Lemma 5 (Factor structure under the substitution measure) *The substitution matrix has the form* (34) *if and only if* $\widehat{\mathbb{E}}[yy^{\mathrm{T}}] = dI + B\Lambda B^{\mathrm{T}}$ *for some* $d > 0$ *and some* $K \times K$ *matrix* Λ.

The two decompositions can differ sharply even when the physical factor fit is good. Appendix G gives a three-state example in which two assets are uncorrelated under the physical measure but have correlation 0.98 under the substitution measure, and in which the physical measure identifies the wrong asset as the closest substitute.

[24] To recover the full substitution matrix, it must be invariant in consumption plans. While a single cross-section can recover $\mathcal{E}^{\text{rel}}$, time-series shocks to portfolios sorted on observables are required to recover the second block. If this block is not constant across shocks, one cannot recover a fixed demand function for any asset. A fixed matrix is thus necessary for estimation, not just for counterfactuals.

The lemma follows from the Woodbury identity: a scalar multiple of the identity plus a matrix in the column space of B inverts to a matrix of the same form, so the factor structure of payoff second moments passes to the substitution matrix. The blocks, however, are not the same. Inverting (34) gives $\mathcal{E}^{\text{rel}} = 1/(\kappa d)$: the relative elasticity is the reciprocal of the idiosyncratic second moment of payoffs under $\hat{\pi}$. The factor block Γ is a transformation of Λ that captures the remaining substitution through common exposures.

The difficulty is now apparent: substitution matrix and controls are constructed under different measures and they can disagree when the change of measure is not zero. Specifically, for any two assets j and k, equation (9) implies the wedge

$$\widehat{\mathbb{E}}[y_j y_k] - \mathbb{E}_\pi[y_j y_k] = \sum_z (\hat{\pi}_z - \pi_z)\, y_j(z)\, y_k(z) = \frac{\mathbb{E}_\pi\big[\big(u''(\tilde{c}_1) - \bar{u}''\big)\, y_j y_k\big]}{\bar{u}''}. \tag{36}$$

Under the factor model (35), the substitution matrix has the form (34) at a given consumption plan if the wedge (36) is equal to zero for every factor–residual and residual–residual pair and takes the same value for every residual variance. But there is a good reason for a non-zero wedge. The physical measure captures statistical co-movement; the substitution measure is concerned with how strongly the *marginal value of consumption* responds to payoffs in each state of the world. Two assets which are uncorrelated may be close substitutes under the substitution measure because the few states in which they pay off together are the ones in which the investor's marginal utility is most curved. And since the substitution measure depends on the investor's preferences and (latent) consumption plan, it is not observable and cannot be used to construct the controls directly.

Controls thus create the desired conditional substitution matrix only if second moments under the physical measure conform to those under the substitution measure, or if (36) is zero because payoff do not overlap. The next result shows the required conditions.

Proposition 5 (Controls require mean-variance optimization or disjoint payoffs) *Assume factor model* (35) *with* $\sigma^2 > 0$. *Under the factor-structure condition of Lemma* 5, *controls deliver a substitution matrix that is common up to scale across consumption plans if and only if payoff second-moment matrices under the substitution measure are common up to scale across those plans. By* (36), *this can happen in three ways.*

(i) Quadratic utility. $\hat{\pi} = \pi$ *at every plan if and only if* u'' *is constant. Then payoff second moments under the substitution measure are fixed, and the substitution matrix is common up to scale for any payoff distribution with the maintained factor structure. The scale of the substitution matrix can vary with the marginal value of wealth.*

(ii) CARA-Normal. *Under CARA utility, the substitution measure is the risk-neutral measure:* $\hat{\pi} = Q$. *With risk-free asset traded at price* p_f, *the substitution matrix for risky assets is*

$$M = -\frac{1}{\gamma p_f} \mathrm{Cov}_Q(y)^{-1}.$$

Risk-neutral covariances are independent of risky portfolio holdings on an open set if and only if payoffs are Gaussian. CARA–Normal therefore delivers a fixed substitution matrix up to scale, where the scale is determined by the observable risk-free price.

(iii) Disjoint payoffs. *For non-negative payoffs,* $H_{jk} = 0$ *at every plan if and only if* $y_j(z)y_k(z) = 0$ *in every state. Disjoint payoffs thus eliminate cross-asset spillovers.*

Proof. See Appendix A. ■

Controls thus work in three highly restrictive cases.[25] The first is quadratic utility, where the substitution measure *is* the physical measure. But quadratic utility implies satiation and increasing absolute risk aversion, which is not well-suited for counterfactuals.

Part (ii) is the CARA-Normal model. Under CARA, the substitution measure is the risk-neutral measure. When payoffs are Gaussian, this change of measure leaves covariances unchanged. With a riskless asset, physical covariances are thus sufficient to build the substitution matrix. But it is important to see what this means: controls are not a flexible approach that is *separate* from structural restrictions on preferences and payoffs; *they require them.* The CARA–Normal analysis in Appendix B further shows that controlled regressions require *exact* measurement of the observables that govern substitution with third assets—there, expected payoffs—which is difficult under realistic data constraints: small and unobservable differences in expected payoffs can lead to substantial spillover bias, and the bias diverges as the assets become close substitutes.

Part (iii) eliminates cross-asset interactions by a restrictions on payoffs: two assets must have entirely disjoint payoff processes, so no third-price effects can arise between them under any utility. But the condition is exact, not approximate—a single state of overlap reintroduces the wedge. This is a very tight restriction on payoffs that is violated by essentially all financial assets. Moreover, if payoffs are disjoint across asset groups, H is block diagonal, eliminating cross-group substitution at fixed marginal value of wealth under any preferences. However, within-group matrices need not be common up to scale.

[25] The relevant requirement is that the structure hold at every consumption plan, not only at the plan observed in a given cross-section. The block $B\Gamma B^{\mathrm{T}}$ is recovered from time-series variation, which pools across dates with different consumption plans, and any external use of the estimates, including counterfactuals, evaluates demand at plans other than those observed. If the structure holds only at a single plan, it can at most deliver the cross-sectional relative elasticity at that date.

Importantly, mean-variance optimization cannot be invoked as an innocuous approximation to a given decision problem. A second-order expansion of utility approximates the pricing kernel to first order in consumption risk, but it approximates the substitution measure to zeroth order: it replaces $u''(\tilde{c}_1(z))$ by a constant and thus sets $\hat{\pi} = \pi$ by assumption. This changes the fundamental properties of the decision problem, and thus directly alters the substitutability of different assets through a priori restrictions.

Illustration. We now illustrate these issues in the benchmark three-asset economy, starting with the simplest case, $\phi = 0$, in which red and green payoffs are disjoint from those of the outside asset. The logit estimator failed in this setting because it assumes that the outside asset is as substitutable with each inside asset as the inside assets are with each other, which is plainly violated for any $\epsilon < 1$. HHHKL25 offer a solution by requiring no third-price effect only for *specific asset pairs*, such as the two inside assets. Because inside and outside assets have disjoint payoffs, the substitution matrix is block diagonal. As such, (A1) holds between inside and outside assets, and the estimator recovers the relative elasticity. (Since not all payoffs are disjoint, the inside-asset block is not fixed.)

Now consider a small perturbation, $\phi > 0$. Since markets are complete and budget sets are unchanged, optimal consumption plans are as before. But the substitution matrix is different because the red tree and the outside asset no longer have disjoint payoffs. As a result, the conditional substitution matrix is no longer of the required form, the relative elasticity is subject to spillover bias, and the resulting estimate cannot be interpreted as the slope of a relative demand function controlling for factor exposures. This can be shown analytically. Denote HHHKL25's estimator around a green supply shock ψ by

$$\hat{\mathcal{E}}^{\text{rel}} = -\frac{(d\omega_g - d\omega_r)/d\psi}{(d\log p_g - d\log p_r)/d\psi}.$$

Then spillover bias in the relative elasticity satisfies the decomposition

$$\mathcal{B}^{\text{rel}} \equiv \mathcal{E}^{\text{rel}} - \hat{\mathcal{E}}^{\text{rel}} = \underbrace{\left[(\mathcal{E}_{gg} - \mathcal{E}_{rg}) + (\mathcal{E}_{gr} - \mathcal{E}_{rr})\right]}_{\text{asymmetric relative substitution}} \underbrace{\frac{(d\log p_r)/d\psi}{(d\log p_r - d\log p_g)/d\psi}}_{\text{spillover to } p_r}, \tag{37}$$

The next result formally characterizes the bias as a function of ϕ and ϵ. Figure 4 illustrates.

Proposition 6 (Relative elasticity bias) *Under the payoffs in Table 1 and log utility:*

1. *The relative elasticity is biased down: $\mathcal{B}^{\text{rel}} \geq 0$, with equality if and only if $\phi = 0$ or $\epsilon = 1$.*
2. *For any fixed $\phi > 0$, the bias is decreasing in ϵ and diverges as $\epsilon \to 0$.*

3. *If both ϕ and ϵ converge to zero, the bias continues to diverge as long as ϵ converges to zero faster than $\sqrt{\phi}$.*

Proof. See Appendix A. ■

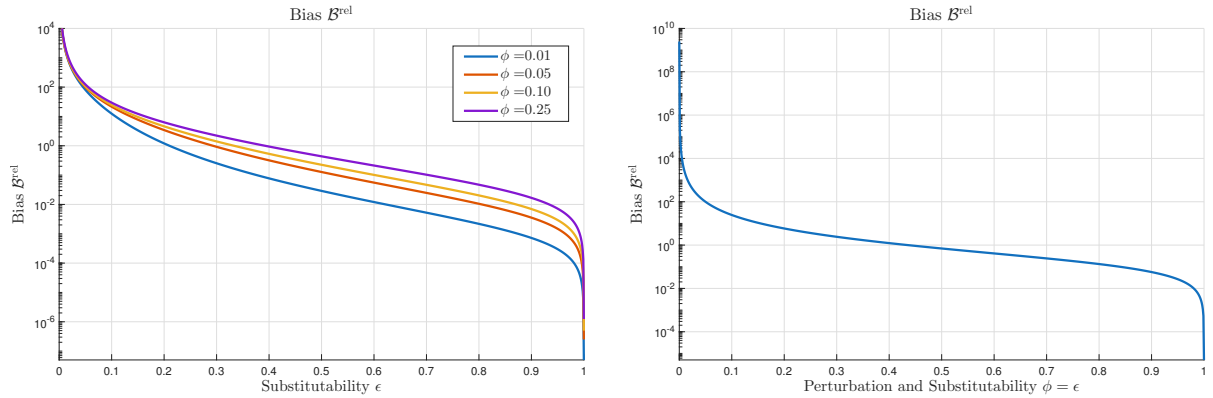


Figure 4: Left panel: bias in the relative elasticity $\mathcal{B}^{\text{rel}}$ as a function of ϵ, holding $\phi > 0$ fixed. Right panel: bias in the relative elasticity as a function of ϵ given $\phi = \epsilon$. Other parameters: $(\rho, \pi_1) = (\frac{1}{2}, \frac{1}{2})$.

For any $\epsilon < 1$, the estimator is unbiased only if $\phi = 0$. For a small perturbation $\phi > 0$, the relative elasticity is underestimated and the bias diverges when the inside assets are close substitutes ($\epsilon \approx 0$). The right panel shows that this divergence obtains even if ϕ and ϵ go to zero at the same rate. Payoff overlap is generic for essentially all assets of interest, and small payoff perturbations cannot be detected from realistic data. This bias is structural rather than specific to a particular estimator or instrument: it obtains because of a misspecified substitution matrix, not because of weak instruments.

5.2 Identifying the substitution measure from state prices: a sketch

We have shown that the substitution measure, combined with asset payoffs, is the critical object determining investor asset demand. The fundamental problem is that this measure is investor-specific: it depends on the investor's preferences and on her consumption plan, neither of which is observed. Future research should therefore seek methods that allow the substitution measure to be constructed under minimal assumptions.

We now present one such method, which is to employ the assumption of CARA preferences to turn the substitution measure into the risk-neutral measure. If option prices can be used to construct the second-moment matrix of asset payoffs under the risk-neutral measure, one can then recover the substitution matrix from market data.

Proposition 7 (Substitution under CARA) *Let investor i have CARA utility with coefficient γ^i, no asset-specific tastes, and no binding constraints on the assets she holds, and let q denote the*

vector of state prices implied by asset prices. Then, for any distribution of payoffs,

$$M^i = -\frac{1}{\gamma^i}\left(Y^{\mathrm{T}}\mathrm{diag}(q)\,Y\right)^{-1}; \tag{38}$$

with a traded riskless asset, the substitution matrix among risky assets is $-(\gamma^i p_f)^{-1}\mathrm{Cov}_Q(y)^{-1}$. *The substitution matrix is therefore identified up to the scalar* γ^i *from risk-neutral second moments of payoffs, and a single exogenous trade identifies* γ^i.

Proof. See Appendix A. ■

We emphasize that this method has several shortcomings. It cannot accommodate rich investor heterogeneity, such as fixed differences in tastes or in beliefs about asset-level payoffs, since state prices do not reveal marginal valuations in this case. And it is tied to preferences that admit no wealth effects. At the same time, it offers a simple framework that accommodates payoff-based demand complementarities for any distribution of payoffs, as well as heterogeneity in risk aversion and in beliefs about state probabilities, which enter only through a state-price density that the investor's own first-order conditions equate to the market's. The price response to any trade is then predicted across all assets up to a single parameter. This presents an improvement over logit asset demand.

6 Are Elasticities Structural Parameters?

The previous section discussed the measurement and interpretation of demand elasticities, but stopped short of establishing whether asset demand elasticities are structural parameters that are invariant to counterfactuals, even if they are well-measured. This is a critical concern if asset demand systems are to be used to inform policy.

We discuss two main challenges. First, investors care about resale value, so demand elasticities reflect not only individual tastes for an asset, but also expectations about *other* investors' future valuations. Second, it is difficult to separately identify unobserved tastes and constraints. However, many counterfactuals are sensitive to the precise micro-foundation. We analyze these issues using simple variants of our baseline model.

6.1 Dynamic trading: whose preferences are being measured?

We begin by discussing the role of resale considerations. Forward-looking demand makes it difficult to separately identify individual and "market-wide" tastes because investors may buy an asset they personally dislike if they expect other investors will pay a high price for it (Keynes, 1936; Harrison and Kreps, 1978). However, counterfactuals involving shifts in the wealth distribution require knowledge of individual preferences.

In Appendix E we formally construct a simple two-period variant of our baseline framework for the special case where green and red trees have identical cash flows ($\epsilon = 0$), but variation in tastes creates variation in prices. Here, we only report the key equation determining asset demand in the first period.

Let $p_{j,1}$ denote the price of asset j at date 1, R_j the gross return of asset j between dates 1 and 2, and W_2^i the investor's wealth at date 2. Given discount factor δ, an interior choice of $a_{j,1}^i$, investor i's holdings of asset j at date 1, must satisfy

$$\frac{\pi_1 y(1)\theta_{j,1}^i}{p_{j,1}} u'\left(\tilde{c}_1^i(1)\right) = \pi_2 y(2) u'\left(\tilde{c}_1^i(2)\right) + \delta \mathbb{E}^i\left[\frac{R_2 - R_j}{W_2^i}\right]. \tag{39}$$

The left-hand side is the date-1 marginal benefit of buying asset j at current price $p_{j,1}$ in state 1. The parameter $\theta_{j,1}^i$ is investor i's taste for asset j at date 1. The right-hand side consists of two components: the marginal loss from consuming less in state 2 at date 1, and the expected return reduction from carrying wealth forward in the form of tree j rather than tree 2. Hence demand is increasing in both private tastes and expected market returns. Since market returns are determined by the tastes of tomorrow's marginal investor, observed demand elasticities do not reveal whether an investor is buying based on her tastes or her beliefs over market-wide tastes.

6.2 Tastes versus constraints

Next we consider the interpretation of observed elasticities when investors differ in both tastes and unobservable investment mandates. To this end, we study our baseline economy but assume that there are two types of investors: some prefer green to red, and the rest prefer red to green. As we extend our baseline three-asset economy, let γ denote the share of the aggregate endowment of every tree owned by "green investors." This allows us to model counterfactual wealth distributions.

To ensure equilibrium existence, we assume that each investor faces short-sale constraints. Moreover, we assume that a share of each type faces a strict mandate to only invest in their preferred trees (i.e., a green investor with a mandate cannot buy red trees). Denote the share by m. This mandate is unobserved to the econometrician. All derivations are in Appendix F.

Since investors have different tastes, the equilibrium may feature sorting. In particular, if diversification motives are sufficiently small ($\epsilon \approx 0$) and green investors are not too wealthy ($\gamma \leq \overline{\gamma}$ for some $\overline{\gamma}$), all green investors buy only green assets. In a sorting equilibrium, green investors without a mandate are observationally equivalent to green

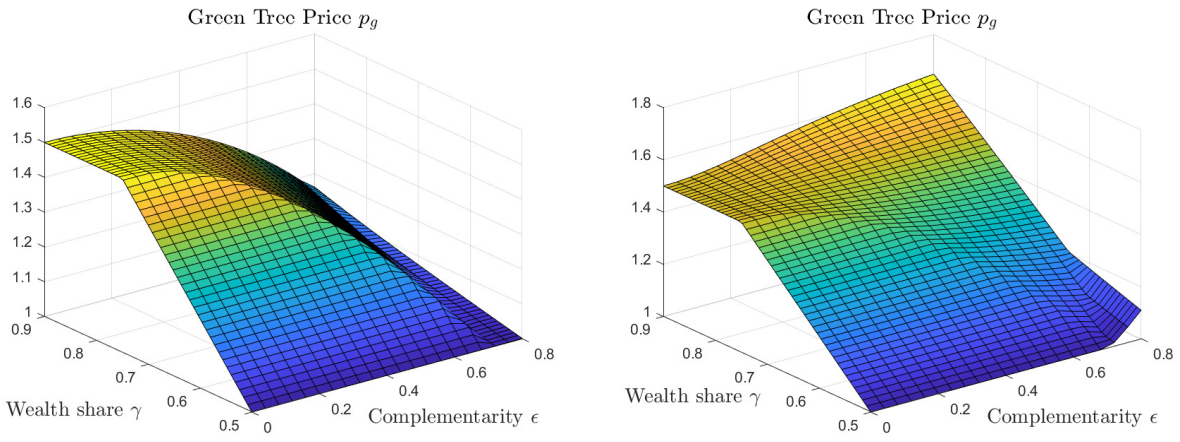


Figure 5: Green Price. Left: Mandate Share $m = 0.01$. Right: Mandate Share $m = 0.85$.

investors with a mandate. However, sorting breaks down when assets are complementary ($\epsilon \gg 0$) or when green investors are wealthy ($\gamma > \overline{\gamma}$), pushing up the green price.

Figure 5 shows the equilibrium green price p_g as a function of the wealth share γ and complementarity ϵ for two economies: one in which very few investors face a mandate (left panel), and the other with many constrained investors (right panel).

The two economies are observationally equivalent near the origin where unconstrained investors choose to specialize in their preferred color to the same extent as mandate investors. However, they differ sharply under counterfactual wealth distributions or payoff structures. For unconstrained investors, a shock to ϵ creates more demand for diversification. In the left panel, the price of green trees is thus decreasing in ϵ when green investors choose to hold inside assets. In contrast, mandate investors do not buy red trees at any price. In the right panel, the price of green trees thus *increases* in ϵ. Hence, the two economies are observationally equivalent for some parameters, but qualitatively different under counterfactuals. Thus, researchers should be cautious when assessing counterfactuals using demand systems that cannot separately identify tastes and constraints.

7 Conclusion

We present an analysis of demand systems for financial assets. Our results highlight the critical role of demand complementarities and equilibrium price spillovers. Specifically, we show that a failure to properly account for these effects can lead to severe bias in measured elasticities. This offers a simple reconciliation of the striking difference between demand elasticities estimated in demand-system approaches and canonical benchmarks.

In complementary work, Fuchs, Fukuda, and Neuhann (2026) show that these concerns reflect a more fundamental identification problem. When investors value assets for

their payoffs and prices satisfy no arbitrage, the mapping that governs demand complementarities is latent and cannot be recovered model-free from observational data. The substitution measure of Definition 2 makes the latency precise: substitution is governed by payoff second moments under a measure that depends on the investor's preferences and portfolio, and the physical second moments the econometrician can measure are the right ones only under quadratic utility or CARA with Gaussian payoffs. Identification of a local elasticity and its use across counterfactual equilibria require distinct assumptions. This implies we must rely on structural models and/or richer data. The proposed models must be evaluated on first principles or based on how they perform along other dimensions. As we have highlighted in this paper, very restrictive assumptions, such as those implied by the logit framework, can lead to highly biased estimates. Our analysis shows that the critical next step is developing structural models of asset demand which allow for payoff-based demand complementarities.

We see three main paths for future work. The first is to develop new frameworks that can account for richer cross-asset interactions. Section 5.2 gives one such framework under strong assumptions; an open question is which of these assumptions can be relaxed at low cost. The second is to develop bounds on the estimation biases that we discuss. The third is to consider alternative estimands which do not suffer from the spillover bias.

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A Proofs

A.1 Section 3

Proof of Lemma 1. Totally differentiating both sides of $F^i = \mathbf{0}$, we obtain:

$$H^i da^i - \frac{p}{\lambda^i} d\lambda^i - dp = \mathbf{0}.$$

We show that H^i is invertible. Then,

$$da^i = \left(H^i\right)^{-1} \left[dp + \frac{p}{\lambda^i} d\lambda^i\right]. \tag{40}$$

To see that H^i is invertible, denote by $Y^i \equiv (\theta^i_j y_j(z))_{z,j}$ the $Z \times J$ matrix of investor i's payoff-augmented cash flows, and denote by K^i the $Z \times Z$ diagonal matrix of curvature weights $\delta \pi_z u^{i\prime\prime}(\tilde{c}^i_1(z))/\lambda^i < 0$. Then, we can write (5) as

$$H^i = (Y^i)^{\mathrm{T}} K^i Y^i.$$

Then, H^i is negative definite and consequently invertible if and only if Y^i has full column rank, which follows from the non-redundancy hypothesis of Lemma 1.

Letting $b^i \equiv (H^i)^{-1} p$ and $q^i \equiv p^{\mathrm{T}} (H^i)^{-1} p < 0$, we can express (40) as

$$da^i = (H^i)^{-1} dp + \frac{b^i}{\lambda^i} d\lambda^i, \quad \text{and thus} \quad p^{\mathrm{T}} da^i = p^{\mathrm{T}} (H^i)^{-1} dp + \frac{q^i}{\lambda^i} d\lambda^i.$$

Setting $dp = \mathbf{0}$ in the first of these shows that b^i is the direction in which the portfolio moves when the marginal value of wealth changes at fixed prices.

It remains to determine $d\lambda^i$. With $\lambda^i = u^{i\prime}(c^i_0)$ and $c^i_0 = w^i_0 - p^{\mathrm{T}}(a^i - e^i)$, we have

$$\frac{d\lambda^i}{\lambda^i} = \frac{u^{i\prime\prime}(c^i_0)}{u^{i\prime}(c^i_0)} dc^i_0 = -c^i \left[p^{\mathrm{T}} da^i + (a^i - e^i)^{\mathrm{T}} dp\right], \quad \text{where} \quad c^i \equiv \frac{u^{i\prime\prime}(c^i_0)}{u^{i\prime}(c^i_0)} < 0.$$

Substituting $p^{\mathrm{T}} da^i$ and solving the resulting scalar equation gives

$$\frac{d\lambda^i}{\lambda^i} = -(v^i)^{\mathrm{T}} dp \quad \text{with} \quad v^i = \frac{c^i \left[(a^i - e^i) + b^i\right]}{1 + c^i q^i}, \tag{41}$$

the denominator $1 + c^i q^i > 0$ being the fixed-point adjustment of the feedback from re-optimization to date-0 spending.[26] Hence, $v^i_k = -\frac{\partial \log \lambda^i}{\partial p_k}$ has two ingredients: (i) the

[26] Note that, in the pure portfolio choice case in Section 4.3, date-0 consumption cannot move and the

net trade $a_k^i - e_k^i$, the direct revaluation of the investor's position, and (ii) b^i, the re-optimization channel.

Substituting $\frac{d\lambda^i}{\lambda^i} = -(v^i)^{\mathrm{T}} dp$ into (40) gives (10). The elasticity expressions in (11) follow from

$$\log \omega_j^i = \log p_j + \log a_j^i - \log W^i \quad \text{and} \quad \frac{\partial \log W^i}{\partial \log p_k} = \omega_k^i - \sum_j \omega_j^i \mathcal{E}_{jk}^{a,i}.$$

The proof is complete. ■

Proof of Proposition 1. First, it follows from (the s-th row of) equation (14) that

$$\frac{da_s^i}{d\chi_s} = \frac{\partial a_s^i}{\partial p_s}\frac{dp_s}{d\chi_s} + \sum_{j \neq s} \frac{\partial a_s^i}{\partial p_j}\frac{dp_j}{d\chi_s} + \frac{\partial a_s^i}{\partial \chi_s}.$$

Second, multiplying both sides by $-\frac{1}{\frac{dp_s}{d\chi_s}}\frac{p_s}{a_s^i}$ and noting the definitions of $\hat{\mathcal{E}}_{ss}^{a,i}$, $\mathcal{E}_{ss}^{a,i}$, $\mathcal{S}_{js}$, and $\mathcal{S}_{ss}$, we have:

$$\hat{\mathcal{E}}_{ss}^{a,i} = \mathcal{E}_{ss}^{a,i} - \left(\sum_{j \neq s} \frac{\partial a_s^i}{\partial p_j}\frac{\mathcal{S}_{js}}{\mathcal{S}_{ss}}\frac{p_s}{a_s^i} + \frac{\frac{\partial a_s^i}{\partial \chi_s}}{\mathcal{S}_{ss}}\frac{p_s}{a_s^i} \right).$$

Third, then, rearranging, we obtain equation (15). Fourth, similarly, equation (16) holds, where the income effects (i.e., the last term in (15)) disappear when the utility function is iso-elastic (as portfolio shares are then wealth-independent). ■

A.2 Section 4

Proof of Lemma 2. Under Assumption 1, the right-hand side of (17) depends on the price vector only through p_j. Thus, writing $v_j \equiv -\beta_0 \log p_j + c_j$ for inside assets and $v_o \equiv 0$ gives

$$\omega_j^i = \frac{\exp(v_j)}{\sum_{\ell \in \mathcal{J}} \exp(v_\ell)},$$

with $\frac{\partial v_j}{\partial \log p_k} = -\beta_0 \mathbf{1}_{j=k}$ for inside k, and $\frac{\partial v_j}{\partial \log p_o} = 0$ for every j. Thus, we obtain:

$$\mathcal{E}_{jo}^{\omega,i} = 0.$$

budget $p^{\mathrm{T}}(a^i - e^i) = w_0^i$ is itself an equation of the system. Thus, $p^{\mathrm{T}} da^i = -(a^i - e^i)^{\mathrm{T}} dp$ and

$$v^i = \frac{(a^i - e^i) + b^i}{q^i}.$$

Differentiating,

$$\mathcal{E}^{\omega,i}_{jk} = -\frac{\partial \log \omega^i_j}{\partial \log p_k} = \beta_0 \left[\mathbf{1}_{j=k} - \omega^i_k\right],$$

which yields (18). For $j, \ell \neq k$ both elasticities equal $-\beta_0 \omega^i_k$, so their difference is zero. ■

Proof of Proposition 2. Write $v_j \equiv -\beta_0 \log p_j + c_j$ for each inside asset and normalize $v_o \equiv 0$. Since equation (17) reads $\omega_j/\omega_o = \exp(v_j)$, the demand system satisfies:

$$\omega_j = \frac{\exp(v_j)}{\sum_{\ell \in \mathcal{J}} \exp(v_\ell)},$$

and thus we have:

$$\frac{\partial \omega_j}{\partial v_k} = \omega_j \left(\mathbf{1}_{j=k} - \omega_k\right). \tag{42}$$

Equilibrium prices. Market clearing for asset j requires $\omega_j W = p_j E_j$, where W is aggregate portfolio value. Applied to the outside asset it gives $\omega_o W = E_o$ with $p_o = 1$. Dividing and using $\omega_j/\omega_o = \exp(v_j)$, we have:

$$\exp(c_j) p_j^{-\beta_0} = \frac{p_j E_j}{E_o}, \quad \text{that is,} \quad p_j^{1+\beta_0} = \frac{E_o \exp(c_j)}{E_j},$$

which is (19); the solution exists and is unique, and one verifies directly that the implied shares then satisfy $\sum_j \omega_j = 1$. Since $c_A = c_B$ by hypothesis, (20) follows, and $p_A \neq p_B$ whenever $E_A \neq E_B$. Note that the argument uses only that the ratio $\omega^i_A/\omega^i_B = \exp(v_A - v_B)$ is common across investors, which the hypothesis $\zeta(A) = \zeta(B)$ delivers; heterogeneity in wealth and in tastes for other assets is immaterial.

The arbitrage. Suppose $E_A \neq E_B$ and, without loss of generality, $p_A > p_B$. Consider the long-short portfolio $a \equiv e_B - e_A$ that buys one unit of B and shorts one unit of A. Since $\theta^i_A y_A(z) = \theta^i_B y_B(z)$ for every i and z, its taste-adjusted payoff is $v^i(a) = \theta^i_B y_B(z) - \theta^i_A y_A(z) = 0$ in every state and for every investor, while its price is $P(a) = p_B - p_A < 0$.[27]

The true model. Write $\overline{y}(z) \equiv \theta^i_A y_A(z) = \theta^i_B y_B(z)$, so that

$$\tilde{c}^i_1(z) = \overline{y}(z)\left(a^i_A + a^i_B\right) + \sum_{j \neq A,B} \theta^i_j y_j(z) a^i_j + w^i_1(z)$$

[27] See Appendix C for the formal definition of $v^i(a)$. Definition 5 in Appendix C requires a strictly positive payoff in some state, and so does not literally cover a portfolio with zero payoff and negative price. Adding an arbitrarily small position in the outside asset restores strictness: $a_\varepsilon \equiv e_B - e_A + \varepsilon e_o$ satisfies $v^i(a_\varepsilon) > 0$ and $P(a_\varepsilon) < 0$ for any $0 < \varepsilon < p_A - p_B$.

depends on (a_A^i, a_B^i) only through their sum. Holding that sum and all other positions fixed, transferring one unit from B to A leaves every $\tilde{c}_1^i(z)$ unchanged and changes date-0 consumption by $-(p_A - p_B)$. Whenever $p_A \neq p_B$ each investor therefore gains strictly from repeating the transfer in the profitable direction without bound, and does so without altering any other position. Since aggregate demand for A and B is then unbounded while supplies are finite, no equilibrium with $p_A \neq p_B$ exists, for any supply vector $(E_j)_j$.

Bounded response (footnote 18). The perturbation sets $dv_A = -\beta_0 d\log p_A$, $dv_B = \beta_0 d\log p_A$ and $dv_\ell = 0$ otherwise. By (42),

$$d\omega_j = -\beta_0 \omega_j \left[\left(\mathbf{1}_{j=A} - \mathbf{1}_{j=B} \right) - (\omega_A - \omega_B) \right] d\log p_A.$$

At $j = \ell \notin \{A, B\}$ the indicators vanish, which gives the expression in the main text; it is zero for every such ℓ if and only if $\omega_A = \omega_B$. Evaluating instead at $j \in \{A, B\}$ with $\omega_A = \omega_B \equiv w$ gives $d\omega_A = -\beta_0 w \cdot d\log p_A = -d\omega_B$, so that $d(\omega_A - \omega_B) = -2\beta_0 w \cdot d\log p_A$, which is finite for every $w \in (0, 1)$. ■

Proof of Proposition 3. First, the structural elasticity in terms of the relative portfolio shares is given by:

$$-\frac{\partial \log\left(\frac{\omega_j^i}{\omega_o^i}\right)}{\partial \log p_j} = -\left(\frac{\partial \log \omega_j^i}{\partial \log p_j} - \frac{\partial \log \omega_o^i}{\partial \log p_j} \right).$$

Second, the observed elasticity in terms of the relative portfolio shares is given by:

$$\begin{aligned} -\frac{d\log\left(\frac{\omega_j^i}{\omega_o^i}\right)}{d\log p_j} &= -\left(\frac{d\log \omega_j^i}{d\log p_j} - \frac{d\log \omega_o^i}{d\log p_j} \right) \\ &= -\left(\frac{\partial \log \omega_j^i}{\partial \log p_j} + \sum_{k\neq j} \frac{\partial \log \omega_j^i}{\partial \log p_k} \frac{d\log p_k}{d\log p_j} - \frac{\partial \log \omega_o^i}{\partial \log p_j} - \sum_{k\neq j} \frac{\partial \log \omega_o^i}{\partial \log p_k} \frac{d\log p_k}{d\log p_j} \right). \end{aligned}$$

Third, the bias is:

$$\begin{aligned} \mathcal{B}_{jj}^i &= \sum_{k\neq j} \frac{\partial \log \omega_j^i}{\partial \log p_k} \frac{d\log p_k}{d\log p_j} - \sum_{k\neq j} \frac{\partial \log \omega_o^i}{\partial \log p_k} \frac{d\log p_k}{d\log p_j} \\ &= \sum_{k\neq j} \left\{ \left(\frac{\partial \log \omega_j^i}{\partial \log p_k} - \frac{\partial \log \omega_k^i}{\partial \log p_k} \right) - \left(\frac{\partial \log \omega_o^i}{\partial \log p_k} - \frac{\partial \log \omega_k^i}{\partial \log p_k} \right) \right\} \frac{d\log p_k}{d\log p_j} \\ &= \sum_{k\neq j} \left\{ \left(\frac{\partial \log \omega_j^i}{\partial \log p_k} - \frac{\partial \log \omega_k^i}{\partial \log p_k} \right) - \left(\frac{\partial \log \omega_o^i}{\partial \log p_k} - \frac{\partial \log \omega_k^i}{\partial \log p_k} \right) \right\} \frac{\mathcal{S}_{kj} \frac{1}{p_k}}{\mathcal{S}_{jj} \frac{1}{p_j}}, \end{aligned}$$

as desired. ■

Proof of Lemma 3. With $\theta_j^i = 1$ and the payoffs of Table 1 at $\phi = 0$, and with λ^i the Lagrange multiplier on the date-0 budget constraint of the pure portfolio choice problem of Definition 4, Definition (5) reads

$$H_{jk}^i = \frac{1}{\lambda^i} \sum_z \pi_z u^{i\prime\prime}(\tilde{c}_1^i(z)) y_j(z) y_k(z).$$

The outside asset pays only in state 2 and the trees only in state 1, so $y_g(\cdot)y_2(\cdot) = y_r(\cdot)y_2(\cdot) = 0$ in every state and $H_{g2}^i = H_{r2}^i = 0$, while $y_g(\cdot)y_r(\cdot) = (1+\epsilon)(1-\epsilon) = 1-\epsilon^2$ in each sub-state of state 1, which gives (22).

For (23), set $\rho = \pi_1 = \frac{1}{2}$ and $u^i = \log$. Identifying a shock $\iota \in \{g, r\}$ with a state, states $(g, r, 2)$ have probabilities $(\frac{1}{4}, \frac{1}{4}, \frac{1}{2})$ and the representative investor consumes the aggregate endowment, $\tilde{c}_1^i \equiv 1$, and thus $u^{i\prime\prime} = -1$. Market clearing gives $p = (1,1,1)$, $a^i = e^i = (\frac{1}{2}, \frac{1}{2}, 1)$, $W^i = 2$, $\omega^i = (\frac{1}{4}, \frac{1}{4}, \frac{1}{2})$, and the outside asset's first-order condition gives

$$\lambda^i = \frac{\pi_2 u^{i\prime}(1)}{p_2} = \frac{1}{2}.$$

Hence $H_{jk}^i = -2\sum_z \pi_z y_j(z) y_k(z)$, that is,

$$H^i = -\begin{pmatrix} 1+\epsilon^2 & 1-\epsilon^2 & 0 \\ 1-\epsilon^2 & 1+\epsilon^2 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \text{and thus} \quad \left(H^i\right)^{-1} = -\frac{1}{4\epsilon^2}\begin{pmatrix} 1+\epsilon^2 & -(1-\epsilon^2) & 0 \\ -(1-\epsilon^2) & 1+\epsilon^2 & 0 \\ 0 & 0 & 4\epsilon^2 \end{pmatrix}.$$

At a no-trade point $a^i = e^i$, by the proof of Lemma 1, $v^i = b^i/q^i$ with

$$b^i = (H^i)^{-1} p = -\left(\frac{1}{2}, \frac{1}{2}, 1\right) \quad \text{and} \quad q^i = p^{\mathrm{T}} b^i = -2,$$

giving $v^i = (\frac{1}{4}, \frac{1}{4}, \frac{1}{2}) = \omega^i$. The green column of (10) is therefore

$$\frac{\partial a^i}{\partial p_g} = \left(H^i\right)_{\cdot g}^{-1} - v_g^i b^i = \left(-\frac{1+\epsilon^2}{4\epsilon^2} + \frac{1}{8}, \frac{1-\epsilon^2}{4\epsilon^2} + \frac{1}{8}, \frac{1}{4}\right)^{\mathrm{T}}.$$

With $p_g/a_g^i = p_g/a_r^i = 2$ and $p_g/a_2^i = 1$, the quantity elasticities in (11) are

$$\mathcal{E}_{gg}^{a,i} = \frac{\epsilon^2+2}{4\epsilon^2}, \quad \mathcal{E}_{rg}^{a,i} = \frac{\epsilon^2-2}{4\epsilon^2}, \quad \text{and} \quad \mathcal{E}_{2g}^{a,i} = -\frac{1}{4}.$$

Since the investor is at her endowment, $W^i = \sum_j p_j e_j^i$ and $\frac{\partial \log W^i}{\partial \log p_g} = \omega_g^i = \frac{1}{4}$. Substituting into the second equation in (11) gives (23), and (24) follows by subtraction. ■

Proof of Lemma 4. Observe that, under $y(1) = y(2) = 1$, the first-order conditions of investor i's portfolio choice problem with respect to a_g^i and a_r^i reduce to:

$$p_g = \frac{\pi_1(1-\rho)}{\pi_2} \frac{(1+\epsilon)\theta_g^i a_2^i}{(1+\epsilon)\theta_g^i a_g^i + (1-\epsilon)\theta_r^i a_r^i} + \frac{\pi_1 \rho}{\pi_2} \frac{(1-\epsilon)\theta_g^i a_2^i}{(1-\epsilon)\theta_g^i a_g^i + (1+\epsilon)\theta_r^i a_r^i}; \tag{43}$$

$$p_r = \frac{\pi_1(1-\rho)}{\pi_2} \frac{(1-\epsilon)\theta_r^i a_2^i}{(1+\epsilon)\theta_g^i a_g^i + (1-\epsilon)\theta_r^i a_r^i} + \frac{\pi_1 \rho}{\pi_2} \frac{(1+\epsilon)\theta_r^i a_2^i}{(1-\epsilon)\theta_g^i a_g^i + (1+\epsilon)\theta_r^i a_r^i}. \tag{44}$$

Substituting the budget constraint

$$a_2^i = p_g e_g^i + p_r e_r^i + e_2^i - p_g a_g^i - p_r a_r^i$$

and solving for the demand functions (a_g^i, a_r^i), we obtain:

$$a_g^i = \theta_r^i \pi_1 \cdot \frac{(p_g e_g^i + p_r e_r^i + e_2^i)\left((\theta_r^i p_g + \theta_g^i p_r)\epsilon^2 - (\theta_r^i p_g - \theta_g^i p_r) + 2\theta_g^i p_r \epsilon(1-2\rho)\right)}{(\theta_r^i p_g + \theta_g^i p_r)^2\epsilon^2 - (\theta_r^i p_g - \theta_g^i p_r)^2}; \tag{45}$$

$$a_r^i = \theta_g^i \pi_1 \cdot \frac{(p_g e_g^i + p_r e_r^i + e_2^i)\left((\theta_r^i p_g + \theta_g^i p_r)\epsilon^2 + (\theta_r^i p_g - \theta_g^i p_r) - 2\theta_r^i p_g \epsilon(1-2\rho)\right)}{(\theta_r^i p_g + \theta_g^i p_r)^2\epsilon^2 - (\theta_r^i p_g - \theta_g^i p_r)^2}. \tag{46}$$

Substituting $\theta_j^i = 1$, we obtain the portfolio weights (ω_g^i, ω_r^i) in equations (25) and (26), respectively. ■

Proof of Proposition 4. First, (27) is a special case of (21) when there are two inside assets (i.e., trees j and $-j$) and the outside asset (tree 2).

Second, the representative agent (for ease of notation, we suppress the superscript i) consumes the aggregate endowment in equilibrium. Under the assumptions that $y(1) = y(2) = 1$, $\pi_1 = \frac{1}{2}$, and no tastes, the first-order conditions yield:

$$p_g = (1-\rho)\frac{1+\epsilon}{1+(1+\epsilon)\psi} + \rho\frac{1-\epsilon}{1+(1-\epsilon)\psi};$$
$$p_r = (1-\rho)\frac{1-\epsilon}{1+(1+\epsilon)\psi} + \rho\frac{1+\epsilon}{1+(1-\epsilon)\psi}.$$

Then, the change in prices $\Delta p_g \equiv p_g - p_g^0$ satisfies:

$$\Delta p_g \equiv p_g - p_g^0 = -\frac{1+\epsilon^2+(1-\epsilon^2)\psi + (1-2\rho)(2+(1-\epsilon^2)\psi)\epsilon}{(1+(1+\epsilon)\psi)(1+(1-\epsilon)\psi)}\psi.$$

When $\epsilon = 0$, we have $p_g = \frac{1}{1+\psi}$, $p_g^0 = 1$, and $\Delta p_g = -\frac{\psi}{1+\psi}$.

Third, substituting the equilibrium prices into the portfolio weight function ω_g yields the equilibrium portfolio weight of the green tree:

$$\begin{aligned}
\omega_g &= \frac{1+2\psi}{4} \cdot \frac{1+\psi+\epsilon(1-2\rho-\epsilon\psi)}{(1+(1+\epsilon)\psi)(1+(1-\epsilon)\psi)}; \\
\omega_g^0 &= \frac{1}{4}\left(1+(1-2\rho)\epsilon\right); \\
\Delta\omega_g \equiv \omega_g - \omega_g^0 &= \frac{1}{4} \cdot \frac{(1-\epsilon^2)\psi(1+\psi-\epsilon(1-2\rho)\psi)}{(1+(1-\epsilon)\psi)(1+(1+\epsilon)\psi)}.
\end{aligned}$$

In the limit as $\epsilon \to 0$, we have $\omega_g = \frac{1+2\psi}{4(1+\psi)}$, $\omega_g^0 = \frac{1}{4}$, and $\Delta\omega_g = \frac{\psi}{4(1+\psi)}$. Note that, since $\omega_2 = 1 - \pi_1$, $\mathcal{E}_{2g} = \hat{\mathcal{E}}_{2g} = 0$.

Fourth, the observed elasticity reduces to

$$-\frac{\Delta\omega_g}{\Delta p_g}\frac{p_g^0}{\omega_g^0} = \frac{(1-\epsilon^2)(1+\psi-\epsilon(1-2\rho)\psi)}{1+\epsilon^2+(1-\epsilon^2)\psi+(1-2\rho)(2+(1-\epsilon^2)\psi)\epsilon}. \tag{47}$$

In the limit as $\psi \to 0$, we obtain the first equation in (28). Also, in the limit as $\epsilon \to 0$, the observed elasticity tends to the first equation in (29).

Fifth, to compute the structural elasticity, we can compute

$$-\frac{\partial\omega_g}{\partial p_g}\frac{p_g}{\omega_g}$$

as a function of (p_g, p_r) when $(e_g, e_r, e_2) = (\frac{1}{2}+\psi, \frac{1}{2}, 1)$. Note that, since ω_g does not depend on (e_g, e_r, e_2) (recall equations (25) and (26)), this term does not depend on (e_g, e_r, e_2), i.e., ψ. Then, evaluating this at the initial equilibrium prices (p_g^0, p_r^0), the structural elasticity is given by the second equation in (28), which goes to infinity as $\epsilon \to 0$, establishing the second equation in (29).

Sixth, as in Proposition 1, we can decompose the measured demand response (in the limit as $\psi \to 0$) as the true (i.e., structural) response and the bias term:

$$\frac{\frac{d\omega_g}{d\psi}}{\frac{d\log p_g}{d\psi}} = \frac{\frac{\partial\omega_g}{\partial\log p_g}\frac{d\log p_g}{d\psi} + \frac{\partial\omega_g}{\partial\log p_r}\frac{d\log p_r}{d\psi}}{\frac{d\log p_g}{d\psi}} = \frac{\partial\omega_g}{\partial\log p_g} + \frac{\partial\omega_g}{\partial p_r}\frac{\frac{dp_r}{d\psi}}{\frac{d\log p_g}{d\psi}}.$$

Then, we can decompose the observed elasticity (in the limit as $\psi \to 0$) into the structural

elasticity and the bias term:

$$-\frac{\frac{d\omega_g}{d\psi}}{\frac{d\log p_g}{d\psi}}\frac{1}{\omega_g}=-\frac{\partial\omega_g}{\partial\log p_g}\frac{1}{\omega_g}-\frac{\partial\omega_g}{\partial p_r}\frac{\frac{dp_r}{d\psi}}{\frac{d\log p_g}{d\psi}}\frac{1}{\omega_g}.$$

Consequently, the observed elasticity in the limit as $\psi\to 0$ is:

$$-\frac{\frac{d\omega_g}{de_g}}{\frac{dp_g}{de_g}}\frac{p_g}{w_g}=\frac{(1-\epsilon^2)}{(1+\epsilon)^2-4\epsilon\rho},$$

which coincides with the limit of equation (47) as $\psi\to 0$. In the limit $\psi\to 0$, the bias is:

$$\frac{\partial\omega_g}{\partial p_r}\frac{\frac{dp_r}{d\psi}}{\frac{d\log p_g}{d\psi}}\frac{1}{\omega_g}=\frac{(1-\epsilon^2)(1-\epsilon(1-2\rho))}{8\epsilon^2\rho(1-\rho)}\frac{(1-\epsilon^2)\big(1+(1-2\rho)\epsilon\big)}{\big(1-(1-2\rho)\epsilon\big)\big((1+\epsilon)^2-4\epsilon\rho\big)},\tag{48}$$

which establishes equations (30) and (31), using $p_g^0=1+(1-2\rho)\epsilon$, $p_r^0=1-(1-2\rho)\epsilon$, $\frac{dp_g}{d\psi}\big|_{\psi=0}=-\big((1+\epsilon)^2-4\epsilon\rho\big)$ and $\frac{dp_r}{d\psi}\big|_{\psi=0}=-(1-\epsilon^2)$ from the equilibrium prices above.

It follows from equation (48) that the bias is positive (when $\epsilon<1$) and goes to infinity as $\epsilon\to 0$. Also, the derivative of the bias term with respect to ϵ is

$$-\frac{(1-\epsilon^2)Q(\epsilon,\rho)}{8\epsilon^3\rho(1-\rho)\left((1+\epsilon)^2-4\epsilon\rho\right)^2}<0,$$

where

$$Q(\epsilon,\rho)\equiv(1+\epsilon)^3(\epsilon^2+\epsilon+2)-2\epsilon(1+\epsilon)(7+\epsilon)(1+\epsilon^2)\rho+16\epsilon^2(1+\epsilon^2)\rho^2.$$

We have $Q>0$ on $(0,1)^2$ because it is quadratic in ρ whose smaller root satisfies

$$\frac{1+\epsilon}{16\epsilon}\left((7+\epsilon)-(1-\epsilon)\sqrt{\frac{17+\epsilon^2}{1+\epsilon^2}}\right)>1\quad\text{for every}\quad\epsilon\in(0,1),$$

which establishes that the bias is strictly decreasing in ϵ. Finally, the bias is zero if and only if $\epsilon=1$. ■

Proof of Lemma 5. By the Woodbury identity, if $M=mI+B\Gamma B^{\mathrm{T}}$ then

$$M^{-1}=m^{-1}I-m^{-2}B\big(\Gamma^{-1}+m^{-1}B^{\mathrm{T}}B\big)^{-1}B^{\mathrm{T}}$$

when Γ is invertible, and the same form with Γ restricted to its range otherwise. Thus, $M^{-1} = dI + B\Lambda B^{\mathrm{T}}$ with $d = m^{-1}$, and conversely. Since $M^{-1} = H = \kappa\,\widehat{\mathbb{E}}[yy^{\mathrm{T}}]$ by (7), equation (34) is equivalent to $\widehat{\mathbb{E}}[yy^{\mathrm{T}}] = (d/\kappa)I + B(\Lambda/\kappa)B^{\mathrm{T}}$: a K-factor structure in matrix form, with loadings B, under $\hat{\pi}$. ■

Proof of Proposition 5. The first sentence restates Lemma 5 at every consumption plan, with (34) fixed.

Part (i). By (8), $\hat{\pi} = \pi$ at a plan if and only if $u''(\tilde{c}_1(z))$ is the same in every state; at every plan, $\hat{\pi} = \pi$ if and only if u'' is constant. Then $\widehat{\mathbb{E}}[yy^{\mathrm{T}}] = \mathbb{E}_\pi[yy^{\mathrm{T}}]$, which has the required form by (35).

Part (ii). By (8), $\hat{\pi} = Q$ at a plan if and only if $A(\tilde{c}_1(z))$ is the same in every state; at every plan, if and only if A is constant, i.e., $u(c) = -e^{-\gamma c}$ up to an affine transformation. With a traded riskless asset the substitution matrix among risky assets is

$$\kappa^{-1}\widehat{\mathrm{Cov}}(y)^{-1} = \kappa^{-1}\mathrm{Cov}_Q(y)^{-1},$$

so (34) is fixed across plans if and only if $\mathrm{Cov}_Q(y)$ is. Under CARA with $\tilde{c}_1 = a_0 + a^{\mathrm{T}}y$, the risk-neutral density is proportional to $e^{-\gamma a^{\mathrm{T}}y}$ times the physical density, so Q is the exponential tilt of π at $t = -\gamma a$. For the tilt $\pi_t \propto \pi\, e^{t^{\mathrm{T}}y}$ and the cumulant generating function $K(t) \equiv \log \mathbb{E}_\pi[e^{t^{\mathrm{T}}y}]$,

$$\mathbb{E}_{\pi_t}[y] = \nabla K(t) \quad \text{and} \quad \mathrm{Cov}_{\pi_t}(y) = \nabla^2 K(t).$$

Hence $\mathrm{Cov}_Q(y) = \nabla^2 K(-\gamma a)$, which is independent of a on an open set if and only if K is quadratic there; since K is analytic on the interior of its domain, K is then quadratic on its domain, $K(t) = t^{\mathrm{T}}\mu + \frac{1}{2}t^{\mathrm{T}}\Sigma t$, which is the cumulant generating function of a Gaussian, and the moment generating function determines the distribution. Conversely, for Gaussian y, $\nabla^2 K \equiv \Sigma$, and since $-u'' = \gamma u'$ gives $\kappa = -\gamma p_f$, the substitution matrix among risky assets is $-\Sigma^{-1}/(\gamma p_f)$ at every plan, which has the form (34) by Lemma 5 because Σ has the form (35).

Part (iii). At fixed λ,

$$H_{jk} = \sum_z K_z\, y_j(z) y_k(z).$$

If $y_j(z)y_k(z) = 0$ for every z this is identically zero, for every plan and every utility. (Equality of the normalized moments, $\widehat{\mathbb{E}}[y_j y_k] = \mathbb{E}_\pi[y_j y_k]$ under every reweighting, holds more generally whenever $y_j y_k$ is constant across states.) Conversely, suppose H_{jk} vanishes at every plan in an open set for some utility with $u''' \neq 0$ on the relevant range. Varying

$\tilde{c}_1(z_0)$ alone changes H_{jk} by

$$\frac{\delta}{\lambda}\pi_{z_0}u'''(\tilde{c}_1(z_0))y_j(z_0)y_k(z_0)d\tilde{c}_1(z_0),$$

which must vanish, so $y_j(z_0)y_k(z_0) = 0$; as z_0 was arbitrary, the supports are disjoint. For (A1), apply this to every pair (j,k) with j in one group and k in the other: $H_{jk} = 0$ for all such pairs, so H and hence $M = H^{-1}$ are block diagonal across the two groups, and the cross-elasticity of any asset in one group to any price in the other is zero. For (A2),

$$\widehat{\mathbb{E}}[y_j^2] = \kappa^{-1}\sum_z K_z y_j(z)^2$$

varies with the plan whenever $u''' \neq 0$ on the support of y_j, by the same differentiation. For the last sentence, if $y_j(z_0)y_k(z_0) \neq 0$ for some z_0, the same differentiation shows that H_{jk} cannot vanish on any open set of plans, so H is not block diagonal across the groups and (A1) between them fails generically. ■

Proof of Proposition 6. First, similarly to Lemma 4, given relative prices $p = (p_g, p_r)$, one can show that the optimal portfolio shares $(\omega_g^i, \omega_r^i, \omega_2^i)$ for investor i are given by:

$$\omega_g^i(p) = (1-\phi)\pi_1 p_g \cdot \frac{(1+\epsilon)\left((p_g-p_r)-(p_g+p_r)\epsilon+\phi\left(2(1+\epsilon)-(p_g-p_r)+(p_g+p_r)\epsilon\right)\right)-4(2\phi-(1-\phi)p_r)\epsilon\rho}{\left(2\phi+(1-\phi)(p_g-p_r)\right)^2-\left((1-\phi)(p_g+p_r)-2\phi\right)^2\epsilon^2}; \tag{49}$$

$$\omega_r^i(p) = (1-\phi)\pi_1 p_r \cdot \frac{-(1-\epsilon)\left((p_g-p_r)-(p_g+p_r)\epsilon+\phi\left(2(1+\epsilon)-(p_g-p_r)+(p_g+p_r)\epsilon\right)\right)-4(1-\phi)p_g\epsilon\rho}{\left(2\phi+(1-\phi)(p_g-p_r)\right)^2-\left((1-\phi)(p_g+p_r)-2\phi\right)^2\epsilon^2}; \tag{50}$$

$$\omega_2^i(p) = 1-\pi_1+2\phi\pi_1\frac{(1-\epsilon)\left(2\phi(1+\epsilon)+(1-\phi)\left((p_g-p_r)-(p_g+p_r)\epsilon\right)\right)+4(1-\phi)p_g\epsilon\rho}{\left(2\phi+(1-\phi)(p_g-p_r)\right)^2-\left((1-\phi)(p_g+p_r)-2\phi\right)^2\epsilon^2}. \tag{51}$$

Second, the equilibrium prices of green and red trees are:

$$p_g = \frac{1}{1-\phi}\frac{\pi_1}{\pi_2}\frac{1+(1-2\rho)\epsilon+(1-\epsilon^2)\psi}{(1+(1+\epsilon)\psi)(1+(1-\epsilon)\psi)}; \tag{52}$$

$$p_r = \frac{1}{1-\phi}\frac{\pi_1}{\pi_2}\frac{1-(1-2\rho)\epsilon+\psi\left((1-(1-2\rho)\epsilon)^2+4\rho(1-\rho)\epsilon^2\right)}{(1+(1+\epsilon)\psi)(1+(1-\epsilon)\psi)}+\frac{2\phi}{1-\phi}. \tag{53}$$

Then, substituting the equilibrium prices into equations (49), (50), and (51), the equilibrium portfolio shares $(\omega_g, \omega_r, \omega_2)$ (note that from now on we suppress the superscript i)

are:

$$\omega_g = \pi_1 \frac{1+2\psi}{2} \cdot \frac{1+(1-2\rho)\epsilon+(1-\epsilon^2)\psi}{(1+(1+\epsilon)\psi)(1+(1-\epsilon)\psi)}; \tag{54}$$

$$\omega_r = \frac{\pi_1}{2} \cdot \frac{1-(1-2\rho)\epsilon+\psi\left((1-(1-2\rho)\epsilon)^2+4\rho(1-\rho)\epsilon^2\right)}{(1+(1+\epsilon)\psi)(1+(1-\epsilon)\psi)} + (1-\pi_1)\phi; \tag{55}$$

$$\omega_2 = (1-\pi_1)(1-\phi). \tag{56}$$

Third, to compute the bias $\mathcal{B}^{\text{rel}}$, it follows from the definition of the observed relative elasticity $\hat{\mathcal{E}}_{g|r}$ that

$$\hat{\mathcal{E}}_{g|r} = \mathcal{E}_{g|r} + \left\{\left(\frac{\partial\omega_g}{\partial\log p_g} - \frac{\partial\omega_r}{\partial\log p_r}\right) + \left(\frac{\partial\omega_g}{\partial\log p_r} - \frac{\partial\omega_r}{\partial\log p_g}\right)\right\} \frac{d\log p_r}{d\log\left(\frac{p_g}{p_r}\right)}.$$

Thus, the bias satisfies:

$$\mathcal{B}^{\text{rel}} = \left\{\left(\frac{\partial\omega_g}{\partial\log p_g} - \frac{\partial\omega_r}{\partial\log p_r}\right) + \left(\frac{\partial\omega_g}{\partial\log p_r} - \frac{\partial\omega_r}{\partial\log p_g}\right)\right\} \cdot \frac{d\log p_r}{d\log\left(\frac{p_g}{p_r}\right)}. \tag{57}$$

While the structural relative elasticity satisfies

$$\mathcal{E}_{g|r} = \frac{(1-\epsilon^2)(1+(1-2\rho)\epsilon)\left\{\pi_1(1-(1-2\rho)\epsilon)+\phi(1-\pi_1)\right\}}{8\rho(1-\rho)\epsilon^2}, \tag{58}$$

the observed relative elasticity satisfies:

$$\hat{\mathcal{E}}_{g|r} = \pi_1 \frac{(1-\epsilon^2)(1+(1-2\rho)\epsilon)\left\{\pi_1(1-(1-2\rho)\epsilon)+2\phi(1-\pi_1)\right\}}{2\left(4\pi_1\rho(1-\rho)\epsilon^2+\phi(1-\pi_1)\left((1+\epsilon)^2-4\epsilon\rho\right)\right)}. \tag{59}$$

Fourth, the bias $\mathcal{B}^{\text{rel}}$ is given by:

$$\mathcal{B}^{\text{rel}} = \frac{\phi(1-\pi_1)(1-\epsilon^2)(1+(1-2\rho)\epsilon)}{8\rho(1-\rho)\epsilon^2} \frac{\pi_1\left\{(1-(1-2\rho)\epsilon^3)+(1-2\rho)\epsilon(1-(1-2\rho)\epsilon)\right\}+\phi(1-\pi_1)\left((1+\epsilon)^2-4\epsilon\rho\right)}{4\pi_1\epsilon^2\rho(1-\rho)+\phi(1-\pi_1)\left((1+\epsilon)^2-4\epsilon\rho\right)}. \tag{60}$$

Within this proof, it is convenient to abbreviate

$$\mathcal{L} \equiv (1+\epsilon)^2 - 4\epsilon\rho, \quad \mathcal{M} \equiv 4\pi_1\rho(1-\rho)\epsilon^2 + \phi(1-\pi_1)\mathcal{L}, \quad \text{and} \quad \mathcal{P} \equiv \pi_1\mathcal{K} + \phi(1-\pi_1)\mathcal{L},$$

where

$$\mathcal{K} \equiv \left(1-(1-2\rho)\epsilon^3\right) + (1-2\rho)\epsilon\left(1-(1-2\rho)\epsilon\right).$$

Then, (60) can be written as:

$$\mathcal{B}^{\text{rel}} = \frac{\phi(1-\pi_1)(1-\epsilon^2)\left(1+(1-2\rho)\epsilon\right)}{8\rho(1-\rho)\epsilon^2} \cdot \frac{\mathcal{P}}{\mathcal{M}}. \tag{61}$$

We show that, for any $\epsilon \in (0,1)$, $\mathcal{L} > 0$, $\mathcal{K} > 0$, $\mathcal{M} > 0$, and $\mathcal{P} > 0$. Indeed, we have:

$$\mathcal{L} = (1-\epsilon)^2 + 4\epsilon(1-\rho) > 0.$$

We also have:

$$\mathcal{K} = 1 + (1-2\rho)\epsilon(1-\epsilon^2) - (1-2\rho)^2\epsilon^2 \geq (1-\epsilon)(1-\epsilon^2) > 0,$$

as $|1-2\rho| \leq 1$. Then, $\mathcal{M} > 0$ and $\mathcal{P} > 0$ follow. Hence, the bias $\mathcal{B}^{\text{rel}}$ in (61) is non-negative, and it is 0 if and only if the first numerator is 0, i.e., $\phi = 0$ or $\epsilon = 1$.

Fifth, we show that the bias $\mathcal{B}^{\text{rel}}$ is decreasing in ϵ when $\phi > 0$. At the equilibrium prices (52)–(53) the two cross semi-elasticities coincide,

$$\frac{\partial \omega_g}{\partial \log p_r} = \frac{\partial \omega_r}{\partial \log p_g}. \tag{62}$$

Then, we can rewrite (57) as:

$$\mathcal{B}^{\text{rel}} = \left(\left(\frac{\partial \omega_g}{\partial \log p_g} - \frac{\partial \omega_r}{\partial \log p_r}\right)\epsilon\right) \cdot \left(\frac{d \log p_r}{d \log\left(\frac{p_g}{p_r}\right)}\frac{1}{\epsilon}\right).$$

We show that both factors are positive and decreasing in ϵ. For the first factor,

$$\left(\frac{\partial \omega_g}{\partial \log p_g} - \frac{\partial \omega_r}{\partial \log p_r}\right)\epsilon = \frac{\phi(1-\pi_1)\mathcal{P}}{4\pi_1\rho(1-\rho)\epsilon} > 0,$$

and its derivative in ϵ is

$$-\frac{\phi(1-\pi_1)\left\{\pi_1\left(1-\epsilon^4+\epsilon^2\left((1-2\rho)+\epsilon\right)^2\right)+(1-\pi_1)\phi(1-\epsilon^2)\right\}}{4\pi_1\rho(1-\rho)\epsilon^2} < 0,$$

where we used the identity $1+(1-2\rho)^2\epsilon^2+2(1-2\rho)\epsilon^3 = 1-\epsilon^4+\epsilon^2\left((1-2\rho)+\epsilon\right)^2$,

which is positive because $\epsilon < 1$. For the second factor,

$$\frac{d\log p_r}{d\log\left(\frac{p_g}{p_r}\right)}\frac{1}{\epsilon} = \frac{\pi_1(1-\epsilon^2)\left(1+(1-2\rho)\epsilon\right)}{2\epsilon\mathcal{M}} > 0,$$

and, after some algebra, its derivative in ϵ is

$$-\pi_1 \cdot \frac{(1-\pi_1)\phi\mathcal{A} + 4\pi_1\rho(1-\rho)\epsilon^2\left((1-\epsilon^2)+2\left(1+(1-2\rho)\epsilon\right)\right)}{2\epsilon^2\mathcal{M}^2},$$

where

$$\mathcal{A} \equiv 8\epsilon^2(1+\epsilon^2)\rho^2 - 8\epsilon(1+\epsilon)(1+\epsilon^2)\rho + (1+\epsilon)^4 = 8\epsilon^2(1+\epsilon^2)\left(\rho - \frac{1+\epsilon}{2\epsilon}\right)^2 - (1-\epsilon^2)^2.$$

Since $\frac{1+\epsilon}{2\epsilon} > 1$, this quadratic in ρ is decreasing on $(0,1]$ and equals $(1-\epsilon)^4 > 0$ at $\rho = 1$; hence $\mathcal{A} > 0$. Since $1+(1-2\rho)\epsilon \geq 1-\epsilon > 0$, both terms in the numerator are positive and the derivative is negative. Hence, the bias is decreasing in ϵ.

Sixth, fix $\phi > 0$ and let ϵ go to 0. By (59) the observed relative elasticity converges to a finite number,

$$\lim_{\epsilon\to 0}\hat{\mathcal{E}}_{g|r} = \frac{\pi_1\left(\pi_1 + 2\phi(1-\pi_1)\right)}{2\phi(1-\pi_1)} < \infty,$$

while the structural relative elasticity (58) diverges to infinity because its denominator is of order ϵ^2. Thus,

$$\lim_{\epsilon\to 0}\mathcal{B}^{\text{rel}} = \infty.$$

Seventh, suppose $\epsilon = o(\phi^{\frac{1}{2}})$, that is, $\frac{\epsilon^2}{\phi} \to 0$ as $\phi \to 0$. Multiplying the numerator and the denominator of (61) by $\frac{\phi}{\epsilon^2(1-\pi_1)}$ yields the identity

$$\mathcal{B}^{\text{rel}} = \frac{(1-\epsilon^2)\left(1+(1-2\rho)\epsilon\right)\mathcal{P}}{8\rho(1-\rho)} \cdot \frac{1}{\frac{4\pi_1\rho(1-\rho)}{1-\pi_1}\frac{\epsilon^4}{\phi} + \epsilon^2\mathcal{L}}. \tag{63}$$

Since $\epsilon^2 = \frac{\epsilon^2}{\phi}\cdot\phi \to 0$, we have $\epsilon \to 0$, so the first factor of (63) converges to $\frac{\pi_1}{8\rho(1-\rho)} > 0$, while both terms in the denominator of the second factor vanish: $\frac{\epsilon^4}{\phi} = \frac{\epsilon^2}{\phi}\cdot\epsilon^2 \to 0$ and $\epsilon^2\mathcal{L} \to 0$. Hence

$$\lim_{\phi\to 0}\mathcal{B}^{\text{rel}} = \infty.$$

The proof is complete.[28] ■

[28] The condition $\epsilon = o(\phi^{\frac{1}{2}})$ is indeed for ease of exposition. Along $\epsilon = \phi^a$ with $a > 0$, (63) shows that the

Remark 3 (The bias $\mathcal{B}^{\text{rel}}$ is increasing in ϕ) *We remark that the bias $\mathcal{B}^{\text{rel}}$ is increasing in ϕ. By dropping the terms of (61) that do not depend on ϕ, it is sufficient to show that*

$$\frac{\partial}{\partial \phi}\left(\frac{\phi}{2\rho(1-\rho)}\frac{\mathcal{P}}{\mathcal{M}}\right) > 0. \tag{64}$$

After some algebra, the left-hand side reduces to:

$$\frac{1}{2\rho(1-\rho)} + \frac{2\pi_1^2\epsilon^2(1-\epsilon^2)\left(1+(1-2\rho)\epsilon\right)}{\mathcal{M}^2} > 0. \tag{65}$$

Thus, the bias $\mathcal{B}^{\text{rel}}$ is increasing in ϕ.

A.3 Section 5.2

Proof of Proposition 7. With $u_i(c) = -e^{-\gamma^i c}$ we have $u_i'' = -\gamma^i u_i'$, so the curvature weights of Lemma 1 are

$$K_z^i = \frac{\delta \pi_z^i u_i''(\tilde{c}_1^i(z))}{\lambda^i} = -\gamma^i \frac{\delta \pi_z^i u_i'(\tilde{c}_1^i(z))}{\lambda^i} = -\gamma^i \tilde{\Lambda}^i(z),$$

where $\tilde{\Lambda}^i(z)$ is the investor's personal state price for state z. With no asset-specific tastes and no binding constraints on the assets she holds, her first-order conditions are $\sum_z \tilde{\Lambda}^i(z)\, y_j(z) = p_j$ for every such asset, that is, $Y^{\mathrm{T}}\tilde{\Lambda}^i = p = Y^{\mathrm{T}}q$: her personal state prices agree with the market's on the span of payoffs she trades, and under complete markets $\tilde{\Lambda}^i = q$. Hence

$$H^i = \sum_z K_z^i\, y(z)y(z)^{\mathrm{T}} = -\gamma^i\, Y^{\mathrm{T}}\mathrm{diag}(\tilde{\Lambda}^i)\, Y = -\gamma^i\, Y^{\mathrm{T}}\mathrm{diag}(q)\, Y,$$

where the last equality uses only agreement on the span, since $Y^{\mathrm{T}}\mathrm{diag}(v)Y$ depends on v only through the prices it assigns to the payoffs $y_j(z)y_k(z)$; these are observed from derivative prices under the hypothesis of the proposition, and are priced by the market and by the investor alike when she can trade the relevant derivatives. Inverting gives (38). Her beliefs π^i entered only through $\tilde{\Lambda}^i$, which the first-order conditions replaced by q; this is why heterogeneity in beliefs about state probabilities is accommodated, and why disagreement about payoffs, which changes Y on the left-hand side of $Y^{\mathrm{T}}\tilde{\Lambda}^i = p$, is not.

With a traded riskless asset, $\sum_z q_z = p_f$ and $q_z/p_f = Q_z$, so $H^i = -\gamma^i p_f\, \mathbb{E}^Q[yy^{\mathrm{T}}]$. Partition H^i with the riskless asset first. The riskless row of $H^i x = e_j$, for a risky asset j, reads $-\gamma^i p_f\left(x_0 + \mathbb{E}^Q[y]^{\mathrm{T}}x_{\text{risky}}\right) = 0$; substituting $x_0 = -\mathbb{E}^Q[y]^{\mathrm{T}}x_{\text{risky}}$ into the risky

bias diverges if and only if $a > \frac{1}{4}$.

rows removes the mean term from $\mathbb{E}^Q[yy^T]$ and leaves $-\gamma^i p_f \operatorname{Cov}_Q(y)\, x_{\text{risky}} = e_j$, so the substitution matrix among risky assets is $-(\gamma^i p_f)^{-1}\operatorname{Cov}_Q(y)^{-1}$.

Finally, under CARA there are no wealth effects on risky demand, so for any price change dp at fixed beliefs the holdings change is $da^i = -(\gamma^i p_f)^{-1}\operatorname{Cov}_Q(y)^{-1}dp$; given $\operatorname{Cov}_Q(y)$ from derivative prices, any single component of this equation determines γ^i, and the remaining components are over-identifying restrictions. ■

A.3.1 Additional Result

HHHKL25 measure demand in semi-elasticities of portfolio shares, $\mathcal{E}^i_{jk} = -\partial\omega^i_j/\partial \log p_k$ (retaining our sign convention), which is the natural unit in the benchmark economy with CRRA preferences. The following lemma is stated in those units.

Lemma 6 (Restricted demand complementarities) *Suppose (A1) and (A2) hold in semi-elasticity units, and that portfolio shares are homogeneous of degree zero in the price vector, as in the benchmark economy where all wealth is invested. Then for any inside assets j and k with the same observables, the difference of portfolio shares depends only on the price ratio: there exists a function G with $\omega_j - \omega_k = G(p_j/p_k)$, and since $\mathcal{E}^{\text{rel}}$ is a parameter, G is affine in the log price ratio:*

$$\omega_j - \omega_k = \alpha - \mathcal{E}^{\text{rel}} \log \frac{p_j}{p_k}. \tag{66}$$

Lemma 6 is the logit restriction conditional on observables. In logit, $\omega_j/\omega_k = \exp(\eta_j - \eta_k)(p_j/p_k)^{-\beta_0}$ for every pair; in HHHKL25, (66) holds for pairs with identical observables, in differences rather than ratios. The economic content is the same: conditional on observables, the relative demand for two assets may depend on neither the level of prices nor any third price, even though both dependences arise generically in portfolio choice.

Proof of Lemma 6. The proof consists of three steps. The first step shows that $\omega_j - \omega_k$ depends only on p_j and p_k. Assumption (A1) gives $\mathcal{E}_{j\ell} = \mathcal{E}_{k\ell}$ for every $\ell \neq j,k$. Recalling $\mathcal{E}_{j\ell} = -\frac{\partial \omega_j}{\partial \log p_\ell}$, we have:

$$\frac{\partial(\omega_j - \omega_k)}{\partial \log p_\ell} = -\left(\mathcal{E}_{j\ell} - \mathcal{E}_{k\ell}\right) = 0 \quad \text{for every } \ell \notin \{j,k\}.$$

Thus, $\omega_j - \omega_k$ depends on p_j and p_k alone.

The second step shows that $\omega_j - \omega_k$ depends only on the ratio $\frac{p_j}{p_k}$. Portfolio shares are homogeneous of degree zero in the price vector, and thus $\sum_s \frac{\partial \omega_j}{\partial \log p_s} = 0$ for each asset

and hence $\sum_s \frac{\partial(\omega_j-\omega_k)}{\partial \log p_s} = 0$. With Step 1 this leaves

$$\frac{\partial(\omega_j - \omega_k)}{\partial \log p_j} + \frac{\partial(\omega_j - \omega_k)}{\partial \log p_k} = 0,$$

and thus $\omega_j - \omega_k$ is constant along the direction $(1,1)$ in $(\log p_j, \log p_k)$. Integrating along that direction gives $(\omega_j - \omega_k)(\lambda p_j, \lambda p_k) = (\omega_j - \omega_k)(p_j, p_k)$, and setting $\lambda = 1/p_k$ completes the proof of the second step.

The third step shows that G is affine in $\log \frac{p_j}{p_k}$. For ease of notation, write $\mathcal{R} = \frac{p_j}{p_k}$. Then, we have:

$$\frac{\partial(\omega_j - \omega_k)}{\partial \log p_j} = \frac{dG}{d \log \mathcal{R}}.$$

Then,

$$\mathcal{E}^{\text{rel}} = \mathcal{E}_{jj} - \mathcal{E}_{kj} = -\frac{\partial(\omega_j - \omega_k)}{\partial \log p_j} = -\frac{dG}{d \log \mathcal{R}}.$$

Since Assumption (A2) holds this constant, it follows that

$$G(\mathcal{R}) = -\mathcal{E}^{\text{rel}} \log \mathcal{R} + \alpha,$$

where α is a constant. Thus, we obtain (66). ■

Demand-System Asset Pricing: Theoretical Foundations

Online Supplementary Appendix

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B The CARA-Normal model

In the main text, we characterized spillover bias both generally and within a tractable three-asset model designed to highlight the underlying economic mechanisms. We now show that these mechanisms extend naturally to a more general—and arguably more realistic—benchmark framework: the "CARA-Normal" setting, which features CARA utility, an arbitrary number of risky assets and a risk-free asset, and normally distributed payoffs. This framework is particularly relevant because it serves as a standard workhorse model in the literature. Indeed, KY25 claim their cross-sectional estimator overcomes spillover bias in this setting. However, we show that demand estimation in this setting remains generically subject to spillover bias. Furthermore, even under the most favorable *knife-edge* parameter restrictions, the KY25 estimator identifies, at best, the relative slope between two assets rather than the absolute slope of any individual demand curve. By Section 5.1, this relative parameter is generically insufficient for conducting counterfactuals.

Setup. We closely follow the setup of KY25. The representative agent has CARA utility $u(c) = -\exp(-\gamma c)$ (where $\gamma > 0$ is the coefficient of absolute risk aversion) and consumes at two dates, $t = 0, 1$. There are $N \geq 2$ risky assets with payoff vector

$$Y = \mu + F\beta + \nu,$$

where $\mu \in \mathbf{R}^N$ is the vector of mean payoffs, $\beta \in \mathbf{R}^N$ is the vector of factor loadings, and $F \sim \mathcal{N}(0, \sigma_F^2)$ is a common factor independent of $\nu \sim \mathcal{N}(0, \sigma^2 \mathbb{I}_N)$. To follow standard notation in the literature, in this appendix we use β_j to denote factor loadings as opposed to the coefficient on the own price in KY19's demand function.

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While KY25 assume factor loadings are identical and equal to one, $\beta = \mathbf{1}$, we begin with the general case to characterize the mechanics of cross-asset spillovers. To that end, we impose the following regularity conditions: first, as $N \to \infty$, $\sup_j |\mu_j| < \infty$ and $\overline{\mu}_N \equiv \frac{1}{N}\sum_{j=1}^N \mu_j \to \overline{\mu}$; second, as $N \to \infty$, $\sup_j |\beta_j| < \infty$ and $\overline{\beta}_N \equiv \frac{1}{N}\sum_{j=1}^N \beta_j \to \overline{\beta}$; and third, $\lim_{N\to\infty} \frac{1}{N}\beta^{\mathrm{T}}\beta \in (0, \infty)$. The covariance matrix simplifies to

$$\Sigma \equiv \mathrm{Cov}[Y, Y^{\mathrm{T}}] = \sigma^2 \left(\mathbb{I}_N + \frac{\sigma_F^2}{\sigma^2} \beta\beta^{\mathrm{T}} \right).$$

By the Woodbury formula, Σ^{-1} can be written as:

$$\Sigma^{-1} = \frac{1}{\sigma^2}\left(\mathbb{I}_N - \tau(\beta) \cdot \beta\beta^{\mathrm{T}} \right), \quad \text{where} \quad \tau(\beta) \equiv \frac{\sigma_F^2}{\sigma^2 + \sigma_F^2(\beta^{\mathrm{T}}\beta)} \in \left(0, \frac{1}{\beta^{\mathrm{T}}\beta}\right) \tag{67}$$

measures the share of the covariance structure attributable to the common factor. When $\beta = \mathbf{1}$, we have $\tau(\mathbf{1}) = \frac{\sigma_F^2}{\sigma^2 + N\sigma_F^2}$.

Investors can also trade a risk-free asset. KY25 assume this asset is exogenously and perfectly elastically supplied at some fixed price p_f. We instead assume that the risk-free asset is in fixed supply (zero for convenience) and its rate of return is endogenously determined. In our specification, all assets are thus consistently priced by a single market-wide stochastic discount factor. KY25 instead assume that the risk-free asset price is set exogenously, which is inconsistent with general equilibrium market clearing in an endowment economy. While this assumption may appear innocuous in a large economy with many assets, we show below that this is not the case.

We normalize each asset supply by N to hold the size of the economy fixed as we change the number of assets. This allows us to study the same limit as in KY25. Specifically, we assume a supply of $E_j = \frac{1}{N} + \psi_j$ for each risky asset j. We denote by $\psi \equiv (\psi_j)_{j=1}^N$ a vector of supply shocks used to construct the ideal instrument. In our benchmark analysis, we take $\psi = \mathbf{0}$. As we assume the risk-free asset is in zero net supply, $E_f = 0$. As is standard, consumption is the numeraire with price normalized to one, and the risk-free asset is the "outside asset." The set of "inside assets" thus consists of the N risky assets.

Demand functions and prices. To report prices and demand functions, we introduce two summary statistics. Namely, (i) *aggregate factor exposure*, which measures the quantity

of factor risk the market must absorb, and (ii) the *risk-adjusted mean payoff* of each asset:

$$B(\psi) \equiv \sum_{j=1}^{N} \left(\frac{1}{N} + \psi_j \right) \beta_j \quad \text{and} \quad M_j(\psi) \equiv \mu_j - \gamma \Big(\underbrace{B(\psi)\beta_j \sigma_F^2}_{\text{factor risk}} + \underbrace{(\frac{1}{N} + \psi_j)\sigma^2}_{\text{idiosyncratic risk}} \Big). \tag{68}$$

Under $\beta = \mathbf{1}$, $B(\psi) = 1 + \Psi$, where $\Psi \equiv \sum_{j=1}^{N} \psi_j$ is the total asset supply shock. Also, denote by $p \equiv (p_j)_{j=1}^{N}$ the price vector of risky assets. With these in mind, we derive demand functions and show that they depend on the entire vector of factor loadings and expected returns. We also derive equilibrium asset prices.

Lemma 7 *1. Demand functions for risky assets can be written as:*

$$a_j = \frac{1}{\gamma \sigma^2} \left(\mu_j - \frac{p_j}{p_f} - \beta_j \kappa \right), \quad \textit{where} \quad \kappa \equiv \tau(\beta) \cdot \beta^{\mathrm{T}} \left(\mu - \frac{p}{p_f} \right) \tag{69}$$

is the factor risk premium.

2. *The risk-free asset's price depends on the entire supply vector ψ:*

$$p_f(\psi) = \delta \exp \left(\gamma C_0 - \gamma \overline{\mu}_N - \gamma \sum_{j=1}^{N} \psi_j \mu_j + \frac{\gamma^2}{2} \left[B(\psi)^2 \sigma_F^2 + \sum_{j=1}^{N} \left(\frac{1}{N} + \psi_j \right)^2 \sigma^2 \right] \right), \tag{70}$$

where C_0 is time-0 consumption.

3. *The price of risky asset j can then be written as*

$$p_j(\psi) = p_f(\psi) M_j(\psi). \tag{71}$$

It follows that *any* variation in asset supply which changes a given risky asset price must also change the risk-free price, and consequently alter the price of every other risky asset. We maintain the positivity assumption $p_j > 0$, i.e., $M_j(\psi) > 0$.

Demand complementarities. Equation (69) shows that demand for any asset depends on the supply and payoff properties of all other assets. There are two specific channels of demand complementarities. The first is the *factor risk channel*, captured by the term measured by $\beta_j \kappa$: changes in the quantity of any risky asset alter the total factor risk that must be absorbed, thereby changing demand for any other asset exposed to the same factor. The second is the *valuation channel*: since supply shocks change the total quantity of

consumption in the second period, the relative price of future consumption (i.e., the risk-free rate) must change, thereby altering the valuation of all other assets as well. Hence, demand for asset j depends on the *relative* price p_j/p_f, not just p_j.

The strength of these complementarities generically differs across assets. With respect to factor risk, assets j and k have heterogeneous exposure if factor loadings differ: $\beta_j \neq \beta_k$. With respect to valuation, two assets have different exposure if average payoffs differ: $\mu_j \neq \mu_k$. To see this, it follows from (69) that

$$\frac{\partial a_j}{\partial p_f} = \frac{1}{\gamma\sigma^2 p_f}\left(\frac{p_j}{p_f} - \beta_j \frac{\sigma_F^2}{\sigma^2 + \sigma_F^2 \beta^{\mathrm{T}}\beta}\beta^{\mathrm{T}}\frac{p}{p_f}\right).$$

When $\beta = \mathbf{1}$, exposures differ across assets exactly in proportion to expected payoffs:

$$\frac{\partial a_j}{\partial p_f} - \frac{\partial a_k}{\partial p_f} = \frac{p_j/p_f - p_k/p_f}{\gamma\sigma^2 p_f} = \frac{(\mu_j - \mu_k) - \gamma\sigma^2(\psi_j - \psi_k)}{\gamma\sigma^2 p_f}, \tag{72}$$

which reduces to $(\mu_j - \mu_k)/(\gamma\sigma^2 p_f)$ whenever the two assets are in equal supply, in particular around $\psi = \mathbf{0}$. This shows that asymmetric complementarities are generic in any realistic setting: expected payoffs and factor loadings are latent variables which are, at best, measured with substantial noise, and exact symmetry is measure zero in the space of payoff distributions. Nevertheless, we will show that the estimator in KY25 depends sensitively on the knife-edge restrictions $\beta_j = \beta_k$ and $\mu_j = \mu_k$ for all j and k.

We now discuss how spillover bias appears in both time-series (as in Shleifer (1986)) and cross-sectional demand estimation strategies (as in KY25).

B.1 Within-asset estimator

We first show that demand complementarities create the same fundamental inference problems as in our basic model. We begin by considering the case where researchers observe a "time series" of portfolio holdings prior to and after a purely exogenous supply shock. This is the spirit of Shleifer (1986) and other index inclusion studies, where the demand response is calculated by comparing prices before and after an inclusion event.

While demand is iso-elastic in our benchmark model, the CARA-Normal model yields linear demand. Hence, it is appropriate to describe demand functions using *slopes* rather than elasticities. The *structural slope*

$$\mathcal{J}^i_{jk}(N) \equiv -\frac{\partial a^i_j(p \mid N)}{\partial p_k}$$

is the partial derivative with respect to a single price. In the CARA-Normal model, it follows from (69) that the structural own-price slope is

$$\mathcal{J}^i_{jj}(N) = \frac{1}{\gamma p_f \sigma^2}\left(1 - \frac{\beta_j^2 \sigma_F^2}{\sigma^2 + \sigma_F^2 \beta^{\mathrm{T}} \beta}\right), \tag{73}$$

which reduces to $\frac{1}{\gamma p_f \sigma^2}\left(1 - \frac{\sigma_F^2}{\sigma^2 + N\sigma_F^2}\right)$ when $\beta = \mathbf{1}$. Note that the own-price slope depends on the asset's *own* factor loading: with heterogeneous loadings, no two assets share the same structural slope. The *observed slope* $\hat{\mathcal{J}}^i_{js}$ with respect to a supply shock ψ_s is the total derivative with respect to all endogenous price changes:

$$\hat{\mathcal{J}}^i_{js}(N) \equiv -\frac{\frac{da^i_j(p|N)}{d\psi_s}}{\frac{dp_s}{d\psi_s}}.$$

As discussed in the main text, the bias implied by an estimation strategy depends on the assumed mapping from observed to measured slopes. For example, Shleifer (1986) assumes that the observed slope in response to an index inclusion event *is* the true slope, so that the observed slope is identically equal to measured slope. KY25 instead use a cross-sectional strategy, which we discuss in the next subsection.

In analogy to Proposition 4, Proposition 8 then shows that the structural slope is always higher than the observed slope, and that the difference diverges to infinity as risky assets become perfect substitutes.[29] We also show that observed and structural slopes differ even when the number of risky assets goes to infinity: $N \to \infty$. This is perhaps surprising because cross-asset spillovers operate at least in part through the risk-free asset, and a supply shock to a single asset is typically assumed to have small effects on the risk-free rate. However, spillover bias is the *sum* of effects across all potential substitute assets, and thus remains of at least the same order of magnitude as the direct price change.

Proposition 8 (Slope bias in the CARA-Normal model) *Consider a small supply shock to a single asset j. Then:*

1. *The observed demand slope with respect to the shock is*

$$\hat{\mathcal{J}}^i_{jj}(N) = \frac{1}{\gamma p_f} \frac{1}{\left(M_j(\mathbf{0})\right)^2 + \beta_j^2 \sigma_F^2 + \sigma^2}. \tag{74}$$

[29] As in KY25, we model perfect substitutes as the limit where risky assets' idiosyncratic volatility goes to zero (i.e., $\sigma \downarrow 0$).

2. *The structural slope is higher than the observed slope:* $\mathcal{J}^i_{jj}(N) > \hat{\mathcal{J}}^i_{jj}(N)$ *for all* $\sigma > 0$.

3. *The bias between structural and observed slopes diverges to infinity as risky assets become perfect substitutes, provided that* $\sum_{k \neq j} \beta_k^2 > 0$*:*

$$\lim_{\sigma \downarrow 0} \mathcal{J}^i_{jj}(N) = \infty \quad \textit{while} \quad \lim_{\sigma \downarrow 0} \hat{\mathcal{J}}^i_{jj}(N) < \infty.$$

4. *Structural and observed slopes differ for any number of risky assets: the ratio* $\mathcal{J}^i_{jj}(N) / \hat{\mathcal{J}}^i_{jj}(N)$ *does not converge to one even as* $N \to \infty$.

Proof. See Appendix B.4. ■

Proposition 8 thus mirrors the finding of Proposition 4: the structural slope diverges as assets become perfect substitutes ($\sigma \downarrow 0$, analogous to $\epsilon \downarrow 0$), while the observed slope remains bounded. Notably, this is the case even with many risky assets: since the bias is the *sum* of all price spillovers (cf. equation (21) of Proposition 3), the difference can be unbounded even if the price spillover to any given individual asset is small.[30]

The left panel of Figure 6 confirms this graphically by plotting the structural and observed demand slopes over σ for different values of N. We thus find a strikingly similar pattern to our previous analysis. The right panel confirms that this bias cannot be removed simply by subtracting the observed slope of the risk-free asset, similar in spirit to the logit demand framework which scales all inside asset holdings by the outside asset.

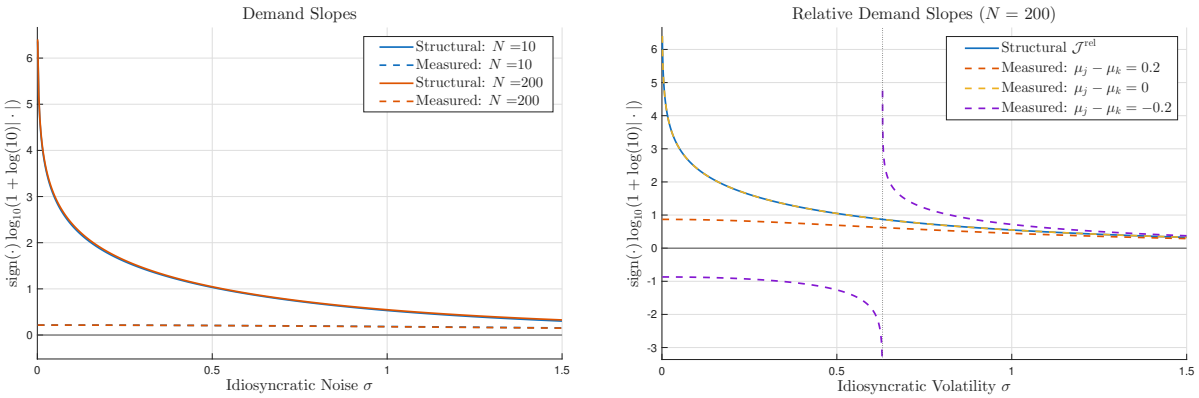


Figure 6: Structural and observed slopes. Left: Structural and observed slopes for various N. Right: Structural and observed *relative* slopes for various $\mu_j - \mu_k$.

These results confirm the challenges presented by cross-asset spillovers in demand measurement. While, for example, Shleifer (1986) *assumes* that a shock to index member-

[30] Similarly, while the effect $\frac{\partial a_j}{\partial p_k}$ vanishes as the number of risky assets N tends to infinity, the bias consists of the *sum* of all price spillovers (there are $N-1$ of them) so that the ratio $\mathcal{J}^i_{jj}(N) / \hat{\mathcal{J}}^i_{jj}(N)$ does not converge to one even as $N \to \infty$ as in Part 4.

ship does not affect other asset prices, Proposition 8 shows that even ideal shocks create spillovers that directly affect the observed slope, thereby contradicting the ceteris paribus assumption required to interpret the observed response as the slope of a fixed demand function.

B.2 Cross-sectional estimators

We next analyze whether cross-sectional estimators can accurately identify slopes of asset demand functions in the CARA-Normal model. We focus in particular on the instrumental-variables estimator proposed by KY25, which aims to identify asset-level demand functions from a single cross-section of holdings and instrumented prices. Specifically, the estimator is based on the linear cross-sectional regression

$$a_j^i = -b_0 p_j + b_1 + \epsilon_j^i,$$

where b_0 is now used to denote the coefficient of interest. As in Section 4.5, let $\mathrm{Cov}_{\mathrm{cs}}$ denote the cross-sectional covariance. Given a vector of supply instruments ψ, the estimator is

$$\hat{\mathcal{J}}^{\mathrm{KY25}} \equiv -\frac{\mathrm{Cov}_{\mathrm{cs}}\left[a_j^i, \psi_j\right]}{\mathrm{Cov}_{\mathrm{cs}}\left[p_j, \psi_j\right]}. \tag{75}$$

By construction, this estimator removes any part of demand constant across all assets in the cross-section. As such, the approach can at best recover the *difference* in demand slopes across inside assets, but not the absolute demand slope for any asset. (This is, of course, the object studied by HHHKL25, which is insufficient for counterfactuals.) The reason is mechanical: the valuation channel operates through p_f, which is fixed within a given cross-section. Hence the demand response $\frac{\partial a_j}{\partial p_f}$ is the coefficient on a regressor without cross-sectional variation and is absorbed into the intercept.

The role of symmetry. By (73), the relative slope of asset j with respect to asset k is

$$\mathcal{J}^{\mathrm{rel}} \equiv \mathcal{J}_{jj}^i - \mathcal{J}_{kj}^i = \frac{1}{\gamma\sigma^2 p_f}\left(1 - \tau(\beta)\cdot\beta_j(\beta_j - \beta_k)\right), \tag{76}$$

which depends on the pair and is not even symmetric whenever $\beta_j \neq \beta_k$. There is thus no single "relative slope" to estimate unless loadings are homogeneous. This is the first of the two symmetry assumptions on which KY25 rely. When $\beta_j = \overline{\beta}$ for all j (without

loss, $\beta = \mathbf{1}$), (76) collapses to a single parameter for every pair:

$$\mathcal{J}^{\text{rel}} = \frac{1}{\gamma\sigma^2 p_f} = (1 + \tau(\mathbf{1}))\, \mathcal{J}^i_{jj}(N), \quad \text{where} \quad \tau(\mathbf{1}) = \frac{\sigma_F^2}{\sigma^2 + (N-1)\sigma_F^2} > 0, \tag{77}$$

and thus the relative slope exceeds the own-price slope by the cross-price slope.

The second assumption is that all assets have identical expected payoffs μ_j. This assumption is unverifiable (as expected returns are not observable) and knife-edge in the space of payoff distributions. This assumption is made to ensure that all price spillovers can be absorbed into a single constant that is the same for every asset. As we show next, even marginal cross-sectional differences in expected payoffs—which generically occur—can induce severe biases, including *wrong signs* in the estimated relative slope.

The exact estimand. We write expected payoffs as $\mu_j = \overline{\mu} + u_j$ where u_j is unobserved by the econometrician, and let σ_μ denote the cross-sectional dispersion in expected payoffs. Exactly consistent with KY19, u_j *is* latent demand in this model: substituting $\mu_j = \overline{\mu} + u_j$ into (69) shows that the residual of the estimating equation is $\epsilon^i_j = u_j/(\gamma\sigma^2)$. Based on KY19, dispersion in expected payoffs thus is the empirically relevant case.

To provide favorable conditions for the estimator, let the instrument be randomly assigned, $\psi \perp u$. For convenience, we also assume a regularity condition which ensures positive asset prices, $\overline{\mu} - \gamma\left(\sigma_F^2 + \frac{\sigma^2}{N}\right) \gg \sigma_\mu$, and Gaussian u. (Because $\hat{\varrho}$ in Lemma 8 is a sample regression coefficient, it is asymptotically normal for any u with finite variance. Our results hold asymptotically for other distributions.) The estimator then admits an exact characterization.

Lemma 8 (Exact estimand) *For any N and any realized supply vector ψ, the estimator is*

$$\hat{\mathcal{J}}^{\text{KY25}} = \frac{1}{p_f(\psi)\,(\gamma\sigma^2 - \hat{\varrho})}, \quad \textit{where } \hat{\varrho} \equiv \frac{\text{Cov}_{\text{cs}}\left[\mu_j, \psi_j\right]}{\mathbb{V}_{\text{cs}}[\psi_j]} \tag{78}$$

is the realized cross-sectional regression coefficient of expected payoffs on the instrument.

Proof. See Appendix B.4. ■

Bias. First, the estimator thus recovers $\mathcal{J}^{\text{rel}}$ only when $\hat{\varrho} = 0$, that is, when the instrument is orthogonal to expected payoffs *in the realized cross-section*. This holds identically under $\mu_j = \overline{\mu}$ but with probability zero otherwise. Second, the estimate changes sign as $\hat{\varrho}$ crosses $\gamma\sigma^2$. The reason is that the estimator infers demand slopes from level differences in prices and quantities, whereas unobserved heterogeneity in expected payoffs can alter

the ranking of assets by price. This can lead to severe biases in estimated slopes, including sign errors, which are particularly large when structural demand functions are steep.

Proposition 9 (Spillover Bias in the CARA-Normal Economy) *Assume $\beta = \mathbf{1}$. If $\sigma_\mu = 0$ (as is guaranteed by the unverifiable assumption that $\mu_j = \overline{\mu}$ for all j), then the cross-sectional estimator exactly identifies the relative slope between inside assets. Thus, assume $\sigma_\mu > 0$. Then, the following hold:*

1. *Denoting the first-stage signal-to-noise ratio by*

$$\lambda \equiv \frac{\gamma\sigma^2\sqrt{N\mathbb{V}_{\mathrm{cs}}[\psi]}}{\sigma_\mu},$$

   we have $\frac{\hat{\mathcal{J}}^{\mathrm{KY25}}}{\mathcal{J}^{\mathrm{rel}}} = \frac{\lambda}{\lambda - Z}$ with $Z \sim \mathcal{N}(0,1)$.

2. *The estimator has the wrong sign with strictly positive probability:*

$$\mathbb{P}\left(\hat{\mathcal{J}}^{\mathrm{KY25}} < 0\right) = 1 - \Phi(\lambda),$$

   which tends to $\frac{1}{2}$ as $\lambda \to 0$, where Φ denotes the cdf of the standard normal distribution.

3. *The estimated relative slope $\hat{\mathcal{J}}^{\mathrm{KY25}}$ exceeds the structural own slope with probability $\Phi(\lambda) - \Phi(-\tau(\mathbf{1})\lambda)$. Moreover, this probability converges to $\frac{1}{2}$ as $N \to \infty$.*

4. *The difference between structural and estimated relative slope can diverge for perfect substitutes. Fix a realization with $\hat{\varrho} \neq 0$. As risky assets become perfect substitutes (i.e., $\sigma \downarrow 0$),*

$$\hat{\mathcal{J}}^{\mathrm{KY25}} \to -\frac{1}{\left(\lim_{\sigma\downarrow 0} p_f\right)\hat{\varrho}} < \infty \quad \text{and} \quad \mathcal{J}^{\mathrm{rel}} \to \infty \quad \text{a.s.}$$

   Also, $\mathbb{P}\left(\hat{\mathcal{J}}^{\mathrm{KY25}} < 0\right) \to \frac{1}{2}$ as $\sigma \downarrow 0$.

Proof. See Appendix B.4. ■

We emphasize that Proposition 9 does not contradict consistency: for fixed $\sigma^2 > 0$ and a randomly assigned instrument, $\hat{\varrho} \xrightarrow{p} 0$ and the estimator converges to $\mathcal{J}^{\mathrm{rel}}$ as $N \to \infty$. Nor is the estimator biased *whenever* expected payoffs differ: controlling for any component of μ_j spanned by observed characteristics repairs the estimand.[31] Instead, the central failure is of uniformity. Taking $N \to \infty$ first, the estimand diverges as assets

[31] However, such an assumption pertains to an unobservable object and can never be verified: see Fuchs, Fukuda, and Neuhann (2026).

become perfect substitutes, exactly as KY25 claim. Taking $\sigma \to 0$ at any finite N—the order of limits relevant to any application, and the one in which close substitutes actually appear—the estimate instead plateaus at a finite value whose sign is a coin flip. Even in the best case, one cannot obtain the full structural demand slope, only the relative slope. Overall, our main conclusions thus are remarkably robust in this alternative setting.

B.3 Relative elasticity

We turn to relative elasticity. To focus on the comparison with HHHKL25 in the context of the CARA-Normal model, we consider the relative elasticity of asset j with respect to asset k in terms of demand slopes, i.e., (76). When $\beta = \mathbf{1}$, the relative slope reduces to the single parameter

$$\mathcal{J}^{\text{rel}} = \frac{1}{\gamma p_f \sigma^2}.$$

Now impose a supply shock ψ_j (which creates variation in relative prices), and let

$$\hat{\mathcal{J}}^{\text{rel}} \equiv -\frac{d(a_j - a_k)/d\psi_j}{d(p_j - p_k)/d\psi_j}$$

be the observed relative slope, with bias $\mathcal{B}^{\text{rel}} \equiv \mathcal{J}^{\text{rel}} - \hat{\mathcal{J}}^{\text{rel}}$.

Proposition 10 *Assume $\beta = \mathbf{1}$. Then:*

1. *The bias $\mathcal{B}^{\text{rel}}$ satisfies:*

$$\mathcal{B}^{\text{rel}} = \left(\frac{\partial a_j}{\partial p_f} - \frac{\partial a_k}{\partial p_f} \right) \frac{dp_f/d\psi_j}{(dp_j - dp_k)/d\psi_j}. \tag{79}$$

2. *At around $\psi = \mathbf{0}$, the observed relative slope and the bias satisfy:*

$$\hat{\mathcal{J}}^{\text{rel}} = \frac{1}{\gamma p_f \left(\sigma^2 + M_j(\mathbf{0})(\mu_j - \mu_k)\right)} \quad \textit{and} \quad \mathcal{B}^{\text{rel}} = \frac{M_j(\mathbf{0})(\mu_j - \mu_k)}{\gamma p_f \sigma^2 \left(\sigma^2 + M_j(\mathbf{0})(\mu_j - \mu_k)\right)}. \tag{80}$$

3. *Whenever $\mu_j \neq \mu_k$:*

$$\lim_{N\to\infty} \mathcal{B}^{\text{rel}} = \frac{\left(\mu_j - \gamma\sigma_F^2\right)(\mu_j - \mu_k)}{\gamma \left(\lim_{N\to\infty} p_f\right) \sigma^2 \left(\sigma^2 + (\mu_j - \gamma\sigma_F^2)(\mu_j - \mu_k)\right)} \neq 0; \tag{81}$$

$$\lim_{\sigma \downarrow 0} \mathcal{B}^{\text{rel}} = +\infty. \tag{82}$$

The right panel of Figure 6 depicts the structural and observed relative slopes $\mathcal{J}^{\text{rel}}$ and $\hat{\mathcal{J}}^{\text{rel}}$. In this general-equilibrium endowment economy, the price of the risk-free asset p_f is not a parameter but a function that depends on the supply shifters ψ. Unless $\frac{\partial a_j}{\partial p_f} = \frac{\partial a_k}{\partial p_f}$ (which is part of the symmetry assumptions of HHHKL25), the relative slope $\mathcal{J}^{\text{rel}}$ under a supply shock is biased. That is, the symmetry assumption regarding the risk-free asset requires that the change in the price of the risk-free asset affects all the risky assets homogeneously. By (72),

$$\left(\frac{\partial a_j}{\partial p_f} - \frac{\partial a_k}{\partial p_f}\right)\bigg|_{\psi=\mathbf{0}} = \frac{\mu_j - \mu_k}{\gamma p_f(\mathbf{0})\sigma^2}. \tag{83}$$

Thus, under $\beta = \mathbf{1}$, the condition for an unbiased relative slope is that two assets have the same expected payoff.

Two remarks are in order. First, when $\mu_j \neq \mu_k$, Part 3 states that, as risky assets become perfect substitutes (i.e., $\sigma \downarrow 0$), the bias goes to infinity. Part 2 and (76) imply that while the observed relative slope is finite, the structural relative slope tends to infinity:

$$\lim_{\sigma\downarrow 0} \mathcal{J}^{\text{rel}} = \infty \quad \text{and} \quad \lim_{\sigma\downarrow 0} \hat{\mathcal{J}}^{\text{rel}} < \infty.$$

Second, as illustrated in the right panel of Figure 6, the bias is not signed.[32] Moreover, the observed relative slope has the wrong sign, i.e., $\hat{\mathcal{J}}^{\text{rel}} < 0 < \mathcal{J}^{\text{rel}}$, if and only if $\mu_k - \mu_j > \sigma^2 / M_j(\mathbf{0})$.

In analogy to Section 5.1, small changes in the underlying economic primitives can therefore bias the relative slope, and this bias is severe when close substitutes are available.

B.4 Proofs for Appendix B

Proof of Lemma 7. The representative agent consumes $C_1 = \sum_{j=1}^N (\frac{1}{N} + \psi_j) Y_j$ in the next period. Since $Y_j = \mu_j + \beta_j F + \nu_j$, it follows that

$$C_1 = \left(\overline{\mu}_N + \sum_{j=1}^N \psi_j \mu_j\right) + B(\psi)F + \sum_{j=1}^N \left(\frac{1}{N} + \psi_j\right)\nu_j.$$

[32] Specifically, $\mathcal{B}^{\text{rel}} > 0$ if $M_j(\mathbf{0})(\mu_j - \mu_k) > 0$ or $M_j(\mathbf{0})(\mu_j - \mu_k) < -\sigma^2$; and $\mathcal{B}^{\text{rel}} < 0$ if $-\sigma^2 < M_j(\mathbf{0})(\mu_j - \mu_k) < 0$.

Then, the period 1 consumption is normally distributed: $C_1 \sim \mathcal{N}(\mathbb{E}[C_1], \mathbb{V}[C_1])$, where

$$\mathbb{E}[C_1] = \overline{\mu}_N + \sum_{j=1}^{N} \psi_j \mu_j \quad \text{and} \quad \mathbb{V}[C_1] = B(\psi)^2 \sigma_F^2 + \sum_{j=1}^{N} \left(\frac{1}{N} + \psi_j\right)^2 \sigma^2.$$

Given CARA utility, the price of the risk-free asset is:

$$\begin{aligned} p_f &= \delta \exp(\gamma C_0) \mathbb{E}[\exp(-\gamma C_1)] \\ &= \delta \exp(\gamma C_0) \exp\left((-\gamma)\mathbb{E}[C_1] + \frac{\gamma^2}{2}\mathbb{V}[C_1]\right), \end{aligned}$$

which coincides with (70), where we write $p_f = p_f(\psi)$ as any supply shock ψ_j affects p_f.

Moving on to the risky assets, we have:

$$p_j = \delta \exp(\gamma C_0) \mathbb{E}[Y_j \exp(-\gamma C_1)].$$

Since Y_j and $-\gamma C_1$ are jointly normal, we can rewrite $\mathbb{E}[Y_j \exp(-\gamma C_1)]$ as:

$$\mathbb{E}[Y_j \exp(-\gamma C_1)] = \mathbb{E}[\exp(-\gamma C_1)](\mathbb{E}[Y_j] + \text{Cov}[-\gamma C_1, Y_j]).^{33}$$

Thus, we obtain:

$$\begin{aligned} p_j &= \delta \exp(\gamma C_0) \mathbb{E}[Y_j \exp(-\gamma C_1)] \\ &= \underbrace{\delta \exp(\gamma C_0) \mathbb{E}[\exp(-\gamma C_1)]}_{=p_f} \left(\underbrace{\mathbb{E}[Y_j]}_{=\mu_j} - \gamma \text{Cov}[C_1, Y_j]\right) \\ &= p_f \left(\mu_j - \gamma \text{Cov}[C_1, Y_j]\right). \end{aligned}$$

Since

$$\begin{aligned} \text{Cov}[C_1, Y_j] &= \text{Cov}\left[B(\psi)F + \sum_{k=1}^{N}\left(\frac{1}{N} + \psi_k\right)\nu_k, \beta_j F + \nu_j\right] \\ &= B(\psi)\beta_j \sigma_F^2 + \left(\frac{1}{N} + \psi_j\right)\sigma^2, \end{aligned}$$

the price $p_j = p_j(\psi)$ of risky asset j satisfies $p_j(\psi) = p_f(\psi) M_j(\psi)$, i.e., (71). We remark that a change in ψ_k affects p_j through two channels: the change in p_f and the change in $\text{Cov}[C_1, Y_j]$ operating through $B(\psi)$.

[33] Generally, if X and Y are jointly normal, it is well-known that $\mathbb{E}[Ye^X] = \mathbb{E}[e^X](\mathbb{E}[Y] + \text{Cov}[X, Y])$.

Moving on to the demand functions, the investor's problem reduces to the following mean-variance problem:

$$\max_{(a,a_f)} \mathbb{E}[W_1] - \frac{\gamma}{2}\mathbb{V}[W_1]$$
$$\text{s.t. } p_f a_f + p^{\mathrm{T}} a = A_0,$$

where A_0 is the investor's initial wealth and

$$W_1 \equiv a_f + a^{\mathrm{T}} Y = \frac{A_0}{p_f} + a^{\mathrm{T}} \left(Y - \frac{p}{p_f} \right).$$

By the first-order condition, the demand functions are:

$$a(p, p_f) = \frac{1}{\gamma} \Sigma^{-1} \left(\mu - \frac{p}{p_f} \right).$$

Thus, by (67), the demand functions can be written as (69). Also, the demand function for the risk-free asset is written as: $a_f(p, p_f) = \frac{A_0 - p^{\mathrm{T}} a(p, p_f)}{p_f}$. ■

Proof of Proposition 8. *Part 1.* We write prices as functions of ψ_j, setting $\psi_k = 0$ for $k \neq j$. From (70), using $\frac{d}{d\psi_j}\mathbb{E}[C_1] = \mu_j$ and $\frac{d}{d\psi_j}\mathbb{V}[C_1] = 2B(\psi)\beta_j\sigma_F^2 + 2(\frac{1}{N} + \psi_j)\sigma^2$, we have:

$$\frac{dp_f}{d\psi_j} = p_f \left(-\gamma\mu_j + \gamma^2 \left(B(\psi)\beta_j\sigma_F^2 + \left(\frac{1}{N} + \psi_j \right) \sigma^2 \right) \right) = -\gamma p_f M_j(\psi). \tag{84}$$

Since $\frac{d}{d\psi_j}B = \beta_j$, (68) gives $\frac{d}{d\psi_j}M_j = -\gamma(\beta_j^2\sigma_F^2 + \sigma^2)$. Thus, the product rule applied to (71) yields

$$\frac{dp_j}{d\psi_j} = p_f \frac{dM_j}{d\psi_j} + M_j \frac{dp_f}{d\psi_j} = -\gamma p_f \left[M_j(\psi)^2 + \beta_j^2\sigma_F^2 + \sigma^2 \right].$$

Combining with $\frac{da_j}{d\psi_j} = 1$, the observed slope $-\frac{da_j/d\psi_j}{dp_j/d\psi_j}$ at $\psi = \mathbf{0}$ is given by (74).

Part 2. For ease of notation, denote by $Q_j \equiv \beta^{\mathrm{T}}\beta - \beta_j^2 = \sum_{k \neq j} \beta_k^2 \geq 0$ the factor exposure of the other assets. By (73) and Part 1, $\mathcal{J}_{jj}^i(N) > \hat{\mathcal{J}}_{jj}^i(N)$ is equivalent to

$$\frac{\sigma^2 + \sigma_F^2 Q_j}{\sigma^2 \left(\sigma^2 + \sigma_F^2 (Q_j + \beta_j^2) \right)} > \frac{1}{M_j(\mathbf{0})^2 + \beta_j^2\sigma_F^2 + \sigma^2}.$$

Since all denominators are positive, cross-multiplying and subtracting yields:

$$\begin{aligned}
&\left(\sigma^2+\sigma_F^2 Q_j\right)\left[M_j(\mathbf{0})^2+\beta_j^2\sigma_F^2+\sigma^2\right]-\sigma^2\left(\sigma^2+\sigma_F^2 Q_j+\sigma_F^2\beta_j^2\right)\\
&=\left(\sigma^2+\sigma_F^2 Q_j\right)M_j(\mathbf{0})^2+\left[\sigma^2\sigma_F^2\beta_j^2+\sigma^4+\sigma_F^4 Q_j\beta_j^2+\sigma^2\sigma_F^2 Q_j\right]-\left[\sigma^4+\sigma^2\sigma_F^2 Q_j+\sigma^2\sigma_F^2\beta_j^2\right]\\
&=\left(\sigma^2+\sigma_F^2 Q_j\right)M_j(\mathbf{0})^2+\sigma_F^4 Q_j\beta_j^2>0,
\end{aligned}$$

where the last inequality follows because $M_j(\mathbf{0})>0$.

Part 3. On the one hand, it follows from (73) that

$$\lim_{\sigma\downarrow 0}\mathcal{J}_{jj}^i(N)=\infty,\quad \text{provided } \sum_{k\neq j}\beta_k^2>0.$$

On the other hand, it follows from part 1 that

$$\lim_{\sigma\downarrow 0}\hat{\mathcal{J}}_{jj}^i(N)=\lim_{\sigma\downarrow 0}\frac{1}{\gamma p_f}\frac{1}{M_j(\mathbf{0})^2+\beta_j^2\sigma_F^2+\sigma^2}<\infty,$$

as we have

$$\lim_{\sigma\downarrow 0}p_f\in(0,\infty)\quad\text{and}\quad\lim_{\sigma\downarrow 0}M_j(\mathbf{0})\in(0,\infty).$$

Part 4. We have:

$$\frac{\mathcal{J}_{jj}^i(N)}{\hat{\mathcal{J}}_{jj}^i(N)}=\frac{\sigma^2+\sigma_F^2 Q_j}{\sigma^2\left(\sigma^2+\sigma_F^2\beta^{\mathrm{T}}\beta\right)}\left(M_j(\mathbf{0})^2+\beta_j^2\sigma_F^2+\sigma^2\right)\to\frac{(M_j^\infty)^2+\beta_j^2\sigma_F^2+\sigma^2}{\sigma^2}>1,$$

where $M_j^\infty\equiv\lim_{N\to\infty}M_j(\mathbf{0})$. ■

Remark 4 (Partial equilibrium) *We remark on estimation in partial equilibrium, where the risk-free rate is held fixed. For ease of presentation and as in KY25, normalize the gross risk-free rate by* 1*. Also, assume that* $\overline{\mu}=\mu_j$ *for all* j *and* $\beta=\mathbf{1}$*. On the one hand, the demand for risky asset* j *depends only on the own price in the limit as* $N\to\infty$*. On the other hand, when the number of assets* N *is fixed (however large it is), considering the limit as* $N\to\infty$ *is equivalent to considering the limit in which the asset payoff matrix exhibits no complementarity.*

As in the previous analysis and KY25, the demand function is written as:

$$a=\frac{1}{\gamma\sigma^2}\Big(\mu-p-\kappa(p)\mathbf{1}\Big),\quad \textit{where}\quad \kappa(p)=\frac{(\sigma_F^2/\sigma^2)}{1+N(\sigma_F^2/\sigma^2)}\mathbf{1}^{\mathrm{T}}(\mu-p)\in\mathbf{R}.$$

In the limit as $N \to \infty$ as in KY25, since

$$\lim_{N\to\infty} \kappa(p) = \lim_{N\to\infty} \frac{(\sigma_F^2/\sigma^2)}{1 + N(\sigma_F^2/\sigma^2)} \mathbf{1}^{\mathrm{T}}(\mu - p) = 0,$$

it follows that the demand a_j depends only on its own price p_j: $a_j(p_j) = \frac{1}{\gamma\sigma^2}(\overline{\mu} - p_j)$. Whenever N is fixed, however large N is (i.e., even if $\kappa(p)$ is already small),

$$\kappa(p) \to 0 \quad \textit{if and only if} \quad \frac{\sigma_F^2}{\sigma^2} \to 0$$

(provided that $\mathbf{1}^{\mathrm{T}}(\mu - p) \neq 0$). Thus, the effect of F has no information relative to the noise. In particular, if σ^2 is fixed, then $\sigma_F^2 \to 0$, in which case the assets do not exhibit any complementarity.

Proof of Lemma 8. By market clearing $a_j^i = \frac{1}{N} + \psi_j$ identically, and thus $\mathrm{Cov}_{\mathrm{cs}}[a_j^i, \psi_j] = \mathbb{V}_{\mathrm{cs}}[\psi_j]$. For the denominator, rewrite (71) using (68) and separate the terms that vary across j:

$$p_j = \underbrace{-\gamma p_f(\psi)\left[(1+\Psi)\sigma_F^2 + \frac{\sigma^2}{N}\right]}_{\text{common across } j} + p_f(\psi)\mu_j - \gamma\sigma^2 p_f(\psi)\psi_j.$$

Then, we have:

$$\begin{aligned}\mathrm{Cov}_{\mathrm{cs}}[p_j, \psi_j] &= p_f(\psi)\mathrm{Cov}_{\mathrm{cs}}[\mu_j, \psi_j] - \gamma\sigma^2 p_f(\psi)\mathbb{V}_{\mathrm{cs}}[\psi_j] \\ &= p_f(\psi)\mathbb{V}_{\mathrm{cs}}[\psi_j]\left(\hat{\varrho} - \gamma\sigma^2\right).\end{aligned}$$

Taking the ratio in (75) and canceling $\mathbb{V}_{\mathrm{cs}}[\psi_j] > 0$ gives (78). ■

Proof of Proposition 9. If $\sigma_\mu = 0$, then $\hat{\varrho} = 0$. By (76) and (78) with $\beta = \mathbf{1}$ and $\hat{\varrho} = 0$, we have

$$\hat{\mathcal{J}}^{\mathrm{KY25}} = \frac{1}{\gamma\sigma^2 p_f(\psi)} = \mathcal{J}^{\mathrm{rel}}.$$

Henceforth, we assume $\sigma_\mu > 0$.

Part 1. Since $\mu_j = \overline{\mu} + u_j$ and constants drop out of a covariance, $\mathrm{Cov}_{\mathrm{cs}}[\mu_j, \psi_j] = \mathrm{Cov}_{\mathrm{cs}}[u_j, \psi_j]$ and

$$\hat{\varrho} = \frac{\sum_j (\psi_j - \overline{\psi})u_j}{\sum_j (\psi_j - \overline{\psi})^2}.$$

Conditional on ψ, this is a linear combination of $(u_j)_j$. With $u_j \sim \mathcal{N}(0, \sigma_\mu^2)$ i.i.d. and $\psi \perp u$,

$$\mathbb{E}\left[\hat{\varrho} \mid \psi\right] = 0 \quad \text{and} \quad \mathbb{V}\left[\hat{\varrho} \mid \psi\right] = \frac{\sigma_\mu^2 \sum_j (\psi_j - \overline{\psi})^2}{\left(\sum_j (\psi_j - \overline{\psi})^2\right)^2} = \frac{\sigma_\mu^2}{N\mathbb{V}_{\mathrm{cs}}[\psi]},$$

and $\hat{\varrho}$ is exactly Gaussian. Hence, $\hat{\varrho} = (\gamma\sigma^2/\lambda)Z$ with $Z \sim \mathcal{N}(0,1)$, and (78) together with $\mathcal{J}^{\text{rel}} = 1/(\gamma\sigma^2 p_f)$ gives

$$\frac{\hat{\mathcal{J}}^{\text{KY25}}}{\mathcal{J}^{\text{rel}}} = \frac{\gamma\sigma^2}{\gamma\sigma^2 - \hat{\varrho}} = \frac{1}{1 - Z/\lambda} = \frac{\lambda}{\lambda - Z}. \tag{85}$$

Part 2. Since $p_f(\psi) > 0$, (78) shows $\hat{\mathcal{J}}^{\text{KY25}} < 0$ if and only if $\gamma\sigma^2 < \hat{\varrho}$ if and only if $Z > \lambda$. Hence, $\mathbb{P}(\hat{\mathcal{J}}^{\text{KY25}} < 0) = 1 - \Phi(\lambda)$, which tends to $1 - \Phi(0) = \frac{1}{2}$ as $\lambda \to 0$.

Part 3. By (77), $\hat{\mathcal{J}}^{\text{KY25}} \geq \mathcal{J}^i_{jj}(N)$ is equivalent to $\hat{\mathcal{J}}^{\text{KY25}}/\mathcal{J}^{\text{rel}} \geq 1/(1 + \tau(\mathbf{1}))$. If $Z > \lambda$ then by (85) the left side is negative and the inequality fails. If $Z < \lambda$ then $\lambda - Z > 0$ and, multiplying through by $(\lambda - Z)(1 + \tau(\mathbf{1})) > 0$, the inequality is equivalent to $\lambda(1 + \tau(\mathbf{1})) \geq \lambda - Z$, i.e., to $Z \geq -\tau(\mathbf{1})\lambda$, with probability $\Phi(\lambda) - \Phi(-\tau(\mathbf{1})\lambda)$.

For the limit, hold $\mathbb{V}_{\text{cs}}[\psi]$ and σ_μ fixed as N grows. Then, $\tau(\mathbf{1}) = O(\frac{1}{N})$ while $\lambda = O(\sqrt{N})$. Thus, $\tau(\mathbf{1})\lambda = O(N^{-1/2}) \to 0$ and $\Phi(-\tau(\mathbf{1})\lambda) \to \Phi(0) = \frac{1}{2}$, while $\Phi(\lambda) \to 1$. Hence the probability converges to $\frac{1}{2}$.

Part 4. Fix a realization of (μ, ψ) with $\hat{\varrho} \neq 0$; since $\sigma_\mu > 0$ and $\hat{\rho}$ is continuously distributed, this holds almost surely. The statistic $\hat{\varrho}$ is a function of μ and ψ alone and does not depend on σ^2. By (70),

$$p_f(\psi) \to p_f^0 \equiv \delta \exp\left(\gamma C_0 - \gamma\overline{\mu}_N - \gamma\sum_j \psi_j\mu_j + \frac{\gamma^2}{2}(1+\Psi)^2\sigma_F^2\right) \in (0,\infty) \quad \text{as } \sigma \downarrow 0.$$

Hence (78) yields

$$\hat{\mathcal{J}}^{\text{KY25}} = \frac{1}{p_f(\psi)\,(\gamma\sigma^2 - \hat{\varrho})} \to -\frac{1}{p_f^0\hat{\varrho}} < \infty \quad \text{and} \quad \mathcal{J}^{\text{rel}} = \frac{1}{\gamma\sigma^2 p_f} \to \infty \quad \text{a.s.}$$

Finally, since $\lambda = \gamma\sigma^2\sqrt{N\mathbb{V}_{\text{cs}}[\psi]}/\sigma_\mu \to 0$ as $\sigma \downarrow 0$, by Part 2 the probability of a negative estimate tends to $\frac{1}{2}$. ■

Proof of Proposition 10. *Part 1.* Following the logic of Proposition 1 of HHHKL25, at around the equilibrium prices, for any risky assets j and k,

$$\frac{da_j}{d\psi_j} = \frac{\partial a_j}{\partial p_j}\frac{dp_j}{d\psi_j} + \frac{\partial a_j}{\partial p_k}\frac{dp_k}{d\psi_j} + \sum_{\ell\neq j,k}\frac{\partial a_j}{\partial p_\ell}\frac{dp_\ell}{d\psi_j} + \frac{\partial a_j}{\partial p_f}\frac{dp_f}{d\psi_j};$$
$$\frac{da_k}{d\psi_j} = \frac{\partial a_k}{\partial p_j}\frac{dp_j}{d\psi_j} + \frac{\partial a_k}{\partial p_k}\frac{dp_k}{d\psi_j} + \sum_{\ell\neq j,k}\frac{\partial a_k}{\partial p_\ell}\frac{dp_\ell}{d\psi_j} + \frac{\partial a_k}{\partial p_f}\frac{dp_f}{d\psi_j}.$$

Under $\beta = \mathbf{1}$, since

$$\frac{\partial a_j}{\partial p_\ell} = \frac{1}{\gamma p_f \sigma^2} \frac{\sigma_F^2}{\sigma^2 + N\sigma_F^2} = \frac{\partial a_k}{\partial p_\ell}$$

and $\mathcal{J}^{\text{rel}}$ reduces to the single parameter, we obtain:

$$-\frac{da_j - da_k}{d\psi_j} = \mathcal{J}^{\text{rel}} \frac{dp_j - dp_k}{d\psi_j} - \left(\frac{\partial a_j}{\partial p_f} - \frac{\partial a_k}{\partial p_f} \right) \frac{dp_f}{d\psi_j}.$$

Dividing through by $(dp_j - dp_k)/d\psi_j \neq 0$ and using the definitions of $\hat{\mathcal{J}}^{\text{rel}}$ and $\mathcal{B}^{\text{rel}} = \mathcal{J}^{\text{rel}} - \hat{\mathcal{J}}^{\text{rel}}$ gives (79).

Part 2. We compute the three ingredients of (79) at $\psi = \mathbf{0}$. First, $\left(\frac{\partial a_j}{\partial p_f} - \frac{\partial a_k}{\partial p_f} \right)\Big|_{\psi=\mathbf{0}}$ is given by (83) (recall (72)). Second, by (84), $\frac{dp_f}{ds_j}\Big|_{\psi=\mathbf{0}} = -\gamma p_f M_j(\mathbf{0})$.

Third, since $p_\ell = p_f(\psi) M_\ell(\psi)$ with $M_\ell(\psi) = \mu_\ell - \gamma((1+\Psi)\sigma_F^2 + (\frac{1}{N} + \psi_\ell)\sigma^2)$, shocking only ψ_j gives $\frac{dM_\ell}{d\psi_j} = -\gamma(\sigma_F^2 + \sigma^2 \mathbf{1}_{\ell=j})$. By the product rule,

$$\frac{dp_\ell}{d\psi_j}\bigg|_{\psi=\mathbf{0}} = p_f \left.\frac{dM_\ell}{d\psi_j}\right|_{\psi=\mathbf{0}} + M_\ell \left.\frac{dp_f}{d\psi_j}\right|_{\psi=\mathbf{0}} = -\gamma p_f \left[\sigma_F^2 + \sigma^2 \mathbf{1}_{\ell=j} + M_\ell(\mathbf{0}) M_j(\mathbf{0})\right].$$

Differencing across $\ell \in \{j,k\}$, the common factor-risk term σ_F^2 cancels and, since $M_j(\mathbf{0}) - M_k(\mathbf{0}) = \mu_j - \mu_k$ under $\beta = \mathbf{1}$,

$$\frac{dp_j - dp_k}{d\psi_j}\bigg|_{\psi=\mathbf{0}} = -\gamma p_f \left[\sigma^2 + M_j(\mathbf{0})(\mu_j - \mu_k)\right]. \tag{86}$$

Substituting (83), $\frac{dp_f}{ds_j}$ and (86) into (79),

$$\mathcal{B}^{\text{rel}} = \frac{\mu_j - \mu_k}{\gamma p_f \sigma^2} \cdot \frac{-\gamma p_f M_j(\mathbf{0})}{-\gamma p_f \left[\sigma^2 + M_j(\mathbf{0})(\mu_j - \mu_k)\right]} = \frac{M_j(\mathbf{0})(\mu_j - \mu_k)}{\gamma p_f \sigma^2 \left(\sigma^2 + M_j(\mathbf{0})(\mu_j - \mu_k)\right)},$$

which is the second expression in (80). The first expression in (80) follows because

$$\hat{\mathcal{J}}^{\text{rel}} = -\frac{d(a_j - a_k)/d\psi_j}{d(p_j - p_k)/d\psi_j} = \frac{1}{\gamma p_f \left(\sigma^2 + M_j(\mathbf{0})(\mu_j - \mu_k)\right)}.$$

Part 3. The first limit follows because, by (68), $\lim_{N\to\infty} M_j(\mathbf{0}) = \mu_j - \gamma\sigma_F^2 < \infty$ and $\lim_{N\to\infty} p_f < \infty$. The second limit follows because, by (68), $\lim_{\sigma\downarrow 0} M_j(\mathbf{0}) = \mu_j - \gamma\sigma_F^2 \in (0,\infty)$ and $\lim_{\sigma\downarrow 0} p_f < \infty$. ■

C No Arbitrage with Tastes

In many applications, researchers aim to control for certain risk exposures or use a low-dimensional factor representation of the matrix of expected returns to model demand. No arbitrage is used to ensure that assets are priced according to their risk exposures. We now discuss the relation between taste heterogeneity and the principle of no arbitrage.

In the standard definition, investors care only about cash flows and an arbitrage is "an investment strategy that guarantees a positive payoff in some contingency with no possibility of a negative payoff and no initial net investment" (Ross, 2004). When investors differ in tastes, they have subjective views on the payoffs of a given trade.

To define no arbitrage with tastes, we provide two preliminary definitions. First, letting a vector subspace $\mathcal{A}$ of $\mathbf{R}^J$ denote the set of feasible portfolios and letting $(p_j)_{j\in\mathcal{J}}$ be a price vector of individual assets, the pricing function $P : \mathcal{A} \to \mathbf{R}$ maps a portfolio $a = (a_j)_{j\in\mathcal{J}}$ into its price according to $P(a) \equiv \sum_{j\in\mathcal{J}} p_j a_j$. Second, investor i has a linear taste function $v^i : \mathcal{A} \to \mathbf{R}^Z$ that maps a portfolio a into a vector $v^i(a)$ of state-contingent taste-augmented payoffs for investor i. Specifically, let $Y^i \equiv (\theta_j^i y_j(z))_{z,j}$ be the $Z \times J$ matrix of investor i's payoff-augmented cash flows, where $y_j(z) \geq 0$ and $(y_j(z))_z \neq \mathbf{0}$ for each j and z. Then, let $v^i(a) \equiv Y^i a$. This generalizes the neoclassical approach in which $\theta_j^i = 1$ for all j. We then have the following definition.

Definition 5 (No arbitrage with tastes) *Let taste functions v^i be given for all investors i. The pricing function P leaves no arbitrage opportunities if, for any investor i and any portfolio $a \in \mathcal{A}$ such that the effective payoff is weakly positive almost surely (i.e., $v^i(a) \geq 0$) and strictly positive with strictly positive probability (i.e., $v^i(a) > 0$), the associated price is positive: $P(a) > 0$.*

That is, pricing function P leaves no arbitrage opportunities given taste functions $(v^i)_i$ if and only if, for every i, the pricing function P leaves no arbitrage opportunities in the standard sense if the cash-flow matrix is replaced by the taste-augmented payoff matrix. The key difficulty is that this payoff matrix is investor-specific. The main restriction is that taste functions are linear, as they are in KY19.

Proposition 11 (Generic arbitrage opportunities with tastes) *Fix taste functions v^i for all*

i. There does not exist a pricing function P that leaves no arbitrage opportunities if:

$$\text{there exist } a, i, \text{ and } i' \text{ such that } v^i(a) > 0 \text{ and } 0 > v^{i'}(a). \tag{C}$$

A sufficient condition for (C) is that there exist assets j and j' such that

(i) both assets have identical cash flows:

$$y_j(z) = y_{j'}(z) \text{ for all } z \in \mathcal{Z};$$

(ii) there exist investors i and i' with sufficiently heterogeneous tastes with respect to these assets:

$$\theta_j^i > \theta_{j'}^i \text{ and } \theta_j^{i'} < \theta_{j'}^{i'}.$$

Proof of Proposition 11. We first show that there does not exist a pricing function P that leaves no arbitrage opportunities if (C) holds. We then establish the stated sufficiency condition for (C).

Suppose first that (C) holds. Suppose to the contrary that there exists a pricing function P that leaves no arbitrage opportunities. Applying Definition 5 to investor i yields $P(a) > 0$. But applying Definition 5 to investor i' yields $P(-a) > 0$, i.e., $P(a) < 0$. This is a contradiction.

Next, suppose that the stated conditions hold: there exist two assets j and j' and two investors i and i' satisfying the two conditions. Denoting by $v_j^i \equiv (\theta_j^i y_j(z))_z$ investor i's marginal taste with respect to asset j, the two conditions imply:

$$v_j^i > v_{j'}^i \text{ and } v_j^{i'} < v_{j'}^{i'}. \tag{C'}$$

It is then sufficient to show that (C') implies condition (C). Let $a \in \mathbf{R}^J$ be such that

$$a_\ell = \begin{cases} 1 & \text{if } \ell = j \\ -1 & \text{if } \ell = j' \\ 0 & \text{otherwise} \end{cases}.$$

Then, we obtain $v^i(a) = v_j^i - v_{j'}^i > 0$ and $v^{i'}(a) = v_j^{i'} - v_{j'}^{i'} < 0$, as desired. ■

The following example provides a simple illustration consistent with our model.

Example 1 (Green and red assets) *There are a green asset and a red asset with prices denoted by p_g and p_r, respectively. Both assets deliver a unit payoff with certainty. There are two investor*

types, denoted by α and β, that differ in their relative taste for the two assets. For each investor type i, the taste function is given by $v^i(a_g, a_r) = \theta^i_g a_g + \theta^i_r a_r$ with the following properties: while type α's taste-augmented payoffs for green and red assets satisfy $\theta^\alpha_g > \theta^\alpha_r$, type β has $\theta^\beta_g < \theta^\beta_r$. Then there are no prices such that both investors agree on the value of a long-short portfolio selling one unit of the green asset and buying one unit of the red asset.

D The Absolute Elasticity Implied by β_0

In the baseline logit model, the estimating equation is written in terms of portfolio shares relative to the outside asset, so that β_0 is identified from the *difference* of observed elasticities: $\hat{\beta}_0 = \hat{\mathcal{E}}^i_{jj} - \hat{\mathcal{E}}^i_{oj}$. We show that the logit demand system nevertheless determines the *absolute* elasticity $\mathcal{E}^{\omega,i}_{jj} = -\frac{\partial \log \omega_j}{\partial \log p_j}$, because the response of the outside asset is pinned down by β_0 and observed portfolio shares. We then show that the restriction this imposes on the outside asset fails in the three-asset economy of Definition 4 outside of the knife-edge case $\epsilon = 1$.

Consider the general case with $J-1$ inside assets and one outside asset o, and suppress the investor index i. As in KY19, the outside asset comprises all holdings outside the set of inside assets, so that inside and outside shares sum to one. Under Assumption 1, the logit demand system (17) imposes two restrictions on the elasticity matrix. First, for inside assets $j \neq k$, the relative share $\frac{\omega_j}{\omega_o}$ does not depend on p_k:

$$0 = \frac{\partial}{\partial \log p_k} \log\left(\frac{\omega_j}{\omega_o}\right)(p), \quad \text{that is,} \quad \frac{\partial \log \omega_j(p)}{\partial \log p_k} = \frac{\partial \log \omega_o(p)}{\partial \log p_k}. \tag{87}$$

Second, the own-price coefficient is

$$\beta_0 = -\frac{\partial}{\partial \log p_j} \log\left(\frac{\omega_j}{\omega_o}\right)(p), \quad \text{that is,} \quad \frac{\partial \log \omega_j(p)}{\partial \log p_j} = \frac{\partial \log \omega_o(p)}{\partial \log p_j} - \beta_0. \tag{88}$$

Restrictions (87) and (88) express every element of the elasticity matrix in terms of β_0 and the outside asset's own row $\left(\frac{\partial \log \omega_o}{\partial \log p_k}\right)_k$. We next show that this row is fully determined by budget balance.

Lemma 9 (Budget balance determines the outside asset's response) *In the logit demand system (17), for every inside asset k,*

$$\frac{\partial \log \omega_o(p)}{\partial \log p_k} = \beta_0 \omega_k. \tag{89}$$

Proof of Lemma 9. Portfolio shares satisfy the identity $\sum_{j\neq o}\omega_j(p)+\omega_o(p)=1$ for every price vector p. Differentiating with respect to $\log p_k$ and writing each term as $\frac{\partial\omega_j}{\partial\log p_k}=\omega_j\cdot\frac{\partial\log\omega_j}{\partial\log p_k}$, we have:

$$\sum_{j\neq o}\omega_j\frac{\partial\log\omega_j(p)}{\partial\log p_k}+\omega_o\frac{\partial\log\omega_o(p)}{\partial\log p_k}=0. \tag{90}$$

Write $d_k\equiv\frac{\partial\log\omega_o(p)}{\partial\log p_k}$. Substituting (87) for each inside asset $j\neq k$ and (88) for asset k into (90) gives

$$d_k\Big(\sum_{j\neq o}\omega_j+\omega_o\Big)-\omega_k\beta_0=d_k-\omega_k\beta_0=0,\quad\text{that is,}\quad d_k=\beta_0\omega_k,$$

which is (90). ■

Substituting (89) into (87) and (88) then yields the full elasticity matrix. In the logit demand system (17), for all inside assets j and k, it follows that the elasticity matrix $\mathcal{E}^{\text{logit}}\equiv\left(\mathcal{E}^{\omega,i}_{jk}\right)_{j,k}$ satisfies:

$$\mathcal{E}^{\text{logit}}_{jk}=\beta_0\left(\mathbf{1}_{j=k}-\omega_k\right)\quad\text{and}\quad\mathcal{E}^{\text{logit}}_{ok}=-\beta_0\omega_k. \tag{91}$$

Thus, the absolute own-price elasticity is $\mathcal{E}^{\text{logit}}_{jj}=\beta_0(1-\omega_j)$. In matrix form, this is

$$\left[\mathcal{E}^{\text{logit}}_{jk}\right]_{j,k}=\beta_0\left(I-\mathbf{1}\omega'\right),$$

where I is the identity matrix, $\mathbf{1}$ is the vector of ones, and $\omega\equiv(\omega_1,\ldots,\omega_{J-1})^{\mathrm{T}}$.

This is the IIA property stated in Section 4.1 with β_0 as the substitution parameter. Given β_0 and observed portfolio shares, the logit demand system thus delivers the entire matrix of absolute own- and cross-price elasticities. However, this is possible only because the logit demand system *restricts* the outside asset's response. In levels, (91) requires asset k to release a portfolio share $\beta_0\omega_k(1-\omega_k)$ in response to an increase in p_k, of which every remaining asset j absorbs the fraction $\omega_j/(1-\omega_k)$. The outside asset thus absorbs a fraction $\omega_o/(1-\omega_k)$ of the substitution away from asset k, determined by its own portfolio share alone and irrespective of its payoff overlap with asset k. This follows from the same IIA restriction (87) whose failure generates the bias in Proposition 3.

The three-asset economy. We now evaluate restriction (89) in the economy of Definition 4 with $\phi=0$ and log utility (we suppress the superscript i). Under log utility the expenditure share on each state-contingent claim equals its probability. With $\phi=0$, asset

2 is the only asset that pays in state 2, so $\omega_2(p) = \pi_2$ for every p. Hence the observed elasticity of the outside asset is

$$\hat{\mathcal{E}}_{2g} = \hat{\mathcal{E}}_{2r} = 0. \tag{92}$$

Since $\hat{\beta}_0 = \hat{\mathcal{E}}_{gg} - \hat{\mathcal{E}}_{2g}$, the green asset's observed absolute own-price elasticity is

$$\hat{\mathcal{E}}_{gg} = \hat{\beta}_0. \tag{93}$$

The logit coefficient thus coincides with the *observed* absolute elasticity, not the *structural* absolute elasticity. Portfolio shares sum to one along the equilibrium path induced by the supply shock ψ, so identity (90) applies to the observed responses as well. By (92) and (93), the observed cross-price elasticity in the true model is

$$\hat{\mathcal{E}}_{rg} = -\frac{\omega_g}{\omega_r}\hat{\beta}_0. \tag{94}$$

We can then ask whether the logit model's elasticity matrix can account for these *observed* substitution patterns. Taking as given the observed elasticity $\hat{\beta}_0 = \hat{\mathcal{E}}_{gg}$ (which by Proposition 4 differs from the structural elasticity), logit requires the own- and cross-terms to equal

$$\mathcal{E}^{\text{logit}}_{gg} = \hat{\beta}_0(1-\omega_g), \quad \mathcal{E}^{\text{logit}}_{rg} = -\hat{\beta}_0\omega_g, \quad \text{and} \quad \mathcal{E}^{\text{logit}}_{2g} = -\hat{\beta}_0\omega_g.$$

Comparing these to the actual observed elasticities generated by the model yields

$$\mathcal{E}^{\text{logit}}_{gg} - \hat{\mathcal{E}}_{gg} = \mathcal{E}^{\text{logit}}_{2g} - \hat{\mathcal{E}}_{2g} = -\hat{\beta}_0\omega_g \quad \text{and} \quad \mathcal{E}^{\text{logit}}_{rg} - \hat{\mathcal{E}}_{rg} = \hat{\beta}_0\omega_g\frac{1-\omega_r}{\omega_r}. \tag{95}$$

Each discrepancy is proportional to $\hat{\beta}_0\omega_g$. Since $\omega_g > 0$, logit thus matches the observed elasticity matrix if and only if $\hat{\beta}_0 = 0$. By (28),

$$\hat{\beta}_0 = \frac{1-\epsilon^2}{(1+\epsilon)^2 - 4\epsilon\rho},$$

which is zero if and only if $\epsilon = 1$. Restriction (89) is thus satisfied by the observed responses if and only if $\epsilon = 1$, which is the same knife-edge for which there is no spillover bias.

Remark 5 *The discrepancies in* (95) *concern the matrix of elasticities recovered by the logit system given a (biased) estimate of the substitution parameter* $\hat{\beta}_0$*. As such, they are distinct from (and additive to) the gap between the observed and structural elasticity derived in Proposition* 4*.*

E Dynamic Trading

Consider a two-period variant of our baseline model in Section 4.3. Trees are durable assets which pay dividends in two periods. Investors trade at the beginning of each period, and consume the per-period payoffs generated by their trees at the end of each period. The payoff structure is the same as in our baseline framework with $\phi = 0$ as in the analysis in Section 4.3, and the aggregate state processes $z \in \{1, 2\}$ and $\iota \in \{g, r\}$, which determine payoffs, are i.i.d. across periods. The discount factor is given by $\delta \in (0, 1)$.

We denote by $\mathbf{S}_t$ the aggregate state variable sufficient for determining prices, which naturally includes the aggregate distribution over investor wealth and tastes. We use this structure to introduce potential variations in market prices over time. In particular, investors know the realization of $\mathbf{S}_1$ when forming portfolio allocations at date 1, and their choices are also influenced by their expectations of $\mathbf{S}_2$.

We write the price of asset $j \in \mathcal{J} \equiv \{g, r, 2\}$ as a function of the aggregate state: $p_{j,t} = P_j(\mathbf{S}_t)$ for some endogenous function P_j. The state-contingent gross return on asset $j \in \mathcal{J}$ is $R_j(\mathbf{S}_2) = \frac{P_j(\mathbf{S}_2)}{P_j(\mathbf{S}_1)}$. Given heterogeneity in tastes, a single investor will thus be concerned with the fact that changes in the preferences or wealth of other investors can induce changes in prices and, therefore, her perceptions of expected returns. It is sufficient for our purposes to consider the decision problem of a single investor who takes as given the stochastic process over the aggregate state. The investor's wealth at the beginning of period t is determined by the realized state and previous asset positions: $W_t^i(\mathbf{a}_{t-1}^i, \mathbf{S}_t) = \sum_{j \in \mathcal{J}} P_j(\mathbf{S}_t) a_{j,t-1}^i$.

We denote investor i's purchases of trees at the beginning of period $t \in \{1, 2\}$ by $\mathbf{a}_t^i = (a_{g,t}^i, a_{r,t}^i, a_{2,t}^i)$, and we denote by $\mathbf{a}_0^i$ investor i's exogenous endowment. Investor i's state variable at the beginning of period $t \in \{1, 2\}$ consists of the portfolio of asset positions purchased in the previous period, $\mathbf{a}_{t-1}^i$, and the current-period taste parameters $\Theta_t^i = (\theta_{j,t}^i)_{j \in \{g,r,2\}}$, which are permitted to evolve stochastically over time. The individual state at the beginning of period t is therefore $\mathbf{s}_t^i = (\mathbf{a}_{t-1}^i, \Theta_t^i)$.

We solve the problem by backwards induction, assuming that investors face short-sale constraints. For ease of exposition, assume that green and red trees are perfect substitutes: $\epsilon = 0$. In this case, the second-period choice between green and red trees is bang-bang: the investor buys only green trees if $\theta_{g,2}^i / P_g(\mathbf{S}_2) > \theta_{r,2}^i / P_r(\mathbf{S}_2)$, and only red trees if the inequality is reversed. We then have the following characterization of the second-period value function.

Lemma 10 (Value function) *Let $\epsilon = 0$ and $u = \log$. The second-period value function satis-*

fies:

$$V_2^i(\mathbf{s}_2^i, \mathbf{S}_2) = H(\Theta_2^i, \mathbf{S}_2) + \log\left(W_2^i(\mathbf{a}_1^i, \mathbf{S}_2)\right),$$

where $H(\Theta_2^i, \mathbf{S}_2)$ *depends on investor* i*'s tastes and market prices in period* 2*, but is independent of any investor choices at date* 1*.*

Proof of Lemma 10. Following the discussion of the static optimization problem, the second-period value function can be written as:

$$V_2^i(\mathbf{s}_2^i, \mathbf{S}_2) = \max_{\mathbf{a}_2^i \geq 0} \pi_1 \left[\rho u\left(\theta_{g,2}^i y_g(r) a_{g,2}^i + \theta_{r,2}^i y_r(r) a_{r,2}^i\right) + (1-\rho) u\left(\theta_{g,2}^i y_g(g) a_{g,2}^i + \theta_{r,2}^i y_r(g) a_{r,2}^i\right)\right]$$
$$+\pi_2 u\left(\frac{y(2)}{P_2(\mathbf{S}_2)}\left(W_2^i(\mathbf{a}_1^i, \mathbf{S}_2) - P_g(\mathbf{S}_2) a_{g,2}^i - P_r(\mathbf{S}_2) a_{r,2}^i\right)\right).$$

Since green and red assets are perfect substitutes (i.e., $\epsilon = 0$), the solution to the second-period decision problem is bang-bang, depending on whether $\theta_{g,2}^i / P_g(\mathbf{S}_2) \geq \theta_{r,2}^i / P_r(\mathbf{S}_2)$. Suppose that this inequality holds (i.e., green trees are cheap). Then, the value of the second-period problem is:

$$V_{g,2}^i(\mathbf{s}_2^i, \mathbf{S}_2) = \max_{a_{g,2}^i \geq 0} \pi_1 u\left(\theta_{g,2}^i y(1) a_{g,2}^i\right) + \pi_2 u\left(\frac{y(2)}{P_2(\mathbf{S}_2)}\left(W_2^i(\mathbf{a}_1^i, \mathbf{S}_2) - P_g(\mathbf{S}_2) a_{g,2}^i\right)\right).$$

The first-order condition together with log utility yields

$$a_{g,2}^i = \frac{\pi_1 W_2^i(\mathbf{a}_1^i, \mathbf{S}_2)}{P_g(\mathbf{S}_2)}.$$

This means that

$$V_{g,2}^i(\mathbf{s}_2^i, \mathbf{S}_2) = \pi_1 u\left(\theta_{g,2}^i \frac{\pi_1 y(1) W_2^i(\mathbf{a}_1^i, \mathbf{S}_2)}{P_g(\mathbf{S}_2)}\right) + \pi_2 u\left(\frac{y(2)}{P_2(\mathbf{S}_2)} \pi_2 W_2^i(\mathbf{a}_1^i, \mathbf{S}_2)\right).$$

With log utility this can be written as

$$V_{g,2}^i(\mathbf{s}_2^i, \mathbf{S}_2) = \pi_1 \log\left(\frac{\theta_{g,2}^i}{P_g(\mathbf{S}_2)}\right) + \pi_2 \log\left(\frac{1}{P_2(\mathbf{S}_2)}\right) + \pi_1 \log\left(\pi_1 y(1)\right) + \pi_2 \log\left(\pi_2 y(2)\right)$$
$$+ \log\left(W_2^i(\mathbf{a}_1^i, \mathbf{S}_2)\right).$$

Similarly, if $\theta^i_{g,2}/P_g(\mathbf{S}_2) \leq \theta^i_{r,2}/P_r(\mathbf{S}_2)$, the value of the second-period problem is:

$$V^i_{r,2}(\mathbf{s}^i_2, \mathbf{S}_2) = \pi_1 \log\left(\frac{\theta^i_{r,2}}{P_r(\mathbf{S}_2)}\right) + \pi_2 \log\left(\frac{1}{P_2(\mathbf{S}_2)}\right) + \pi_1 \log\left(\pi_1 y(1)\right) + \pi_2 \log\left(\pi_2 y(2)\right) + \log\left(W^i_2(\mathbf{a}^i_1, \mathbf{S}_2)\right).$$

Hence, the second-period value function is written as

$$V^i_2(\mathbf{s}^i_2, \mathbf{S}_2) = \max\left\{V^i_{g,2}(\mathbf{s}^i_2, \mathbf{S}_2), V^i_{r,2}(\mathbf{s}^i_2, \mathbf{S}_2)\right\},$$

and it satisfies

$$V^i_2(\mathbf{s}^i_2, \mathbf{S}_2) = H(\Theta^i_2, \mathbf{S}_2) + \log\left(W^i_2(\mathbf{a}^i_1, \mathbf{S}_2)\right),$$

where

$$H(\Theta^i_2, \mathbf{S}_2) = \pi_1 \max\left\{\log\left(\frac{\theta^i_{r,2}}{P_r(\mathbf{S}_2)}\right), \log\left(\frac{\theta^i_{g,2}}{P_g(\mathbf{S}_2)}\right)\right\} + \pi_2 \log\left(\frac{1}{P_2(\mathbf{S}_2)}\right) + \pi_1 \log\left(\pi_1 y(1)\right) + \pi_2 \log\left(\pi_2 y(2)\right),$$

as desired. ■

The separability of the value function into a taste component and a wealth component follows from log utility. However, this feature is not essential for our results below. What is important is that investors take into account future market prices when forming portfolios. Indeed, weakening separability would further complicate identification.

Now turn to the first-period decision problem. Maintaining the assumptions of log utility and normalizing $P_2(\mathbf{S}_1) = 1$, we can then write the period 1 decision problem under discount factor δ as:

$$V^i_1(\mathbf{s}^i_1, \mathbf{S}_1) = \max_{\mathbf{a}^i_1 \geq 0} \left[\pi_1 \log\left(y(1)\left(\theta^i_{g,1} a^i_{g,1} + \theta^i_{r,1} a^i_{r,1}\right)\right) + \pi_2 \log\left(y(2)\left(W^i_1(\mathbf{a}^i_0, \mathbf{S}_1) - P_g(\mathbf{S}_1) a^i_{g,1} - P_r(\mathbf{S}_1) a^i_{r,1}\right)\right)\right] + \delta \mathbb{E}^i\left[H(\Theta^i_2, \mathbf{S}_2) + \log\left(\sum_{j \in \mathcal{J}} P_j(\mathbf{S}_2) a^i_{j,1}\right)\right],$$

where the expectation $\mathbb{E}^i$ is taken with respect to private tastes and the aggregate state variable at date 2. Differentiating this object with respect to $a^i_{j,1}$ yields (39).

F Tastes versus Constraints

We use the three-asset economy of Definition 4 with the following simplifying assumptions:

(i) There are two types of investors, $i \in \{G, R\}$, each of which faces short-sale constraints. Tastes satisfy: $\theta_g^G = 1 + t$ and $\theta_r^G = 1 - t$, while $\theta_g^R = 1 - t$ and $\theta_r^R = 1 + t$.

(ii) Type G (i.e., green investors) owns share $\gamma \geq \frac{1}{2}$ of the aggregate endowment of each tree.

(iii) There is no aggregate risk, $\phi = 0$, $y(1) = y(2) = 1$, $\pi_1 = \frac{1}{2}$, and $\rho = \frac{1}{2}$.

We begin with the baseline where investors face only short-sale constraints. In this case, taste differences can lead to endogenous sorting in equilibrium.

First, look for an equilibrium in which type G specializes in green trees while type R investor specializes in red trees. It follows from the first-order conditions with respect to a_g^G and a_r^R and the aggregate resource constraint on tree 2 that

$$\frac{p_r}{p_g} = \frac{1 - a_2^G}{a_2^G} \quad \text{and} \quad p_1 \equiv \frac{p_g + p_r}{2} = \frac{\pi_1}{1 - \pi_1} = 1,$$

where the last equality follows from our simplifying assumptions. Substituting p_1 and the first-order condition with respect to a_g^G into type G investor's budget constraint, together with the aggregate resource constraint, one can show:

$$(a_2^G, a_2^R) = (\gamma, 1 - \gamma).$$

Thus, if sorting occurs in equilibrium, then the prices satisfy

$$(p_g, p_r) = \left(2\gamma \frac{\pi_1}{1 - \pi_1}, 2(1 - \gamma) \frac{\pi_1}{1 - \pi_1}\right) = (2\gamma, 2(1 - \gamma))$$

and the investors' portfolio choices are

$$(a_g^G, a_r^G, a_2^G) = \left(\frac{1}{2}, 0, \gamma\right) \quad \text{and} \quad (a_g^R, a_r^R, a_2^R) = \left(0, \frac{1}{2}, 1 - \gamma\right).$$

To show that this constitutes an equilibrium, we need to show that the first-order conditions with respect to a_r^G and a_g^R hold at the zero holding. It can be seen that these first-order

conditions are met as long as type G's incentive is satisfied:

$$\gamma \leq \overline{\gamma} \equiv \frac{\theta_g^G}{\theta_g^G + \frac{(y(1))^2+\epsilon^2}{(y(1))^2-\epsilon^2}\theta_r^G}.$$

Next, we consider an equilibrium in which one type of investor, denoted by i, holds both green and red trees while the other type specializes in one tree. Guessing that $\tilde{c}^i(g) = \tilde{c}^i(r) = \tilde{c}^i(2)$, first-order conditions with respect to a_g^i and a_r^i imply that

$$(p_g, p_r, p_1) = \left(\theta_g^i, \theta_r^i, 1\right).$$

Combining these conditions with the budget constraint implies that

$$a_2^i = (1-\pi_1)e_2^i + \pi_1 \frac{y(1)}{y(2)}(\theta_g^i e_g^i + \theta_r^i e_r^i) = \frac{e_2^i + \theta_g^i e_g^i + \theta_r^i e_r^i}{2}.$$

We also guess and verify that $i = G$. Since type R specializes in red, $a_g^R = 0$ and $a_g^G = \frac{1}{2}$. Then, by the aggregate resource constraint,

$$a_r^G = \frac{a_2^G - \theta_g^G E_g}{\theta_r^G}.$$

Since the first-order condition with respect to a_r^R yields $a_r^R = \frac{a_2^R}{p_r}$, it follows from the budget constraint that

$$a_2^R = (1-\pi_1)(e_2^R + p_g e_g^R + p_r e_r^R) = 1-\gamma \quad \text{and} \quad a_r^R = (1-\pi_1)\frac{e_2^R + p_g e_g^R + p_r e_r^R}{p_r} = \frac{1-\gamma}{p_r}.$$

Thus, we obtain

$$a_r^R = \frac{1-\gamma}{2}\frac{E_2 + \theta_g^G E_g + \theta_r^G E_r}{\theta_r^G}.$$

When $\gamma > \overline{\gamma}$, it can be seen that $a_r^G > 0$ and that the first-order condition with respect to a_g^R at $a_g^R = 0$ is also met (i.e., $a_g^R = 0$).

G Physical versus Substitution Measure: An Example

This appendix illustrates how far the substitution measure can depart from the physical measure, and what this implies for factor controls built from physical payoff moments.

There are two normal states and one disaster state, and three assets with the following payoffs:

	Normal 1	Normal 2	Disaster
Physical probability π_z	0.495	0.495	0.01
Asset A payoff	0	2	11
Asset B payoff	2	0	11
Asset C payoff	0	2	0

Under the physical measure, assets A and B are uncorrelated and have the same variance, $\mathrm{Cov}_\pi(y_\mathrm{A}, y_\mathrm{B}) = 0$ and $\mathrm{Var}_\pi(y_\mathrm{A}) = \mathrm{Var}_\pi(y_\mathrm{B}) = 1.98$. Asset A also looks much closer to asset C than to asset B: $\mathbb{E}_\pi[(y_\mathrm{A} - y_\mathrm{C})^2] = 1.21$, whereas $\mathbb{E}_\pi[(y_\mathrm{A} - y_\mathrm{B})^2] = 3.96$.

Now let a representative investor have log utility, with consumption equal to 1 in the normal states and $1/100$ in the disaster. Marginal utility in the disaster is 100 times its normal level, but $-u''(c) = 1/c^2$ is 10,000 times its normal level. By (6), the disaster therefore receives weight $\hat{\pi}_\mathrm{D} \approx 0.990$ under the substitution measure, and each normal state receives weight ≈ 0.0049. Under $\hat{\pi}$, the picture reverses:

$$\widehat{\mathbb{E}}[(y_\mathrm{A} - y_\mathrm{C})^2] \approx 119.81, \qquad \widehat{\mathbb{E}}[(y_\mathrm{A} - y_\mathrm{B})^2] \approx 0.039,$$

and the correlation between assets A and B rises from 0 to approximately 0.98. Because the payoff overlap matrix H in (5) is built from *uncentered* second moments, the relevant comparison is the normalized second moment $\widehat{\mathbb{E}}[y_j y_k] / \sqrt{\widehat{\mathbb{E}}[y_j^2]\widehat{\mathbb{E}}[y_k^2]}$. Between assets A and B it rises from 0.38 under π to 0.9998 under $\hat{\pi}$; between assets A and C it falls from 0.79 to 0.013. The physical measure thus misstates not only the degree of substitutability but also its ranking: it identifies asset C as asset A's closest substitute, whereas under the substitution measure that substitute is asset B.

The implication for controls follows directly. A factor model fit to physical payoff moments would treat assets A and B as well-behaved residuals, and would match asset A with asset C on observables. Neither conclusion survives once payoffs are weighted by utility curvature. A good factor fit under π is therefore not sufficient for the structure required by Lemma 5.

The example fixes consumption exogenously and does not solve for an equilibrium; it only illustrates how the substitution measure reweights states. The ranking reversal does not rely on an extreme disaster.